



TATA
CONSULTANCY
SERVICES

DEDUCTIVE REASONING

DEDUCTIVE REASONING

LOGICAL DEDUCTION



IMPORTANCE OF THIS TOPIC

- This is one of the most important topic in logical reasoning.
- It is a very important topic for placements in IT Companies.
- Questions from this topic are asked in almost all the examinations like IT Placements, CAT, NMAT, SNAP, MAT, CMAT, CET etc.

- **STATEMENT & CONCLUSION**
 - **STATEMENT & ARGUMENT**
 - **STATEMENT & ASSUMPTION**
 - **STATEMENT & COURSE OF ACTION**
- 

Q. Statement: An advertisement of TCS NQT - "If you are a software engineer, we want to hire you".



Conclusion:

1. TCS hires no person with other qualification.
2. TCS is in need of software engineers.

- (A) If only conclusion 1 follows
- (B) If only conclusion 2 follows
- (C) If either 1 or 2 follows
- (D) If neither 1 nor 2 follows and
- (E) If both 1 and 2 follow.

Q. Statement: Should Military training be made compulsory in our country?

Arguments:

1. Yes, every child should go through this training.
2. No, it is not compulsory in USA.

- (A) If only argument 1 is strong
- (B) If only argument 2 is strong
- (C) If either 1 or 2 is strong
- (D) If neither 1 nor 2 is strong
- (E) If both 1 and 2 are strong

Q. Statement: “Do not lean out of the door of the bus.” - a warning in a school bus.



Assumptions:

- I. Leaning out of a running bus is dangerous.
- II. Children do not pay any heed to such warnings.

- (A) If only assumption I is implicit
- (B) If only assumption II is implicit
- (C) If either I or II is implicit
- (D) If neither I nor II is implicit

Q. Statement: Many people in city Y are suffering from flu.

Actions:

1. The Municipal Corporation should take immediate steps to control the disease.
2. People in the city should be advised to take certain measures from their end to prevent the disease.

- a) Only 1 follows
- b) Only 2 follows
- c) Either 1 or 2 follows
- d) Neither 1 nor 2 follows
- e) Both 1 and 2 follow

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BASIC CONCEPTS

- In statement-conclusion, a paragraph or a statement is given to us and based on that we must derive certain conclusions.
- Now what is a conclusion? So, a conclusion is a judgement or a decision which is derived after sufficient reasoning. In our context, conclusions are logical facts that are derived or inferred from the passage.





STEPS TO SOLVE

- Read the passage or given statements thoroughly.
- Analyze each statement individually.
- Draw a logical conclusion based on the given statement.

SOME IMPORTANT POINTS

- Conclusion and Inference are one and the same thing.
- Do not use your general knowledge.
- Conclusion should only be as per the given statement, so we do not need to find a conclusion which is true in general, but we do need to find a statement which is true for the given statement.

TIPS AND TRICKS

Statement: Teacher scolded Ram in front of all the students.

Conclusion: The teacher disliked Ram



TIP No. 1 : DO NOT ASSUME

Only Make those assumptions that can be drawn or inferred from the given passage or statements. Do not assume anything which is not related to the passage.

Statement: Teacher scolded Ram in front of all the students.

Conclusion: The teacher disliked Ram

Explanation: Here, given statement is the teacher scolded Ram but no specific reason is mentioned. It can be because Ram was late to attend the class or Ram did not complete his homework. So just by assuming that the teacher disliked Ram and so she scolded him is totally wrong. So, avoid such conclusions.

TIPS AND TRICKS

Statement: With the increase in consumption of petroleum products, it is feared that petrol will be a rare commodity in near future.

Conclusion: Petroleum Products should be used efficiently.

TIP No. 2 : ADVICE OR RESULT

If a given conclusion is in advice form or a certain result can be deduced out of it then such a conclusion will always be true. But of course, the conclusions should be related to the given statement.

Statement: With the increase in consumption of petroleum products, it is feared that petrol will be a rare commodity in near future.

Conclusion: Petroleum Products should be used efficiently.

Explanation: Here the given conclusion is in advice form. So such a conclusion is always true.

TIPS AND TRICKS

Statement: Many people have been admitted in the hospital. It is assumed that junk food is poisonous.

Conclusion: Junk food is always poisonous.



TIP No. 3 : AVOID PRE-ASSUMPTIONS

If a given conclusion contains a "Pre-Assumption" then such a conclusion is always wrong.

Statement: Many people have been admitted in the hospital. It is assumed that junk food is poisonous.

Conclusion: Junk food is always poisonous.

Explanation: Now as per the given conclusion, it is very clear that we are trying to make a "Pre-Assumption" based on the unhealthy effects of junk food. So, such "Pre-Assumed" conclusions are always wrong.

All, Every, Each, None, No one, Everyone, Everybody, Always



Some, Many, Few, May, Might, May be



TIPS AND TRICKS

Statement: Many Medicines contain Fish oil.

Conclusion:

1. All the vegetarians are also Pescatarians.
2. Some vegetarians consume such medicines to cure their diseases.

TIP No. 4 : DO NOT GO BY MORALS / ETHICS

For certain questions, when you go by the options for finding out an appropriate conclusion, sometimes it may happen that the answer option may not be morally correct, still that will be the correct answer. So, if an option is related to the statement and if you find it to be it logically correct then that option should be selected even if it is morally incorrect.

Statement: Many Medicines contain Fish oil.

Conclusion:

1. All the vegetarians are also Pescatarians.
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TIP No. 4 : DO NOT GO BY MORALS / ETHICS

Statement: Many Medicines contain Fish oil.

Conclusion:

1. All the vegetarians are also Pescatarians.
2. Some vegetarians consume such medicines to cure their diseases.

Explanation: Correct option is 2nd one. So even if the option is not morally correct as vegetarians should not consume fish but to cure certain diseases it is necessary to consume such medicines. So, some conclusions like the one given above are correct even if they are morally incorrect but are logically correct and are related to the given statement.

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Conclusion:

1. TCS hires no person with other qualification.
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Correct Option: (B)

Option no 1 is not appropriate as nothing is mentioned in the statement about TCS not hiring people from any other profession.

Q2. Statement: India is a democratic country.

Conclusion:

1. No other country in the world is democratic.
2. There are many other countries in the world that are democratic.

- (A) If only conclusion 1 follows
- (B) If only conclusion 2 follows
- (C) If either 1 or 2 follows
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- (E) If both 1 and 2 follow.

Correct Option: (D)

Neither 1 nor 2 follows. This is because in the statement, nothing is mentioned about other countries. The statement only talks about India.

Q3. Statement: Shyam is one of the probable students for securing 1st rank in the class.

Conclusion:

1. Shyam will secure 1st rank.
2. Shyam will not secure 1st rank.

- (A) If only conclusion 1 follows
- (B) If only conclusion 2 follows
- (C) If either 1 or 2 follows
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Correct Option: (C)

Either 1 or 2 follows. As the statement says, only 2 possible outcomes are possible. One Shyam will be 1st in his class or he won't be. But both the things cannot happen simultaneously. Hence option C is appropriate.

Q4. Statement: An advertisement- "50% off on all Electronic Goods up till 31st January".

Conclusion:

1. After 31st January, no discount will be provided.
2. No sale of electronic goods after 31st January.

- (A) If only conclusion 1 follows
- (B) If only conclusion 2 follows
- (C) If either 1 or 2 follows
- (D) If neither 1 nor 2 follows and
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- (C) If either 1 or 2 follows
- (D) If neither 1 nor 2 follows and
- (E) If both 1 and 2 follow.

Correct Option: (D)

No information has been given in both conclusions regarding what will happen after 31st January, so both conclusions are false.

Q5. Statement: Due to recession, Company X has fired its 200 employees.

Conclusion:

1. Company X is well known for firing people.
2. All other competitors of Company X are also affected because of recession.

- (A) If only conclusion 1 follows
- (B) If only conclusion 2 follows
- (C) If either 1 or 2 follows
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- (D) If neither 1 nor 2 follows and
- (E) If both 1 and 2 follow.

Correct Option: (D)

Neither 1 nor 2 follows. Company X has fired its employees because of current recession. Nothing is mentioned about the history of the company X. So, option 1 is false.

The competitors may or may not be affected because of recession. We are not aware of that fact. So, option no 2 is also false.

Q6. Statements: The average number of persons per household is 5 in urban areas whereas it is 7 in rural areas. The national average is 6.

Conclusions:

1. The population per unit area in the rural areas is higher than in the urban areas.
2. More persons live in the same household in the rural areas as compared to those in the urban areas.

- A. Only conclusion I follows
- B. Only conclusion II follows
- C. Either I or II follows
- D. Neither I nor II follows
- E. Both I and II follow

Correct Answer: Option B

Explanation:

The population per household and not the population per unit area is being talked about in the statement. So, only II follows while I does not.



Q7. Statement: Many people living in the communist nation says that “Capitalism exploits man by man whereas under communism, is opposite of it”.



Conclusions:

1. Communism is an exploitation of man by man.
 2. People argue that capitalism is not so important over communism
-
- A. Only I follows
 - B. Only II follows
 - C. Both I& II follows
 - D. None of these follow

Correct Answer-Option D



Q8. Statement: Jade plant has thick leaves, and it requires little water.



Conclusions:

I: All plants with thick leaves require little water.

II: Jade plants may be grown in places where water is not in abundance.

- A. Only conclusion I follows
- B. Only conclusion II follows
- C. Either I or II follows
- D. Neither I nor II follows
- E. Both I and II follow



Correct Answer : Option B

Explanation –

The statement talks of jade plants only and not 'all plants with thick leaves'. So, I does not follow. Also, since jade plants require little water, so they can be grown in places where water is not in abundance. So, II follows.



Q9. Statement: Use "Kraft" colours. They add colour to our life. - An advertisement.



Conclusions:

I: Catchy slogans do not attract people.

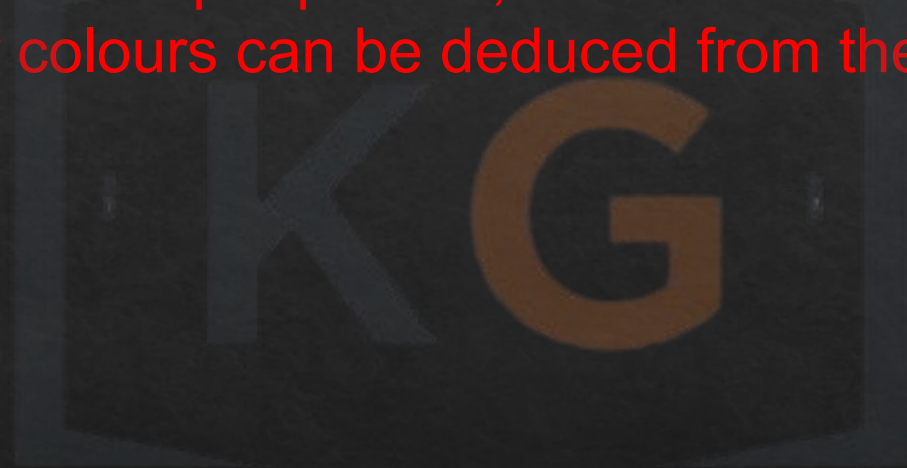
II: People like dark colours.

- A. Only conclusion I follows
- B. Only conclusion II follows
- C. Either I or II follows
- D. Neither I nor II follows
- E. Both I and II follow

Correct Answer : Option D

Explanation –

The slogan given in the statement is definitely a catchy one which indicates that catchy slogans do attract people. So, I does not follow. Nothing about people's preference for colours can be deduced from the statement. Thus, II also does not follow.



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Q. Statement: Should Military training be made compulsory in our country?

Arguments:

1. Yes, every child should go through this training.
2. No, it is not compulsory in USA.

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STATEMENT & ARGUMENT

<u>Strong Argument</u>	<u>Weak Argument</u>
Security of nation	Over exaggeration, Assumption
Development of a country	Words like Each, Nothing, Every, All, Only
Universal Truth	Comparison
Instructions from Supreme Authority	Ambiguous
Facts based on experience	Half explained arguments [KNOWLEDGE GATE]
Educational Development	

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Q2. Statement: Should abortion be legalized in India?

Arguments:

1. Yes, it is legal in many countries.
2. No, Abortion is also a kind of murder.

- (A) If only argument 1 is strong
- (B) If only argument 2 is strong
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Q3. Statement: Should Disarmament Policy be adopted by India?

Arguments:

1. Yes, it will promote peace and would encourage other countries to do the same.
2. No, it will pose a security threat to India.

- (A) If only argument 1 is strong
- (B) If only argument 2 is strong
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Q4. Statement: Should the private sector be allowed to participate in development of railways?

Arguments:

1. Yes, this is one of the ways to provide better services to the public.
2. No, there will be exploitation and fares may also increase.

- (A) If only argument 1 is strong
- (B) If only argument 2 is strong
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Q5. Statement: Should more big industries be started in urban areas?

Arguments:

1. Yes, it will create job opportunities.
2. No, it will increase population and pollution.

- (A) If only argument 1 is strong
- (B) If only argument 2 is strong
- (C) If either 1 or 2 is strong
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Q6. Statement: Should internal assessment in colleges be abolished?

Arguments:

Yes. This will help in reducing the possibility of favoritism.

No, teaching faculty will lose control over students.



- A. Only argument I is strong
- B. Only argument II is strong
- C. Either I or II is strong
- D. Neither I nor II is strong
- E. Both I and II are strong

Correct Answer: Option A

Explanation:

Abolishing the internal assessment would surely reduce favoritism on personal grounds because the teachers would not be involved in examination system so that they cannot extend personal benefits to anyone. So, argument I holds strong. But it will not affect the control of teaching faculty on students because still the teachers would be teaching them. So, argument II is vague.

STATEMENT & ARGUMENT

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Q7. Statement: Should all the annual examinations up to Std. V be abolished?



Arguments:

I: Yes. The young students should not be burdened with such examinations which hampers their natural growth.

II: No. The students will not study seriously as they will get automatic promotion to the next class, and this will affect them in future.

- A. Only argument I is strong
- B. Only argument II is strong
- C. Either I or II is strong
- D. Neither I nor II is strong
- E. Both I and II are strong

Correct Answer: Option C

Explanation:

Clearly, neither the students can be burdened with studies at such a tender age, nor can they be left free to take studies casually, as this shall weaken their basic foundation.



STATEMENT & ARGUMENT

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Q8. Statement: Should education be made compulsory for all children up to the age of 14?

Arguments:

1. Yes. This will help to eradicate the system of forced employment of these children.
2. Yes. This is an effective way to make the entire population educated.
3. No. We do not have adequate infrastructure to educate the entire population.
4. Yes. This would increase the standard of living.

- A. All are strong
- B. Only I, II and III are strong
- C. Only I, II and IV are strong
- D. Only II is strong
- E. Only II and III are strong

Correct Answer: Option D

Explanation:

Clearly, today's children are to make up future citizens of the country and so it is absolutely essential to make them learned, more responsible, more innovative and self-dependent by imparting them education. So, argument II holds strong while I and IV do not. Besides, the goal of literacy cannot be denied for want of infrastructure. So, argument III also does not hold.

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Q. Statement: “Do not lean out of the door of the bus.” - a warning in a school bus.



Assumptions:

- I. Leaning out of a running bus is dangerous.
- II. Children do not pay any heed to such warnings.

- (A) If only assumption I is implicit
- (B) If only assumption II is implicit
- (C) If either I or II is implicit
- (D) If neither I nor II is implicit

RULES

1. Words like EVERY, EACH, NONE, ALL, ONLY etc will always lead to FALSE Assumption.
2. Any assumption which is not stucked to statement is FALSE.
3. Any assumption which talks about past or future is FALSE.
4. Any assumption which goes against the given statement is FALSE.
5. If statement is generalised and assumption is specific or vice-versa, then assumption is FALSE.

Q1. Statement: Electric Vehicles produce less air pollutants.

Assumption: Electric Vehicles will be the only means of transportation in future.



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Q2. Statement: Warning – “Cellphone use is prohibited beyond this point”.

Assumption: This warning is not necessary.



RULES

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3. Any assumption which talks about past or future is FALSE.
4. Any assumption which goes against the given statement is FALSE.
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Q3. Statement: Seats allotted for only physically disabled people.

Assumption: Only females who are physically disabled can occupy such seats.



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Q4. Statement: “Do not lean out of the door of the bus.” - a warning in a school bus.



Assumptions:

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- (A) If only assumption I is implicit
- (B) If only assumption II is implicit
- (C) If either I or II is implicit
- (D) If neither I nor II is implicit

Correct Answer : Option A

Explanation:

Leaning out of a running bus must be dangerous, otherwise, the warning would not have been there. Hence I is implicit. But II is not implicit. If the authorities would have assumed that children do not pay any heed to such warning, they would not have put it up there.

Hence, the correct option is (A).

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Q5. Statement: "A rare opportunity to be a professional while you are at home."
- An advertisement for computer literate housewives by a computer company.

Assumptions:

1. Some housewives simultaneously desire to become professional.
2. Computer industry is growing at a fast pace.
3. It is possible to be a professional as well as a housewife.

- A. Only I and II are implicit
- B. Only II and III are implicit
- C. Only I and III are implicit
- D. Only II is implicit

Correct Answer : Option C

Explanation:

Clearly, a company would highlight only that feature in its advertisement which people crave for and which it can make possible. So, both I and III are implicit. Nothing can be deduced about the growth of computer industry. So, II is not implicit.

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Q6. Statement: Liquid soap should be used in place of soap bars to wash hands.

Assumption: Soap bars are harsher to skin than liquid soaps.



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4. Any assumption which goes against the given statement is FALSE.
5. If statement is generalised and assumption is specific or vice-versa, then assumption is FALSE.

Q7 (VK). **Statement:** Warning – “Paper will get wet because of water”.

Assumption: Water will not make the paper wet.

- A. Given assumption is implicit.
- B. Given assumption is explicit.
- C. Given assumption can be considered as both implicit and explicit.
- D. Given data is insufficient to decide whether the given assumption is implicit or explicit.

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- **STATEMENT & CONCLUSION**
- **STATEMENT & ARGUMENT**
- **STATEMENT & ASSUMPTION**
- **STATEMENT & COURSE OF ACTION** 

BASIC CONCEPTS

- "A course of action is a suitable step adopted to minimize or solve the given problem".
- Statement- Action is one of the main topics in Critical Reasoning. In this topic, a statement is given followed by some actions. Based on appropriate logic you have to identify a correct course of action that will be practically viable and will help in solving the problem.

Q. Statement: Many people in city Y are suffering from flu.

Actions:

1. The Municipal Corporation should take immediate steps to control the disease.
2. People in the city should be advised to take certain measures from their end to prevent the disease.

- a) Only 1 follows
- b) Only 2 follows
- c) Either 1 or 2 follows
- d) Neither 1 nor 2 follows
- e) Both 1 and 2 follow

TIPS AND TRICKS

Statement: The authorities in Society X are cracking down on street hawkers, blaming them for traffic jams near their society.

Actions:

1. Street hawkers should not be allowed during peak hours.
 2. Street hawkers should be warned and asked not to create chaos.
-
- a) Only 1 follows
 - b) Only 2 follows
 - c) Either 1 or 2 follows
 - d) Neither 1 nor 2 follows
 - e) Both 1 and 2 follow

TIP 1: CHOOSE PRACTICAL AND RELEVANT ACTION

1. Without any second thought choose actions which are **Practical in Nature and logically which can be implemented** (should be related to the statement).
2. **Solve or Minimize** the given problem.



TIPS AND TRICKS

Statement: On a particular highway, number of road accidents have increased by many folds.

Action: Personal monitoring should be done by the police officials on each and every vehicle.

TIP No. 2 : ACTION SHOULD NOT GIVE BIRTH TO ANOTHER PROBLEM

Like there will be a case in which-- among the two given actions, if you select a particular action, it might happen that it would help you in solving the problem but indirectly it will give birth to many other problems. So, such actions should be avoided.

Statement: On a particular highway, number of road accidents have increased by many folds.

Action: Personal monitoring should be done by the police officials on each and every vehicle.

Explanation: The mentioned action will definitely yield a good result but if we think practically then it is not at all possible to personally monitor each and every vehicle. This will in turn give birth to many other problems.

TIPS AND TRICKS

Statement: Company Z is incurring huge losses due to many competitors in the market.

Actions:

1. Company Z should offer a huge discount on all its product and attract customers.
2. Company Z should study its competitors, their products and methods.

- a) Only 1 follows
- b) Only 2 follows
- c) Either 1 or 2 follows
- d) Neither 1 nor 2 follows
- e) Both 1 and 2 follow

TIP No. 3 : UNPREDICTED OUTCOMES

Always avoid those actions whose outcomes cannot be predicted. There may be cases where you will not be able to predict the result of the selected action. So, such actions should be ignored.

Statement: Company Z is incurring huge losses due to many competitors in the market

Actions:

1. Company Z should offer a huge discount on all its product and attract customers.
2. Company Z should study its competitors, their products and methods.

TIP No. 3 : UNPREDICTED OUTCOMES

Statement: Company Z is incurring huge losses due to many competitors in the market

Actions:

1. Company Z should offer a huge discount on all its product and attract customers.
2. Company Z should study its competitors, their products and methods.

Answer: Only 2 follows.

Explanation: Now we can also think of option a to be a practical solution but one of the major reasons of option a being wrong is because we cannot predict that even after reducing the price, customers will be attracted towards the company. So, the outcome is unknown and so such actions should be avoided.

TIPS AND TRICKS

Statement: Concerns have risen about air pollution in Indian cities. A recent report by the World Health Organization (WHO) states that of the 20 most polluted cities in the world, 13 are in India.

Actions:

1. Pollution committee should immediately take certain measures to control the pollution levels in India.
 2. Licenses of polluting industries should be cancelled and owners should be put behind the bars.
-
- a) Only 1 follows
 - b) Only 2 follows
 - c) Either 1 or 2 follows
 - d) Neither 1 nor 2 follows
 - e) Both 1 and 2 follow

TIP No. 4 : AVOID NEGATIVE AND HARSH ACTIONS

Avoid Negative and Harsh Actions if possible but if there is no other option available then such actions should be undertaken.



Statement: A train derailed near a station while moving over a bridge and fell into a river, resulting in the death of 65 people.

Courses of Action:

1. The Railway Authorities should clarify the reason of the accident to the Government.
 2. The Government should allocate funds to compensate the destruction caused.
 3. The protection walls of the bridge should be made strong enough to avoid such accidents
-
- A. Only I and II follow
 - B. Only II and III follow
 - C. Only III follows
 - D. All follow
 - E. None of these

TIP No. 5 : LONG TERM ACTIONS

Look for long term and permanent course of actions.



Statement: A train derailed near a station while moving over a bridge and fell into a river, resulting in the death of 65 people.

Courses of Action:

1. The Railway Authorities should clarify the reason of the accident to the Government.
2. The Government should allocate funds to compensate the destruction caused.
3. The protection walls of the bridge should be made strong enough to avoid such accidents

- A. Only I and II follow
- B. Only II and III follow
- C. Only III follows
- D. All follow
- E. None of these

Explanation: What is necessary is the preventive measures to protect the passengers, steps to avoid re-occurrence of such events. So, only course III follows.

Q1. **Statement:** Many people in city Y are suffering from flu.

Actions:

1. The Municipal Corporation should take immediate steps to control the disease.
2. People in the city should be advised to take certain measures from their end to prevent the disease.

- a) Only 1 follows
- b) Only 2 follows
- c) Either 1 or 2 follows
- d) Neither 1 nor 2 follows
- e) Both 1 and 2 follow

Q2. **Statement:** Footpaths of a busy road are crowded with vendors selling cheap items.



Courses of action :

- I. The help of police should be sought to drive them away.
- II. Some space should be provided to them where they can earn their bread without blocking footpaths.

- (A) Only I follows
- (B) Only II follows
- (C) Either I or II follows
- (D) Both I and II follow

Correct Answer : Option D

Explanation –

Crowding on footpaths is a great inconvenience for walkers. So, stern action needs to be taken to remove the vendors. But at the same time these people ought to be provided alternative means of livelihood. So, both the courses follow.

Hence, the correct option is (D).

Q3. Statement: Large number of people have become critically ill after consuming spurious liquor from a local shop.



Courses of Action:

- I. The Government should immediately close down all the shops selling liquor till the stocks are tested for presence of toxicity.
 - II. The owner of the liquor shop should be asked to leave the town and open a shop elsewhere.
 - III. The owner of the liquor shop should immediately be arrested and tried for criminal negligence.
-
- A. Only I and II follow
 - B. Only II and III follow
 - C. Only III follows
 - D. Only I and III follow
 - E. All follow

Answer: Option D

Explanation:

Clearly, the owner of the shop should be punished for selling spurious liquor. Further, attempts should be made to unearth other such unscrupulous liquor sellers by conducting raids and testing samples collected from different shops. So, both I and III follow. However, sending the liquor shop owner to another place may create a problem elsewhere also. So, II does not follow.

Q4. Statement: Major part of the rabi crop in the district is damaged due to unseasonal heavy rains during the last few days.



Courses of Action:

- I. The government should grant relief to the affected farmers to compensate their loss.
 - II. The government should provide free seed and fertilizer to the farmers for the kharif season.
 - III. The government should waive all the loans taken for the rabi crop by the affected farmers.
-
- A. Only I and II follow
 - B. Only II follows
 - C. Only II and III follow
 - D. Only III follows
 - E. All follow

Answer: Option E

Explanation:

The crisis clearly demands extending as much relief to farmers as possible. So, all I, II and III follow.

- **STATEMENT & CONCLUSION**
- **STATEMENT & ARGUMENT**
- **STATEMENT & ASSUMPTION**
- **STATEMENT & COURSE OF ACTION**
- **GROUPING BASED QUESTIONS**
- **MATHEMATICAL QUESTIONS**



Q1. Given question contains six statements. Write as many logical combinations as possible from the set of six.

- A. The newt is an amphibian.
- B. Amphibians can live on land and in water.
- C. The newt is not an amphibian.
- D. The newt lives on land and in water
- E. Amphibians can live on land.
- F. The newt cannot survive on land.

- a) ABD, BFC
- b) ACD, BEF
- c) BCD, AFE
- d) None of these

Correct Answer: Option A

Explanation:

Statement A - The newt is an amphibian.

Statement B - Amphibians can live on land and in water.

Hence, D - The newt lives on land and in water.

Statement B - Amphibians can live on land and in water.

Statement F - The newt cannot survive on land.

Hence C - The newt is not an amphibian.

a) ABD, BFC

b) ACD, BEF

c) BCD, AFE

d) None of these

Q2. Given question contains six statements. Write as many logical combinations as possible from the set of six.

- A. Cells are the basic units of organisms.
- B. Organisms are made of cells.
- C. All organisms contain RNA.
- D. All organisms contain DNA.
- E. All cells contain DNA.
- F. DNA is similar to RNA

- a) CBD, FAE
- b) BCD, AEF
- c) ACB, EDF
- d) None of these

Correct Answer: Option B

Explanation:

Statement B, C, D are talking about organisms, whereas statement A,E are talking about about the cells. Statement F neither talks about organisms nor about cells so it can be clubbed with both BCD & AE, so option B fits best.

- a) CBD, FAE
- b) BCD, AEF
- c) ACB, EDF
- d) None of these

- **STATEMENT & CONCLUSION**
- **STATEMENT & ARGUMENT**
- **STATEMENT & ASSUMPTION**
- **STATEMENT & COURSE OF ACTION**
- **GROUPING BASED QUESTIONS**
- **MATHEMATICAL QUESTIONS**



- **STATEMENT & CONCLUSION**
- **STATEMENT & ARGUMENT**
- **STATEMENT & ASSUMPTION**
- **STATEMENT & COURSE OF ACTION**
- **GROUPING BASED QUESTIONS**
- **MATHEMATICAL QUESTIONS** 

Q1. $P + Q = P$ is greater than Q .
 $P - Q = P$ is either greater than or equal to Q .
 $P * Q = P$ is smaller than Q .
 $P / Q = P$ is either smaller than or equal to Q .
 $P \% Q = P$ is equal to Q .

Statement: $S + T, U * S, T * S, S - M, Q - R$.

Conclusions

1. $T+U$
2. M/T

- A. If conclusion 2 is true
- B. If both conclusion 1 and 2 are true
- C. If either conclusion 1 or 2 are true
- D. If neither conclusion 1 nor 2 is true

Q1. $P + Q = P$ is greater than Q .

$P - Q = P$ is either greater than or equal to Q .

$P * Q = P$ is smaller than Q .

$P / Q = P$ is either smaller than or equal to Q .

$P \% Q = P$ is equal to Q .



Statement: $S + T, U * S, T * S, S - M, Q - R$.

Conclusions

1. $T+U$

2. M/T



Q1. $P + Q = P$ is greater than Q .

$P - Q = P$ is either greater than or equal to Q .

$P * Q = P$ is smaller than Q .

$P / Q = P$ is either smaller than or equal to Q .

$P \% Q = P$ is equal to Q .



Statement: $S + T, U * S, T * S, S - M, Q - R$.

Conclusions

1. $T+U$

2. M/T



Q1. $P + Q = P$ is greater than Q .
 $P - Q = P$ is either greater than or equal to Q .
 $P * Q = P$ is smaller than Q .
 $P / Q = P$ is either smaller than or equal to Q .
 $P \% Q = P$ is equal to Q .

Statement: $S + T, U * S, T * S, S - M, Q - R$.

Conclusions

1. $T+U$
2. M/T

- A. If conclusion 2 is true
- B. If both conclusion 1 and 2 are true
- C. If either conclusion 1 or 2 are true
- D. If neither conclusion 1 nor 2 is true

Correct Answer: Option D

Explanation –

$S > T$, $U < S$, $T < S$, $S \geq M$, $Q \geq R$

Conclusion 1: $T > U$ and $U < S$, which implies UT , that is U can either be $< T$ or $= T$ or $> T$

Conclusion 2: $M \leq T$ and $S \geq M$, which implies $T \geq M \leq S$.

Therefore, nothing can be concluded.

Q2. $P \wedge Q = P$ greater than Q .
 $P @ Q = P$ smaller than or equal to Q .
 $P ! Q = P$ is equal to Q .
 $P * Q = P$ is greater than equal to Q .
 $P \& Q = P$ is smaller than Q .

Statement: $A * B$, $C ! A$, $C * E$

Conclusions:

1. $C * B$
2. $B @ E$

- A. If conclusion 1 is true
- B. If conclusion 2 is true
- C. If both conclusion 1 and 2 are true
- D. If either conclusion 1 or 2 are true

Q2. $P^{\wedge}Q$ = P greater than Q.
 $P@Q$ = P smaller than or equal to Q.
 $P!Q$ = P is equal to Q.
 P^*Q = P is greater than equal to Q.
 $P\&Q$ = P is smaller than Q.



Statement: A^*B , $C!A$, C^*E

Conclusions:

1. C^*B
2. $B@E$



Q2. $P \wedge Q = P$ greater than Q .
 $P @ Q = P$ smaller than or equal to Q .
 $P ! Q = P$ is equal to Q .
 $P * Q = P$ is greater than equal to Q .
 $P \& Q = P$ is smaller than Q .

Statement: $A * B$, $C ! A$, $C * E$

Conclusions:

1. $C * B$
2. $B @ E$

- A. If conclusion 1 is true
- B. If conclusion 2 is true
- C. If both conclusion 1 and 2 are true
- D. If either conclusion 1 or 2 are true

Correct Answer : Option A

Explanation –

$A \geq B, C = B, C > E$

$A \geq B = C > E$

Conclusion 1: $C \geq B$

Conclusion 2: $B \leq E$

Therefore, only 1 is true.



Q3(VK).

+ denotes less than.

- denotes greater than.

* denotes not less than.

/ denotes not greater than.

% denotes equal to.

\$ denotes not equal to

Comment:

Deductive Reasoning / Logical Deduction

Mathematical Questions Q3 (VK): Option __

Statement: $P\%Q$, R/S , $Q\ast R$

Conclusions:

1. $P\%S$

2. $Q-R$

- A. If conclusion 1 is true
- B. If conclusion 2 is true
- C. If both conclusion 1 and 2 are true
- D. If either conclusion 1 or 2 are true

- **STATEMENT & CONCLUSION**
- **STATEMENT & ARGUMENT**
- **STATEMENT & ASSUMPTION**
- **STATEMENT & COURSE OF ACTION**
- **GROUPING BASED QUESTIONS**
- **MATHEMATICAL QUESTIONS** 

Correct Answer: Option B

Explanation –

$P=Q, R \leq S, Q \geq R$

$P=Q \geq R \leq S$ Conclusion 1: $P=S$ Conclusion 2: $Q > R$

Therefore, only 2 is true.



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IDENTIFYING WORD AND NUMERIC PATTERNS

LETTER & NUMBER SERIES

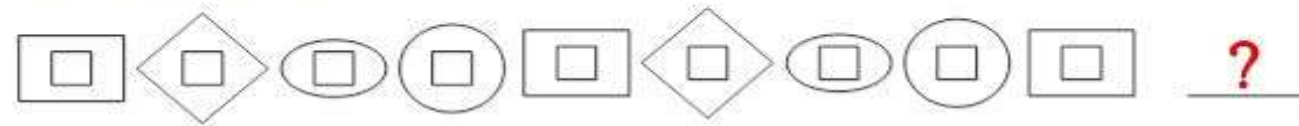
TYPES OF SERIES

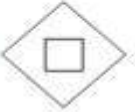
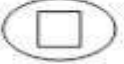
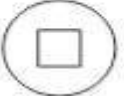
1. Number Series (Example: 4,7,12,19,__)
2. Letter/Alphabet Series (Example: A,E,I,M,__)
3. Alpha-Numeric Series (Example: A1, D4, G9, J16,__)
4. Symbol Series



SYMBOL SERIES

What comes next?



- ☐ A) 
- ☐ B) 
- ☐ C) 
- ☐ D) 

TYPES OF QUESTIONS

1. Find the Next Term / Missing Term in the series
2. Find the Wrong number / Odd number in the series

Eg: Find the odd number in the given series: 4 9 16 25 37



TYPES OF QUESTIONS

Look at the Letter series & Find the **Wrong** term.

AV, CU, DT, GS, IR, KQ.



A: AV

B: CU

C: IR

D: DT

CHOOSE THE CORRECT ANSWER ?

PRE-REQUISITES

Squares of 1 to 50 (or 1 to 100)

Cubes of 1 to 20 (or 1 to 40)

Tables of 1 to 30 (or 1 to 50)

Prime numbers from 1 to 200 (or 1 to 500)



Squares 1 to 50



$1^2 = 1$	$11^2 = 121$	$21^2 = 441$	$31^2 = 961$	$41^2 = 1681$
$2^2 = 4$	$12^2 = 144$	$22^2 = 484$	$32^2 = 1024$	$42^2 = 1764$
$3^2 = 9$	$13^2 = 169$	$23^2 = 529$	$33^2 = 1089$	$43^2 = 1849$
$4^2 = 16$	$14^2 = 196$	$24^2 = 576$	$34^2 = 1156$	$44^2 = 1936$
$5^2 = 25$	$15^2 = 225$	$25^2 = 625$	$35^2 = 1225$	$45^2 = 2025$
$6^2 = 36$	$16^2 = 256$	$26^2 = 676$	$36^2 = 1296$	$46^2 = 2116$
$7^2 = 49$	$17^2 = 289$	$27^2 = 729$	$37^2 = 1369$	$47^2 = 2209$
$8^2 = 64$	$18^2 = 324$	$28^2 = 784$	$38^2 = 1444$	$48^2 = 2304$
$9^2 = 81$	$19^2 = 361$	$29^2 = 841$	$39^2 = 1521$	$49^2 = 2401$
$10^2 = 100$	$20^2 = 400$	$30^2 = 900$	$40^2 = 1600$	$50^2 = 2500$

Cubes 1 to 20



$$1^3 = 1$$

$$2^3 = 8$$

$$3^3 = 27$$

$$4^3 = 64$$

$$5^3 = 125$$

$$6^3 = 216$$

$$7^3 = 343$$

$$8^3 = 512$$

$$9^3 = 729$$

$$10^3 = 1000$$

$$11^3 = 1331$$

$$12^3 = 1728$$

$$13^3 = 2197$$

$$14^3 = 2744$$

$$15^3 = 3375$$

$$16^3 = 4096$$

$$17^3 = 4913$$

$$18^3 = 5832$$

$$19^3 = 6859$$

$$20^3 = 8000$$

Math Tables Horizontal From 1 to 30



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
2	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52	54	56	58	60
3	3	6	9	12	15	18	21	24	27	30	33	36	39	42	45	48	51	54	57	60	63	66	69	72	75	78	81	84	87	90
4	4	8	12	16	20	24	28	32	36	40	44	48	52	56	60	64	68	72	76	80	84	88	92	96	100	104	108	112	116	120
5	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120	125	130	135	140	145	150
6	6	12	18	24	30	36	42	48	54	60	66	72	78	84	90	96	102	108	114	120	126	132	138	144	150	156	162	168	174	180
7	7	14	21	28	35	42	49	56	63	70	77	84	91	98	105	112	119	126	133	140	147	154	161	168	175	182	189	196	203	210
8	8	16	24	32	40	48	56	64	72	80	88	96	104	112	120	128	136	144	152	160	168	176	184	192	200	208	216	224	232	240
9	9	18	27	36	45	54	63	72	81	90	99	108	117	126	135	144	153	162	171	180	189	198	207	216	225	234	243	252	261	270
10	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150	160	170	180	190	200	210	220	230	240	250	260	270	280	290	300

Prime numbers between 1 and 200

2	3	5	7	11	13	17	19
23	29	31	37	41	43	47	53
59	61	67	71	73	79	83	89
97	101	103	107	109	113	127	131
137	139	149	151	157	163	167	173
179	181	191	193	197	199		

ALL POSSIBLE SERIES

1. Difference
2. Double Difference
3. Square
4. Cube
5. Prime
6. Multiples
7. Factors
8. Series with combinations
9. Miscellaneous



Que1: Find the next term/missing term in the given series

2 4 8 14 22 32 ____



ALL POSSIBLE SERIES

1. Difference
2. Double Difference
3. Square
4. Cube
5. Prime
6. Multiples
7. Factors
8. Series with combinations
9. Miscellaneous



Que2: Find the next term/missing term in the given series

1 3 6 11 19 31 ____



Que3: Find the next term in the given series

43, 38, 31, 22, 11, -2, ____



ALL POSSIBLE SERIES

1. Difference
2. Double Difference
3. Square
4. Cube
5. Prime
6. Multiples
7. Factors
8. Series with combinations
9. Miscellaneous



Que4: Find the next term/missing term in the given series

961 1024 1089 ____ 1225



Que: Find the next term/missing term in the given series

961 1024 1089 ____ 1225

Squares 1 to 50



$1^2 = 1$	$11^2 = 121$	$21^2 = 441$	$31^2 = 961$	$41^2 = 1681$
$2^2 = 4$	$12^2 = 144$	$22^2 = 484$	$32^2 = 1024$	$42^2 = 1764$
$3^2 = 9$	$13^2 = 169$	$23^2 = 529$	$33^2 = 1089$	$43^2 = 1849$
$4^2 = 16$	$14^2 = 196$	$24^2 = 576$	$34^2 = 1156$	$44^2 = 1936$
$5^2 = 25$	$15^2 = 225$	$25^2 = 625$	$35^2 = 1225$	$45^2 = 2025$
$6^2 = 36$	$16^2 = 256$	$26^2 = 676$	$36^2 = 1296$	$46^2 = 2116$
$7^2 = 49$	$17^2 = 289$	$27^2 = 729$	$37^2 = 1369$	$47^2 = 2209$
$8^2 = 64$	$18^2 = 324$	$28^2 = 784$	$38^2 = 1444$	$48^2 = 2304$
$9^2 = 81$	$19^2 = 361$	$29^2 = 841$	$39^2 = 1521$	$49^2 = 2401$
$10^2 = 100$	$20^2 = 400$	$30^2 = 900$	$40^2 = 1600$	$50^2 = 2500$

ALL POSSIBLE SERIES

1. Difference
2. Double Difference
3. Square
4. Cube
5. Prime
6. Multiples
7. Factors
8. Series with combinations
9. Miscellaneous



Que5: Find the next term in the given series

30, 34, 43, 59, 84, 120, ____



ALL POSSIBLE SERIES

1. Difference
2. Double Difference
3. Square
4. Cube
5. Prime
6. Multiples
7. Factors
8. Series with combinations
9. Miscellaneous



Que6: Find the next term/missing term in the given series

729 1000 1331 ____ 2197



Que: Find the next term/missing term in the given series

729 1000 1331 ____ 2197

Cubes 1 to 20



$$1^3 = 1$$

$$2^3 = 8$$

$$3^3 = 27$$

$$4^3 = 64$$

$$5^3 = 125$$

$$6^3 = 216$$

$$7^3 = 343$$

$$8^3 = 512$$

$$9^3 = 729$$

$$10^3 = 1000$$

$$11^3 = 1331$$

$$12^3 = 1728$$

$$13^3 = 2197$$

$$14^3 = 2744$$

$$15^3 = 3375$$

$$16^3 = 4096$$

$$17^3 = 4913$$

$$18^3 = 5832$$

$$19^3 = 6859$$

$$20^3 = 8000$$

ALL POSSIBLE SERIES

1. Difference
2. Double Difference
3. Square
4. Cube
5. Prime
6. Multiples
7. Factors
8. Series with combinations
9. Miscellaneous



Que7: Find the next term/missing term in the given series

31 37 ___ 43 47



Que: Find the next term/missing term in the given series

31 37 ___ 43 47

Prime numbers between 1 and 200

2	3	5	7	11	13	17	19
23	29	31	37	41	43	47	53
59	61	67	71	73	79	83	89
97	101	103	107	109	113	127	131
137	139	149	151	157	163	167	173
179	181	191	193	197	199		

Que8: Find the next term/missing term in the given series

6 15 35 ____ 143



Que: Find the next term/missing term in the given series

6 15 35 ____ 143

Prime numbers between 1 and 200

2	3	5	7	11	13	17	19
23	29	31	37	41	43	47	53
59	61	67	71	73	79	83	89
97	101	103	107	109	113	127	131
137	139	149	151	157	163	167	173
179	181	191	193	197	199		

ALL POSSIBLE SERIES

1. Difference
2. Double Difference
3. Square
4. Cube
5. Prime
6. Multiples
7. Factors
8. Series with combinations
9. Miscellaneous



Que9: Find the next term/missing term in the given series

5 15 45 135 405 _____



Que10: Find the next term in the given series

3.25, 6.5, 19.5, 78, 390, ____



ALL POSSIBLE SERIES

1. Difference
2. Double Difference
3. Square
4. Cube
5. Prime
6. Multiples
7. Factors
8. Series with combinations
9. Miscellaneous



Que11: Find the next term/missing term in the given series

524288 65536 8192 1024 128 16 ____



Que12: Find the missing term in the given series

3240, 540, 108, 27, ____, 4.5



ALL POSSIBLE SERIES

1. Difference
2. Double Difference
3. Square
4. Cube
5. Prime
6. Multiples
7. Factors
8. Series with combinations
9. Miscellaneous





Que13: Find the next term/missing term in the given series

1 2 3 4 6 ____



ALL POSSIBLE SERIES

1. Difference
2. Double Difference
3. Square
4. Cube
5. Prime
6. Multiples
7. Factors
8. Series with combinations
9. Miscellaneous



SERIES WITH COMBINATION

1. Square + 1
2. Square - 1
3. Cube + 1
4. Cube - 1
5. Square + n
6. Square - n
7. Cube + n
8. Cube - n
9. Cube + Square
10. Cube - Square



Que14: Find the next term/missing term in the given series

1 10 3 7 5 4 7 ____



Que15: Find the next term/missing term in the given series

96 15 71 8 37 10 41 ____



ALL POSSIBLE SERIES

1. Difference
2. Double Difference
3. Square
4. Cube
5. Prime
6. Multiples
7. Factors
8. Series with combinations
9. Miscellaneous



ALPHABETIC SERIES (LETTER SERIES)

F L R _

A	B	C	D	E	F	G	H
1	2	3	4	5	6	7	8
I	J	K	L	M	N	O	P
9	10	11	12	13	14	15	16
Q	R	S	T	U	V	W	
17	18	19	20	21	22	23	
		X	Y	Z			
		24	25	26			

REVERSE ALPHABETIC SERIES

ZA YB XC ____

A	B	C	D	E	F	G	H
1	2	3	4	5	6	7	8
I	J	K	L	M	N	O	P
9	10	11	12	13	14	15	16
Q	R	S	T	U	V	W	
17	18	19	20	21	22	23	
		X	Y	Z			
		24	25	26			

ALPHA NUMERIC SERIES

W23, T20, __, N14



A	B	C	D	E	F	G	H
1	2	3	4	5	6	7	8
I	J	K	L	M	N	O	P
9	10	11	12	13	14	15	16
Q	R	S	T	U	V	W	
17	18	19	20	21	22	23	
		X	Y	Z			
		24	25	26			

F – 6 (FIX SIX)

G – 7 (GROUP OF SEVEN COUNTRIES)

H – 8 (SEE THE STRUCTURE OF H)

I – 9 (I KNOW)

J-10

K-11 (J&K में 10/11 गोलियां रोज़ चलती हैं)

A	B	C	D	E	F	G	H
1	2	3	4	5	6	7	8
I	J	K	L	M	N	O	P
9	10	11	12	13	14	15	16
Q	R	S	T	U	V	W	
17	18	19	20	21	22	23	
		X	Y	Z			
		24	25	26			

L – 12 (लड़की को देखते ही लड़कों के 12 बज जाते हैं)

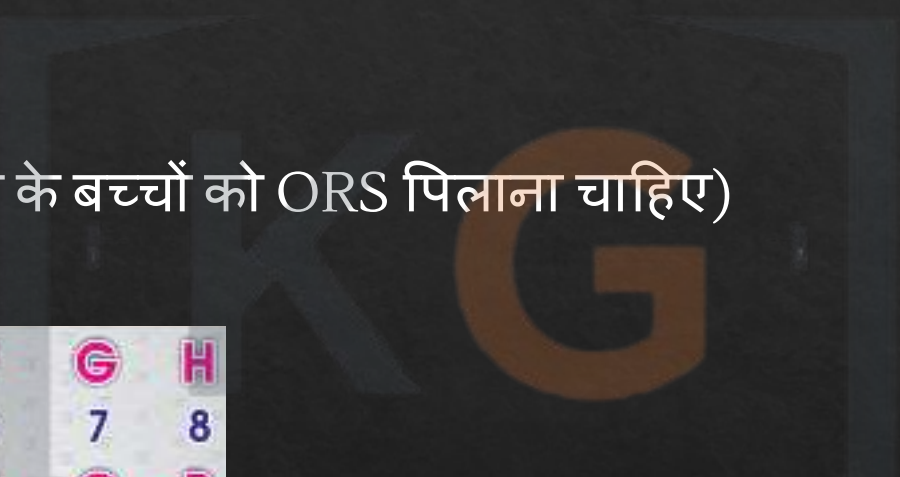
M – 13 (मैं तेरा)

N – 14 (14 February ko को नहाकर बाहर जाना चाहिए)

O – 15

R – 18

S – 19 (15 से 18/19 महीने तक के बच्चों को ORS पिलाना चाहिए)



A	B	C	D	E	F	G	H
1	2	3	4	5	6	7	8
I	J	K	L	M	N	O	P
9	10	11	12	13	14	15	16
Q	R	S	T	U	V	W	
17	18	19	20	21	22	23	
		X	Y	Z			
		24	25	26			

P – 16

Q – 17 (16 साल के Prince ने 17 साल की Queen से शादी की)

T – 20 (T-20 Matches)

U – 21 (U give me A Kiss)

V – 22 (V -> We -> हम दो हमारे दो)



A	B	C	D	E	F	G	H
1	2	3	4	5	6	7	8
I	J	K	L	M	N	O	P
9	10	11	12	13	14	15	16
Q	R	S	T	U	V	W	
17	18	19	20	21	22	23	
		X	Y	Z			
		24	25	26			

W – 23 (24 hour work नहीं करना चाहिए)

X – 24 (X की याद 24 घंटे आती है)

Y – 25 (Yash Sir के पास 25 girlfriends हैं)

Z – 26



A	B	C	D	E	F	G	H
1	2	3	4	5	6	7	8
I	J	K	L	M	N	O	P
9	10	11	12	13	14	15	16
Q	R	S	T	U	V	W	
17	18	19	20	21	22	23	
		X	Y	Z			
		24	25	26			

ALPHA NUMERIC SERIES

W4, T7, __, N13

A	B	C	D	E	F	G	H
1	2	3	4	5	6	7	8
I	J	K	L	M	N	O	P
9	10	11	12	13	14	15	16
Q	R	S	T	U	V	W	
17	18	19	20	21	22	23	
		X	Y	Z			
		24	25	26			

A	B	C	D	E	F	G	H
1	2	3	4	5	6	7	8
I	J	K	L	M	N	O	P
9	10	11	12	13	14	15	16
Q	R	S	T	U	V	W	
17	18	19	20	21	22	23	
		X	Y	Z			
		24	25	26			



Z	Y	X	W	V	U	T	S
1	2	3	4	5	6	7	8
R	Q	P	O	N	M	L	K
9	10	11	12	13	14	15	16
J	I	H	G	F	E	D	C
17	18	19	20	21	22	23	24
B	A						
25	26						

AZIZ is a BOY who has EVS Exam but still he went to watch FALTU Movie and had CRUX & DEW there. He went to watch movie on exam day because his mind is GREAT as his result of HIGHER SECONDARY & INTER was good.

But his father became angry on his this act and hence decided to marry him with JUNGLE QUEEN.

On his marriage he wore KURTA PYJAMA with LOTTO shoes, now after his marriage, he is no more a boy, he is a MAN now.



AZIZ is a BOY who has EVS Exam but still he went to watch FALTU Movie and had CRUX & DEW there. He went to watch movie on exam day because his mind is GREAT as his result of HIGHER SECONDARY & INTER was good.

But his father became angry on his this act and hence decided to marry him with JUNGLE QUEEN.

On his marriage he wore KURTA PYJAMA with LOTTO shoes, now after his marriage, he is no more a boy, he is a MAN now.



AZIZ (A – Z)

BOY (B- Y)

EVS (E – V)

FALTU (F – U)

CRUX (C – X)

DEW (D – W)

GREAT (G – T)

HIGHER SECONDARY (H – S)



INTER (I – R)

JUNGLE QUEEN (J – Q)

KURTA PYJAMA (K – P)

LOTTO (L – O)

MAN (M – N)



A	B	C	D	E	F	G	H
1	2	3	4	5	6	7	8
I	J	K	L	M	N	O	P
9	10	11	12	13	14	15	16
Q	R	S	T	U	V	W	
17	18	19	20	21	22	23	
		X	Y	Z			
		24	25	26			

Z	Y	X	W	V	U	T	S
1	2	3	4	5	6	7	8
R	Q	P	O	N	M	L	K
9	10	11	12	13	14	15	16
J	I	H	G	F	E	D	C
17	18	19	20	21	22	23	24
B	A						
25	26						

ALPHA NUMERIC SERIES

W4, T7, __, N13

A	B	C	D	E	F	G	H
1	2	3	4	5	6	7	8
I	J	K	L	M	N	O	P
9	10	11	12	13	14	15	16
Q	R	S	T	U	V	W	
17	18	19	20	21	22	23	
		X	Y	Z			
		24	25	26			

Que16: Find the next term/missing term in the given series

G, __, O, S, W



Que16: Find the next term/missing term in the given series

G, __, O, S, W



Que17: Find the next term/missing term in the given series

PK, SH, VE, ____



Que18: Find the next term/missing term in the given series

E5 J1 O6 T2 ____



Que19: Find the next term/missing term in the given series

3M 5M 7J ____ 11N



Que20: Jethalal participated in a weight loss program started at the newly opened center named “The Babita Gym”. At the end of every month, the decrease in weight from original weight was measured and noted as 1, 3, 8, 20, 42, ... While Jethalal made a steadfast effort, the weighing machine showed an erroneous weight once. What was that ?

A) 3 B) 8 C) 42 D) 20



*When you send freind request to
a girl of your class*
(Next day in class)



Que: 1, 3, 8, 20, 42, ...



Que: 1, 3, 8, 20, 42, ...



Que21 (VK): QAR, RAS, SAT, TAU, _____

- a) UAV
- b) UAT
- c) TAS
- d) TAT



Correct Answer: Option A

Explanation:

In this series, the third letter is repeated as the first letter of the next segment. The middle letter, A, remains static. The third letters are in alphabetical order, beginning with R.





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DATA SUFFICIENCY



PRE-REQUISITES

Good in Mathematics

Patience



Q: Is X a prime number?

Statement 1: $X=2$

Statement 2: $X=2$

- a) Statement I alone is sufficient
- b) Statement II alone is sufficient
- c) Both statements put together are sufficient
- d) Either of the statements individually is sufficient
- e) Both statements put are not sufficient

Q1: Is X a prime number?

Statement 1: $X=2$

Statement 2: $X=2$

- a) Statement I alone is sufficient
- b) Statement II alone is sufficient
- c) Both statements put together are sufficient
- d) Either of the statements individually is sufficient
- e) Both statements put are not sufficient

Q2: Is X a prime number?

Statement 1: $X=2$

Statement 2: $X=3$

- a) Statement I alone is sufficient
- b) Statement II alone is sufficient
- c) Both statements put together are sufficient
- d) Either of the statements individually is sufficient
- e) Both statements put are not sufficient

Q3: Is X a prime number?

Statement 1: $X=2$

Statement 2: $X=4$

- a) Statement I alone is sufficient
- b) Statement II alone is sufficient
- c) Both statements put together are sufficient
- d) Either of the statements individually is sufficient
- e) Both statements put are not sufficient

Q4: Is X a prime number?

Statement 1: $X=2$

Statement 2: $X=2$ or 4

- a) Statement I alone is sufficient
- b) Statement II alone is sufficient
- c) Both statements put together are sufficient
- d) Either of the statements individually is sufficient
- e) Both statements put are not sufficient

Q5: Is X a prime number?

Statement 1: $X=4$

Statement 2: $X=8$

- a) Statement I alone is sufficient
- b) Statement II alone is sufficient
- c) Both statements put together are sufficient
- d) Either of the statements individually is sufficient
- e) Both statements put are not sufficient

Q6: Is X a prime number?

Statement 1: $X=2$ or 4

Statement 2: $X=10$

- a) Statement I alone is sufficient
- b) Statement II alone is sufficient
- c) Both statements put together are sufficient
- d) Either of the statements individually is sufficient
- e) Both statements put are not sufficient

Q7: Is X a prime number?

Statement 1: $X=2$ or 4

Statement 2: $X=3$ or 10

- a) Statement I alone is sufficient
- b) Statement II alone is sufficient
- c) Both statements put together are sufficient
- d) Either of the statements individually is sufficient
- e) Both statements put are not sufficient

Q8: Is X a prime number?

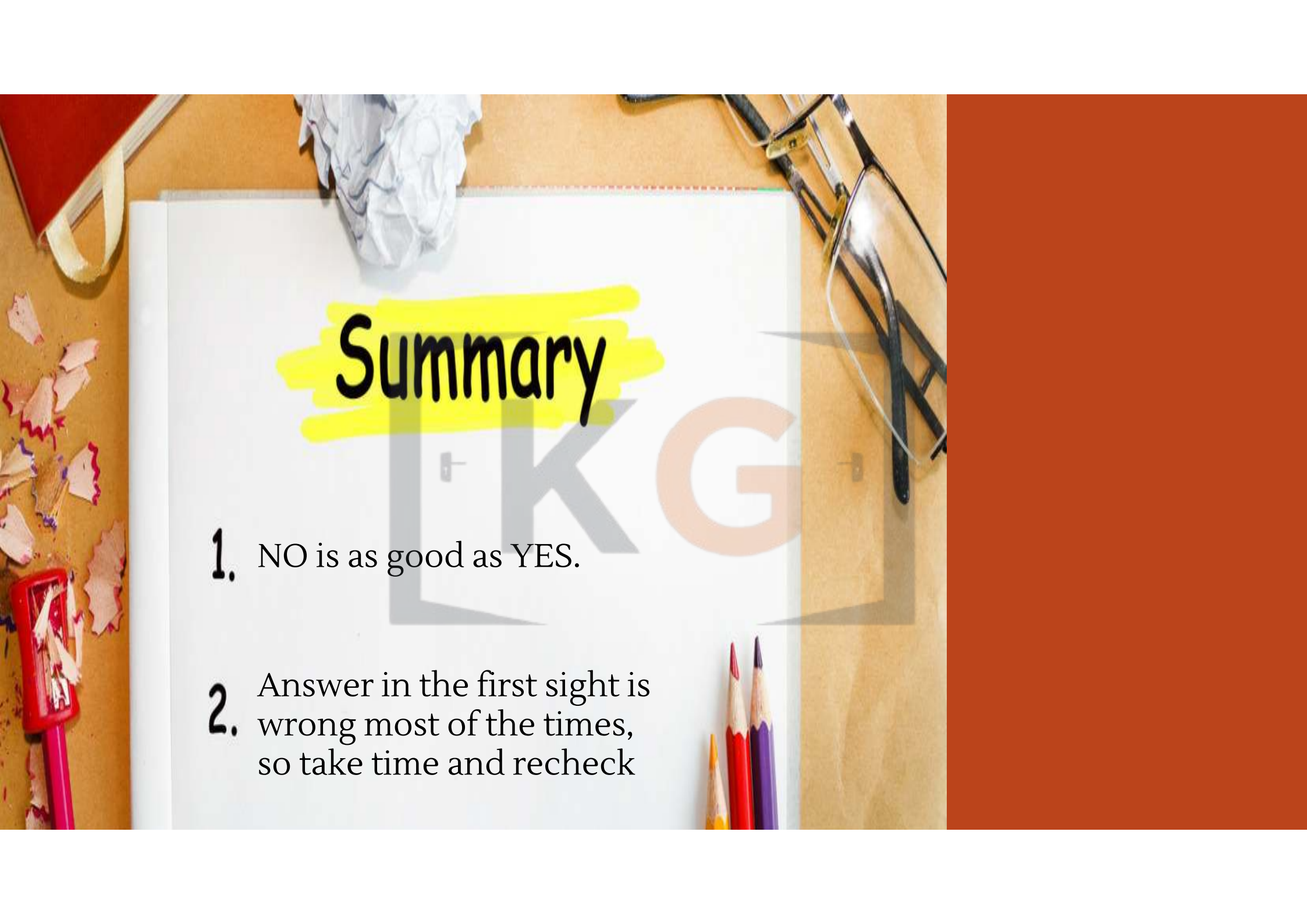
Statement 1: $X=2$ or 3

Statement 2: $X=10$

- a) Statement I alone is sufficient
- b) Statement II alone is sufficient
- c) Both statements put together are sufficient
- d) Either of the statements individually is sufficient
- e) Both statements put are not sufficient

Summary

1. We are looking for unique solution.
2. Search the question type
[i) YES/NO ii) Value Based]
3. When looking at statement 2, forget statement 1 and vice versa

A top-down view of a desk with a light brown wooden surface. In the center is a white notebook with a red cover. On the notebook, the word 'Summary' is written in a black, handwritten-style font and is highlighted with a bright yellow brushstroke. To the left of the notebook is a red pencil sharpener with a pile of pink and white pencil shavings. Above the notebook is a crumpled piece of white paper. To the right is a pair of black-rimmed glasses. Below the notebook are three sharpened pencils: yellow, red, and purple. A large, faint, light-colored watermark 'LKG' is visible across the center of the notebook page.

Summary

1. NO is as good as YES.
2. Answer in the first sight is wrong most of the times, so take time and recheck

Q9. Nibba and Nibbi salaries are in the proportion of 4 : 3 respectively. What is Nibba's salary ?



Statements:

- 1) Nibbi's salary is 75% that of Nibba's salary.
- 2) Nibbi's salary is Rs 4500.

- A. I alone is sufficient while II alone is not sufficient.
- B. II alone is sufficient while I alone is not sufficient.
- C. Either I or II is sufficient.
- D. Both I and II put together are sufficient.
- E. Neither I nor II is sufficient



Q9. Nibba and Nibbi salaries are in the proportion of 4 : 3 respectively. What is Nibba's salary ?

Statements:

- 1) Nibbi's salary is 75% that of Nibba's salary.
- 2) Nibbi's salary is Rs 4500.



Correct Answer : Option B

Explanation –

Statement I is merely an interpretation of the information contained in the question.

However, Nibba's salary can be ascertained from II as follows :

Let Nibba's and Nibbi's salaries be $4x$ and $3x$ respectively. Then, $3x = 4500$ or $x = 1500$. Therefore Nibba's salary $= 4x = \text{Rs. } 6000$.

Q10. Question: What is the code for 'sky' in the code language ?



Statements:

In the code language, 'sky is clear' is written as 'de ra fa'.

In the same code language, 'make it clear' is written as 'de ga jo'.

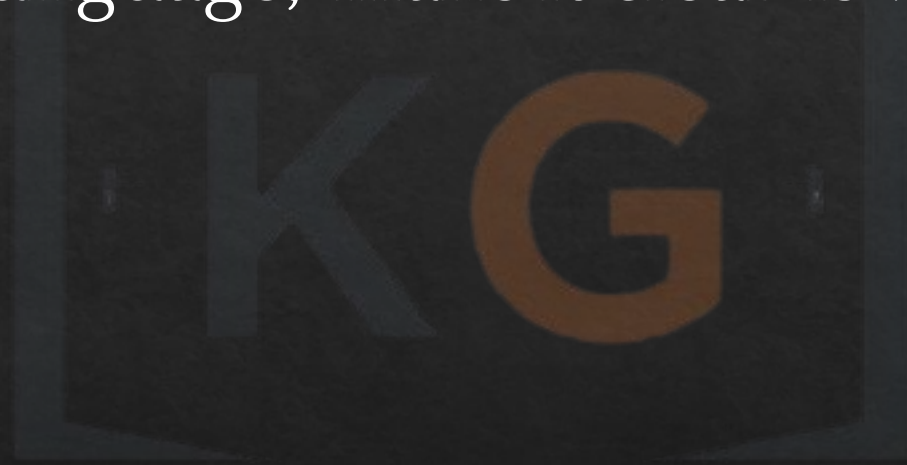
- A. I alone is sufficient while II alone is not sufficient
- B. II alone is sufficient while I alone is not sufficient
- C. Either I or II is sufficient
- D. Neither I nor II is sufficient

Q10. Question: What is the code for 'sky' in the code language ?

Statements:

In the code language, 'sky is clear' is written as 'de ra fa'.

In the same code language, 'make it clear' is written as 'de ga jo'.



Correct Answer : Option D

Explanation –

The only word common to I and II is 'clear' and as such, only the code for 'clear' can be ascertained from the given information.

Q11. Jethalal visits the temple on every festival day. Is today a festival?

Statement A. Jethalal went to the temple today.

Statement B. He does not go to the temple if he is very busy on that day.

(A) if the question can be answered by statement A alone but not by statement B alone

(B) if the question can be answered by either one of the two statements alone

(C) if the question can be answered only if both the statements are taken together.

(D) if the question cannot be answered even if both the statements are taken together



Q11. Jethalal visits the temple on every festival day. Is today a festival?

Statement A. Jethalal went to the temple today.

Statement B. He does not go to the temple if he is very busy on that day.



Correct Answer: Option D

Explanation:

Jethalal visits temple on every festival day, this does not mean that he will not go to temple on other days. He may or may not go to temple on other days. Using both the given statements also, we cannot conclude whether today is a festival day or not.

Q12. The given question has two statements. Find out whether the question can be answered with any of the given two statements.

Who among P, Q, R, S, and T each having a different weight is the lightest?

I. R is heavier than P.

II. S is lighter than T and Q.



- A) If statement I alone is sufficient to answer the question
- B) If statement II alone is sufficient to answer the question
- C) Both Statement put together are not sufficient
- D) If both the statements together are sufficient to answer the question
- E) If the question can't be answered using both the statements together

Q12. Who among P, Q, R, S, and T each having a different weight is the lightest?

I. R is heavier than P.

II. S is lighter than T and Q.



Correct Answer : Option E

Explanation –

If the question can't be answered using both the statements together.



Q13 (VK) Given question has two statements. Find out whether the question can be answered with any of the given two statements.

If n and a are positive integers, what is the unit digit of $n(4a+2) - n(8a)$?

I. $n = 3$

II. a is an odd number

- A. The question can be answered using both the statements together
- B. The question can be answered using the statement II alone
- C. The question can be answered using the statement I alone
- D. The question can't be answered even by using both the

Correct Answer : Option C

Explanation –

The question can be answered using the statement I alone.





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FIGURE AND FACTUAL ANALYSIS

PATTERN RECOGNITION

NON-VERBAL REASONING

TYPES OF QUESTIONS

Series

Classification

Mirror Images

Embedded Images

Figure Matrix

Paper Cutting

Grouping of
Images

Shape

Construction

9. Cubes & Dices

10. Analogy

11. Water Images

12. Pattern Completion

13. Paper Folding

14. Rule Detection

15. Dot Situation

16. Image Analysis

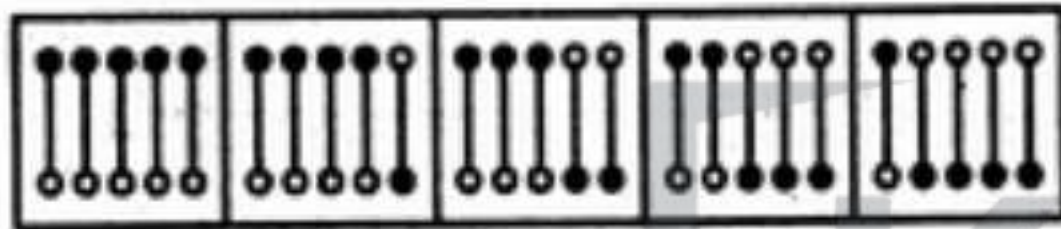
**The perfect trick to solve Non-Verbal Reasoning Questions is to
Eliminate Options.**

TYPE 1: SERIES



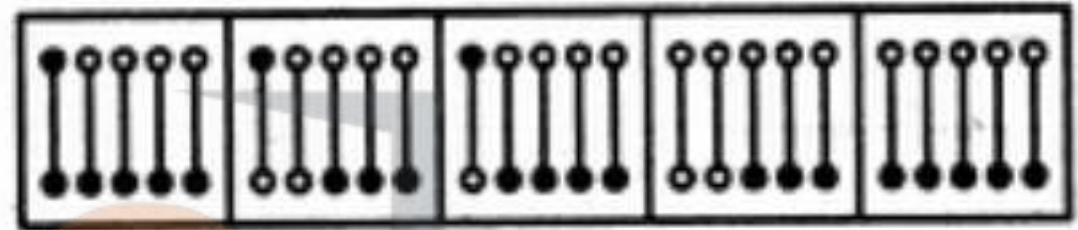
Q1. Select a figure from amongst the Answer Figures which will continue the same series as established by the five Problem Figures.

Problem Figures:



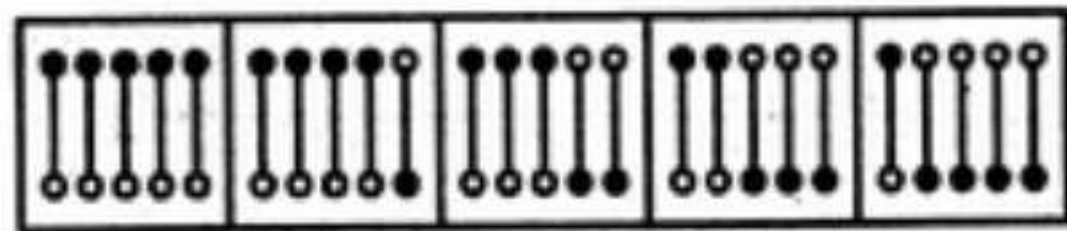
(A) (B) (C) (D) (E)

Answer Figures:



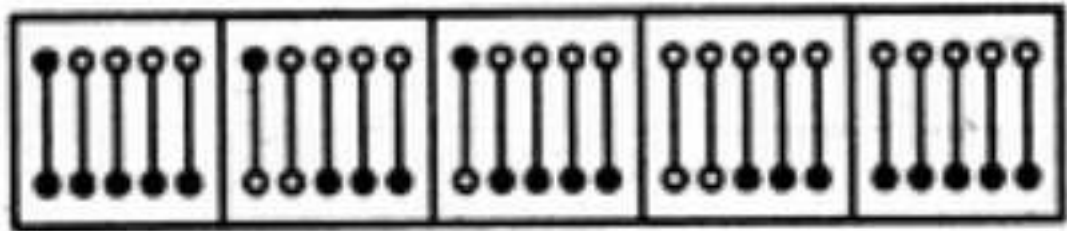
(1) (2) (3) (4) (5)

Problem Figures:



(A) (B) (C) (D) (E)

Answer Figures:



(1) (2) (3) (4) (5)

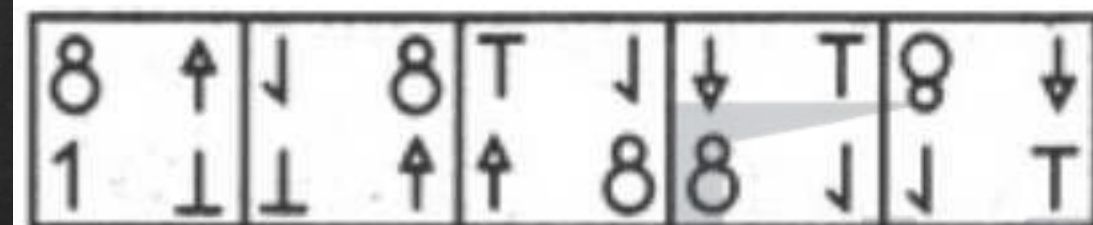
Answer: Option 5

Explanation:

One of the pins gets inverted in each step. The pins get inverted sequentially from right to left.

Q2. Select a figure from amongst the Answer Figures which will continue the same series as established by the five Problem Figures.

Problem Figures:



(A) (B) (C) (D) (E)

Answer Figures:



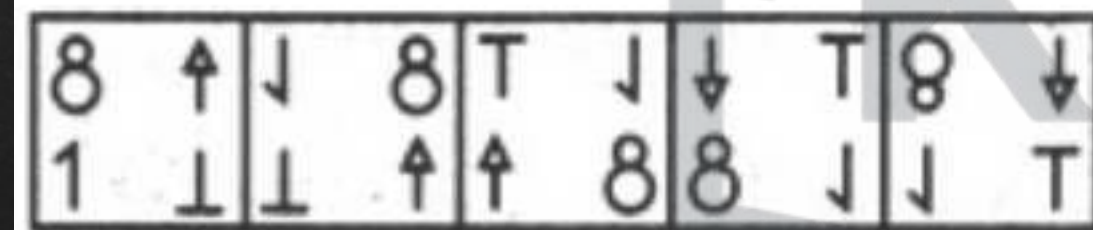
(1) (2) (3) (4) (5)

Correct Answer - Option 4

Explanation :

In each step, all the elements move to the adjacent corner (of the square boundary) in a CW direction and the element that reaches the upper-left corner gets vertically inverted.

Problem Figures:



(A) (B) (C) (D) (E)

Answer Figures:

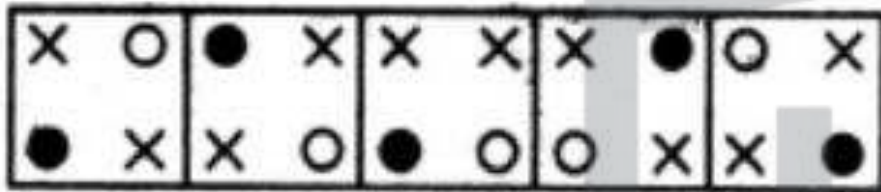


(1) (2) (3) (4) (5)

TYPE 2: CLASSIFICATION



Q3. In each problem, out of the five figures marked (1), (2), (3), (4) and (5), four are similar in a certain manner. However, one figure is not like the other four. Choose the figure which is different from the rest.

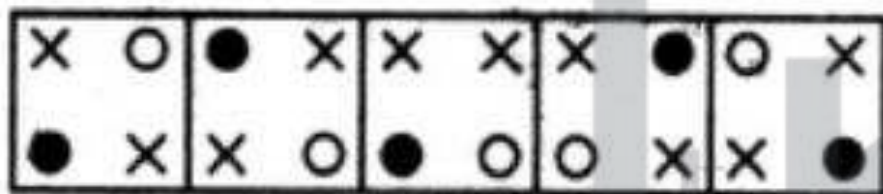


(1) (2) (3) (4) (5)

Correct Answer : Option 3

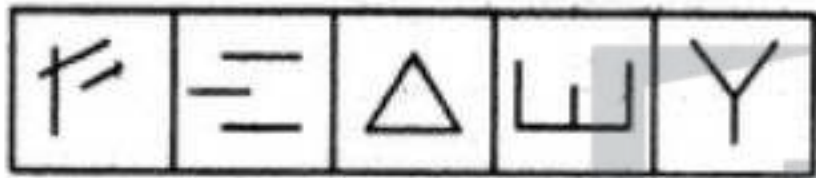
Explanation –

In each one of the figures except fig. (3), the two crosses (x) appear in the diagonally opposite corners.



(1) (2) (3) (4) (5)

Q4. In each problem, out of the five figures marked (1), (2), (3), (4) and (5), four are similar in a certain manner. However, one figure is not like the other four. Choose the figure which is different from the rest.



(1) (2) (3) (4) (5)

Correct Answer : Option 4

Explanation –

Figure (4) is formed by four line segments while each one of the other figures is formed by three line segments.

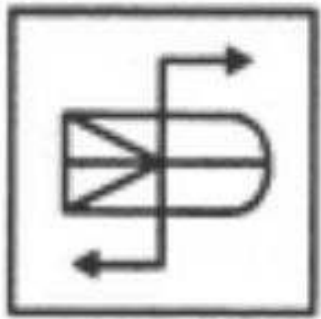


TYPE 3: MIRROR IMAGES

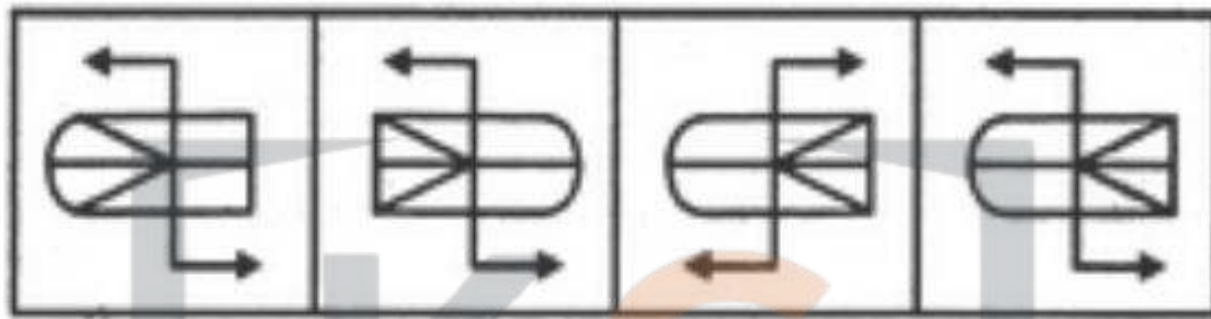


G

Q5. Choose the correct mirror image of the given figure (X) from amongst the four alternatives.



(X)



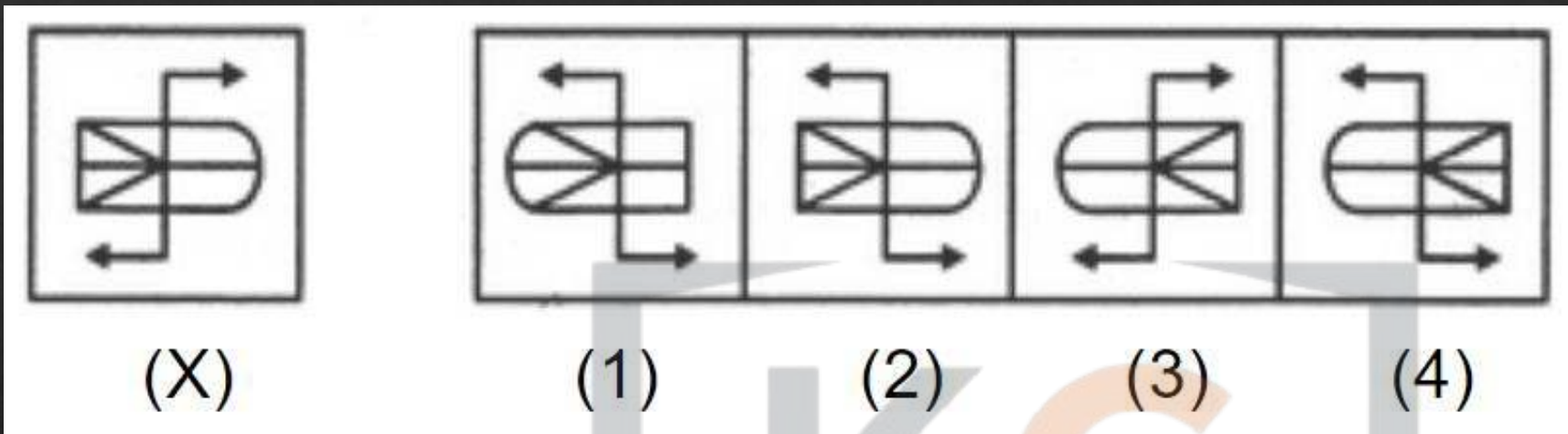
(1)

(2)

(3)

(4)

Correct Answer: Option 4



Q6. Choose the alternative which is closely resembles the mirror image of the given combination.

Nu56p7uR

(1) ИuɹθqɿuЯ

(2) ЯnɿqθɹnИ

(3) ЯuɿqθɹnИ

(4) ЯuɿqθɹnИ

Correct Answer- Option 3

Nu56p7uR

(1) Иu5θq7uЯ

(2) Яn7qθ5nИ

(3) Яu7qθ5uИ

(4) Яu7d95uИ

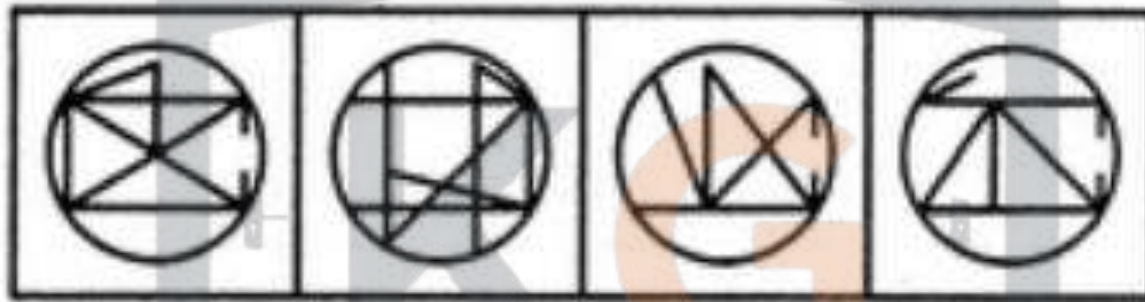
TYPE 4: EMBEDDED IMAGES



Q7. In each of the following questions, you are given a figure (X) followed by four alternative figures (1), (2), (3) and (4) such that figure (X) is embedded in one of them. Trace out the alternative figure which contains fig. (X) as its part Find out the alternative figure which contains figure (X) as its part



(X)



(1)

(2)

(3)

(4)

Correct Answer : Option 2

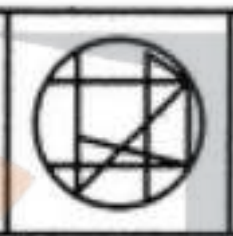
Explanation –



(X)



(1)



(2)



(3)

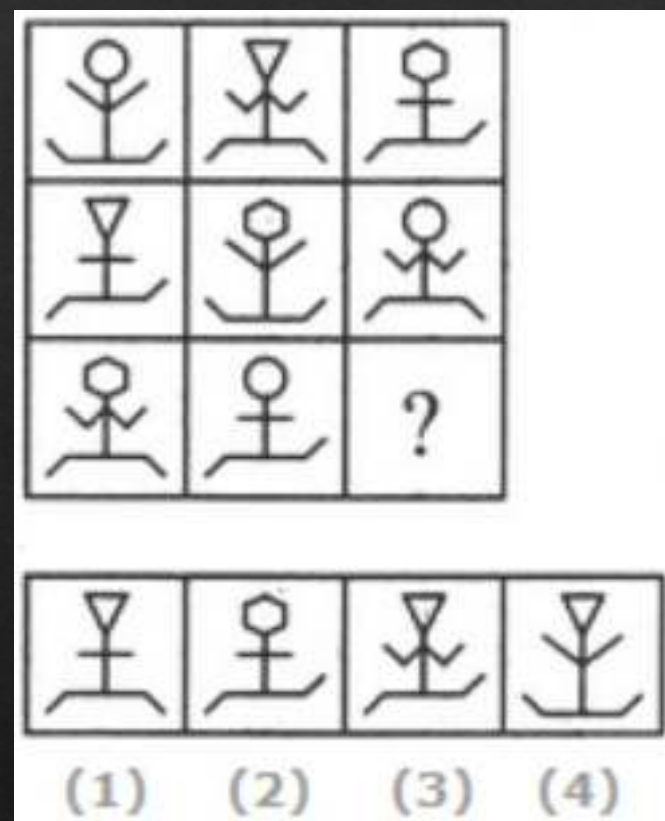


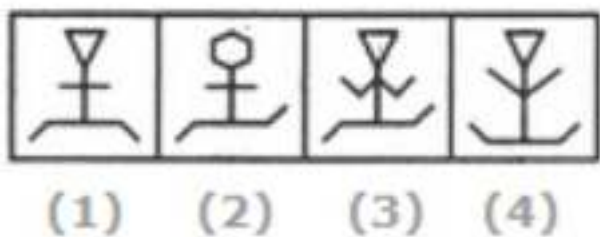
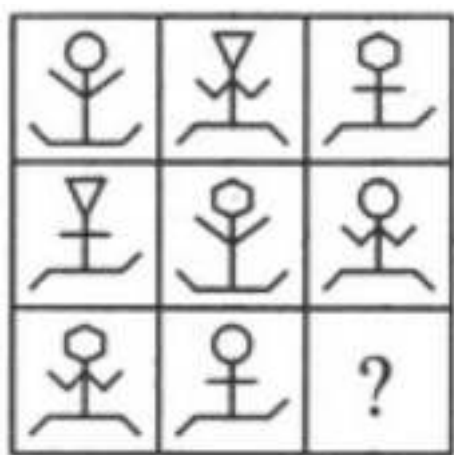
(4)

TYPE 5: FIGURE MATRIX



Q8. Select a suitable figure from the four alternatives that would complete the figure matrix.





Answer: Option 4

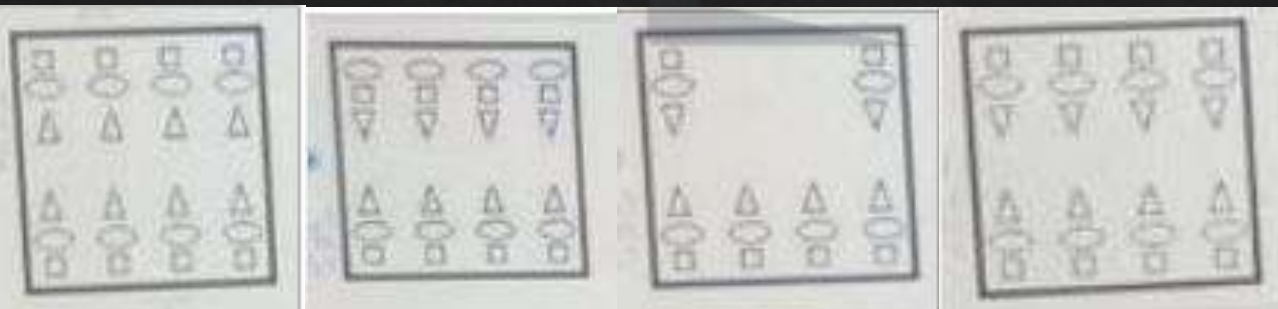
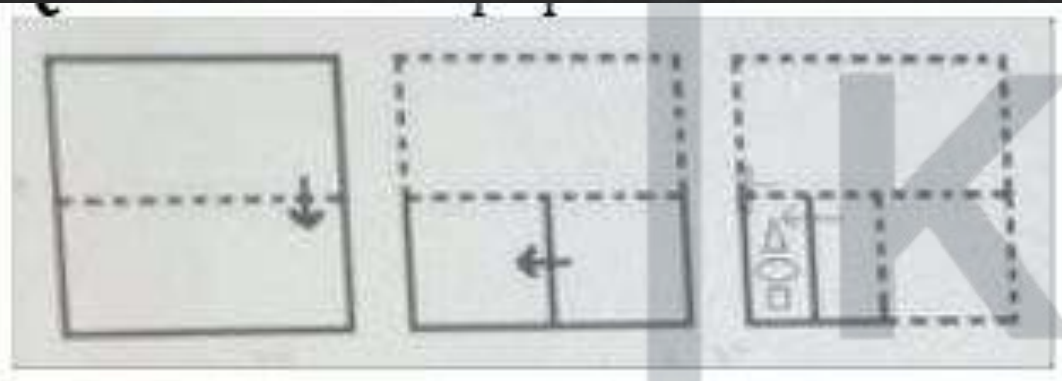
Explanation:

There are 3 types of faces, 3 types of hands and 3 types of legs. Each type is used once in each row! So, the features not used in the first two figures of the third row would together form the missing figure.

TYPE 6: PAPER CUTTING



Q9. Question consists of a set of three figures X, Y and Z showing a sequence of folding of a piece of paper. Figure (Z) shows the manner in which the folded paper has been cut. These three figures are followed by four answer figures from which you have to choose a figure which would most closely resemble the unfolded form of figure (Z).



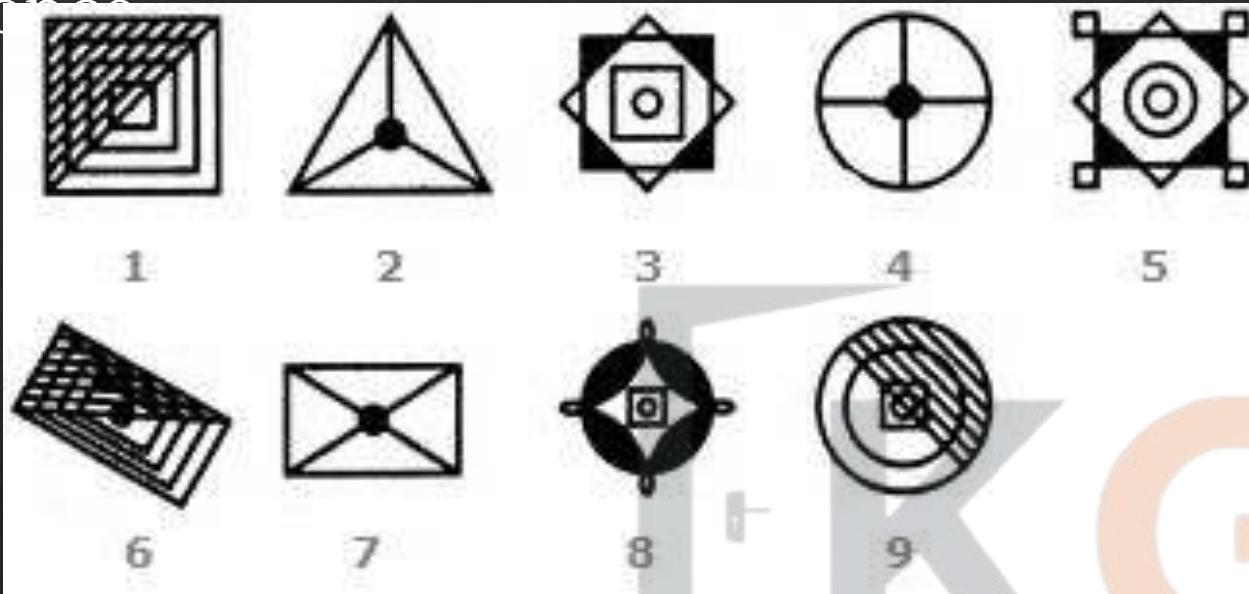
Correct Answer - Option D



TYPE 7: GROUPING OF IMAGES



Q10. Group the given figures into three classes using each figure only once.



- A. 2,4,7 ; 1,6,9 ; 3,5,8
- B. 1,3,5 ; 2,6,7 ; 4,8,9
- C. 1,5,7 ; 2,3,6 ; 4,8,9
- D. 1,3,5 ; 2,4,7 ; 6,8,9

Correct Answer - Option A

Explanation :

1, 6, 9 are figures which are half shaded by slanting lines.

2, 4, 7 are all divided into equal parts (either three or four parts) by straight lines and also have a black circle at the centre.

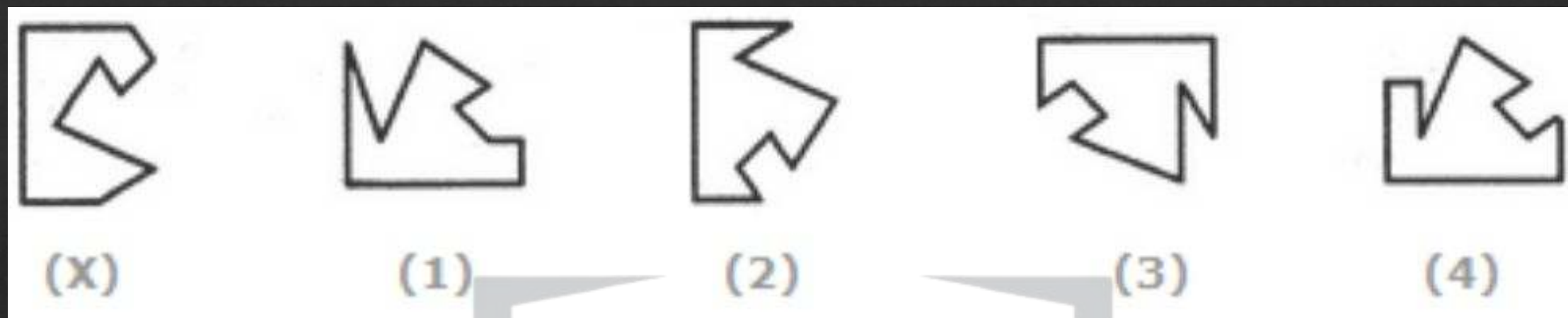
3, 5, 8 have similar designs and have their four corners shaded black.

TYPE 8: SHAPE CONSTRUCTION



Q11. Select a figure from the given four alternatives which fits exactly into Figure-X to form a complete square.



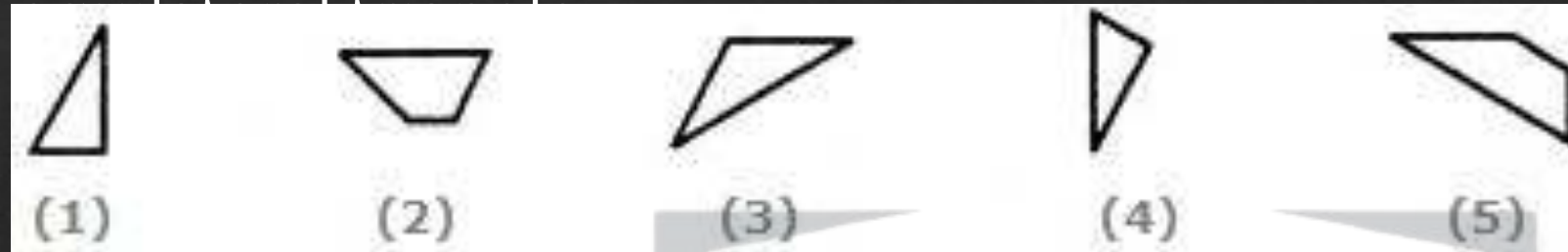


Answer: Option 2

Explanation:

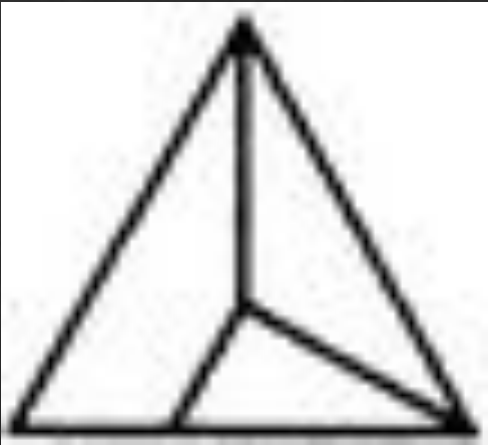


Q12. Select the alternative which represents three out of the five alternative figures which when fitted into each other would form a



- A. 123
- B. 234
- C. 345
- D. 245

Correct Answer - Option C



(1)



(2)



(3)



(4)

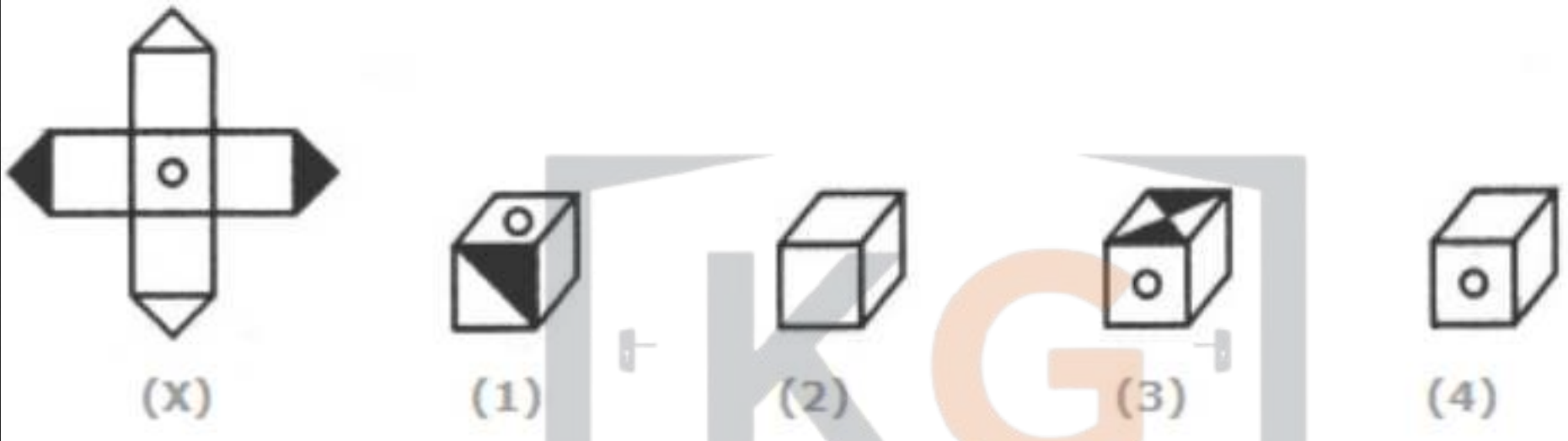


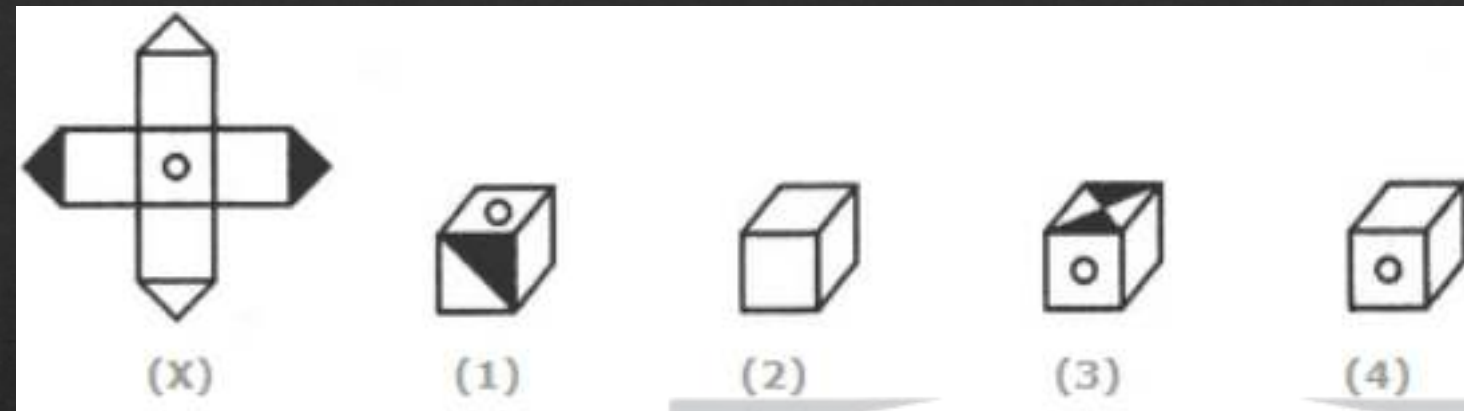
(5)

TYPE 9: CUBES & DICES



Q13. Choose the box that is similar to the box formed from the given sheet of paper (X).





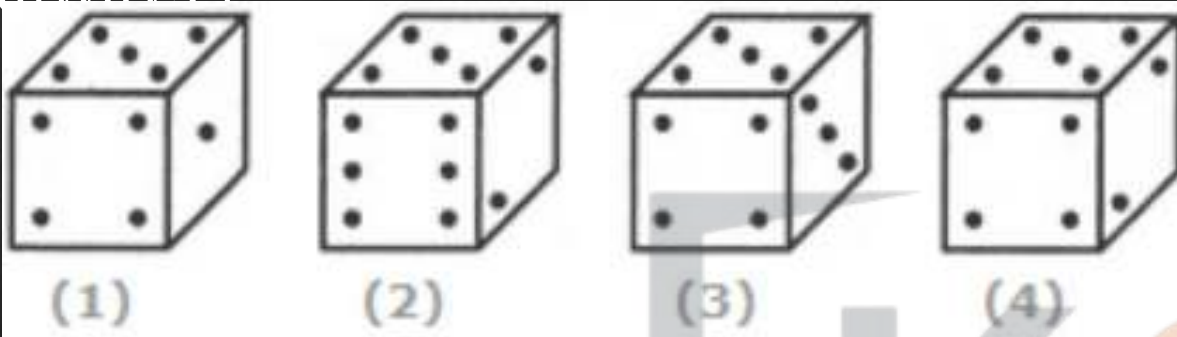
Answer: Option 4

Explanation:

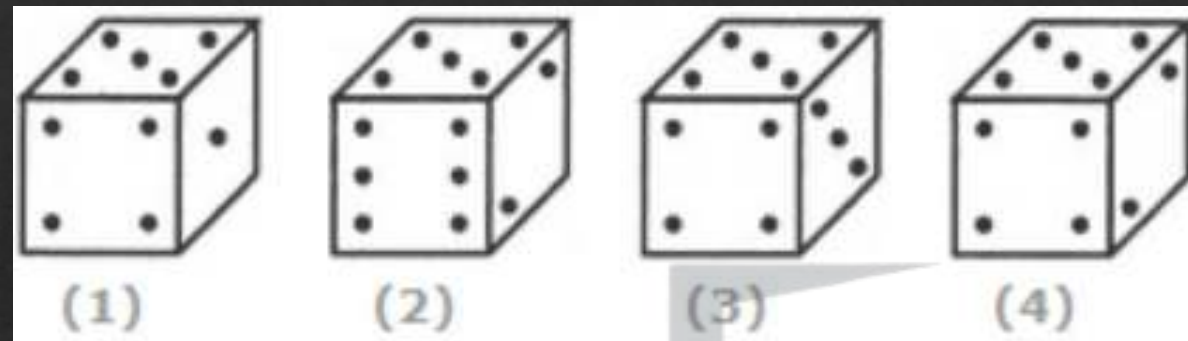
The fig. (X) is similar to the **Form VI**. So, when the cube is formed by folding the sheet shown in fig. (X), then  is one of the faces of the cube and this face lies opposite to the face bearing a circle. Also, one of the blank faces lies opposite to another blank face and yet another blank face lies opposite to the fourth blank face. Thus, out of the four blank faces, no three faces can appear adjacent to each other.

Clearly, the cube in fig. (1) cannot be formed since there is no face of the type , the cube in fig. (2) cannot be formed since it shows three blank faces adjacent to each other and the cube in fig. (3) cannot be formed since the face  cannot appear adjacent to the face bearing the circle. Hence, only the cube in fig. (4) can be formed.

Q14. Amongst the following figures, find the correct one, if it is known that the total number of dots on opposite faces of the cube shown is always 7



Q14. Amongst the following figures, find the correct one, if it is known that the total number of dots on opposite faces of the cube shown is always 7.



Answer: Option 1

Explanation:

Since the total number of dots on opposite faces is always 7, therefore 1 dot must lie opposite 6 dots, 2 dots must lie opposite 5 dots and 3 dots must lie opposite 4 dots. In each of the two figures (2) and (4), 2 dots appear adjacent to 5 dots, and in fig. (3), 3 dots appear adjacent to 4 dots. Hence, these figures are incorrect. Therefore/only fig. (1) is correct.

TYPE 10: ANALOGY



Q15. Look carefully at the sequence of symbols to find the pattern. Select correct pattern.



(1)



(2)



(3)



(4)

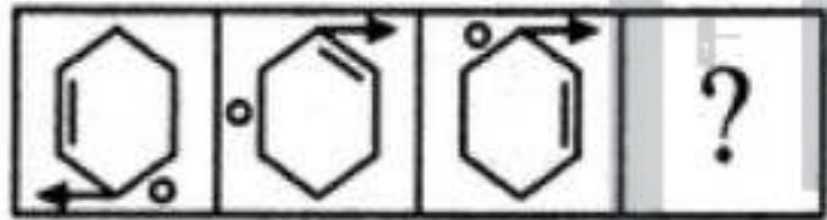
Correct Answer : Option 3

Explanation –

All four segments use the same figures: two squares, one circle, and one triangle. In the first segment, the squares are on the outside of the circle and triangle. In the second segment, the squares are below the other two. In the third segment, the squares are on the inside. In the fourth segment, the squares are above the triangle and circle.

Q16. Each of the following questions consists of two sets of figures. Figures A, B, C and D constitute the Problem Set while figures 1, 2, 3, 4 and 5 constitute the Answer Set. There is a definite relationship between figures A and B. Establish a similar relationship between figures C and D by selecting a suitable figure from the Answer Set that would replace the question mark (?) in fig. (D)

Problem Figures:



(A) (B) (C) (D)

Answer Figures:



(1) (2) (3) (4) (5)

Correct Answer : Option 4

Explanation –

The arrow moves four spaces (each space is equal to a side of the hexagon) in a CW direction while the line segment and the circle move two spaces in a CW direction.

TYPE 11: WATER IMAGES



G

TYPE 11: WATER IMAGES



G

Q17. You are given a combination of alphabets and/or numbers followed by four alternatives (1), (2), (3) and (4). Choose the alternative which is closely resembles the water-image of the given combination.

PQ8AF5BZ9

(1) 608V72BZ0

(2) 608V72BΣ6

(3) 608V79BΣ9

(4) 608V72BΣ0

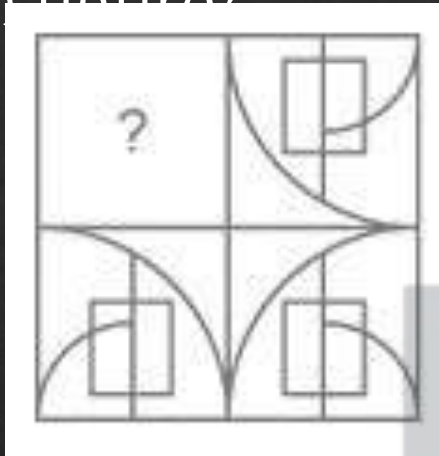
Correct Answer : Option 4



TYPE 12: PATTERN COMPLETION



Q18. Which answer figure will complete the pattern in the following question figure?



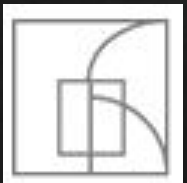
A



C



B

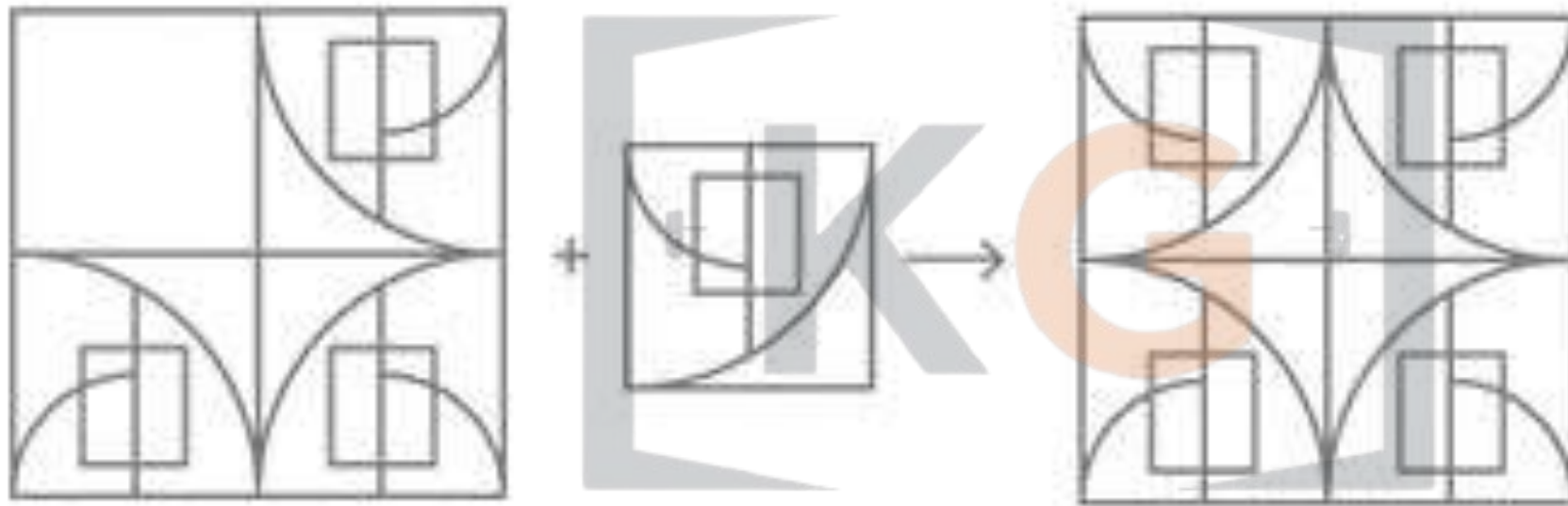


D



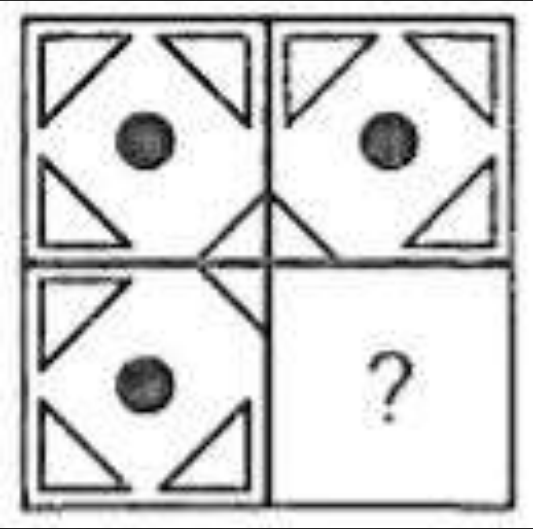
Correct Answer : Option D

Explanation –
The complete figure is shown below:



Hence, the correct option is (D).

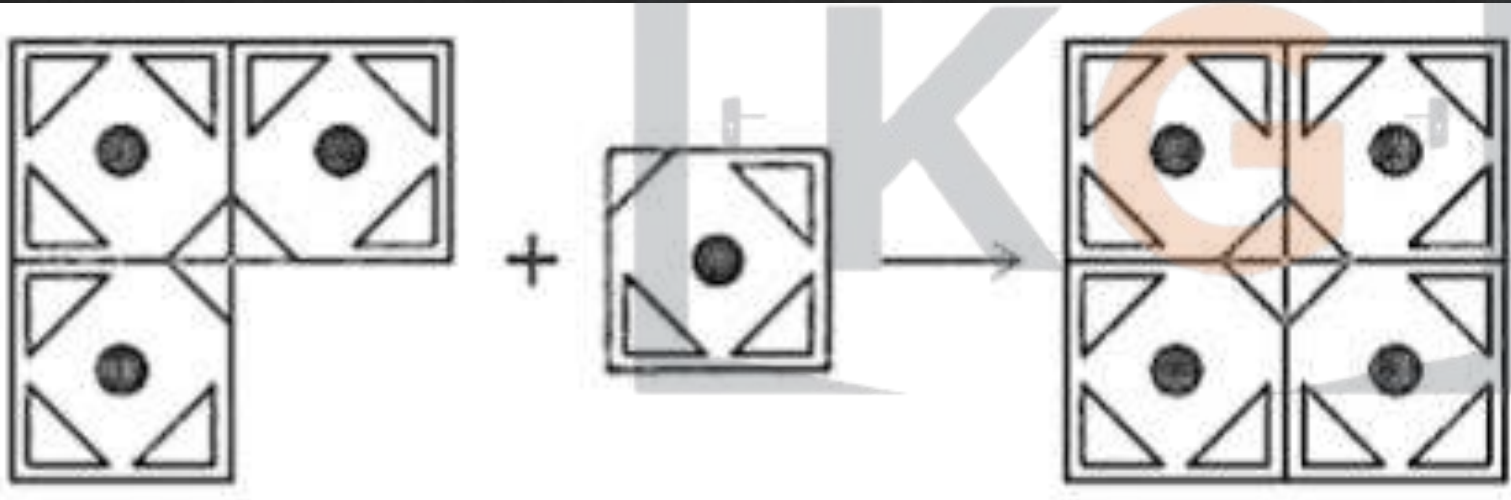
Q19 (VK). Select the option figure that can replace the question mark (?) to complete the pattern in the question figure.



Correct Answer : Option A

Explanation –

The option figure that can replace the question mark (?) to complete the pattern in the question figure is,

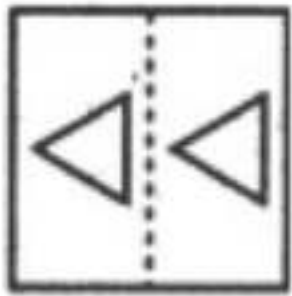


Hence, the correct option is (A).

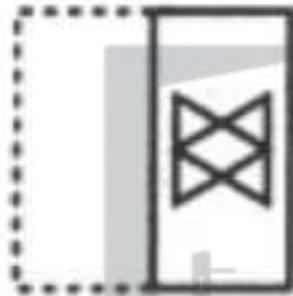
TYPE 13: PAPER FOLDING



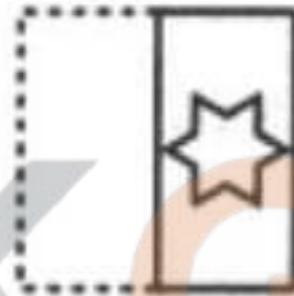
Q20. Find out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded at the dotted line.



(X)



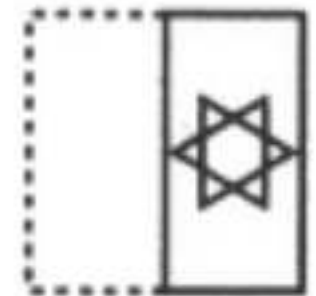
(1)



(2)



(3)



(4)

Correct Answer- Option A

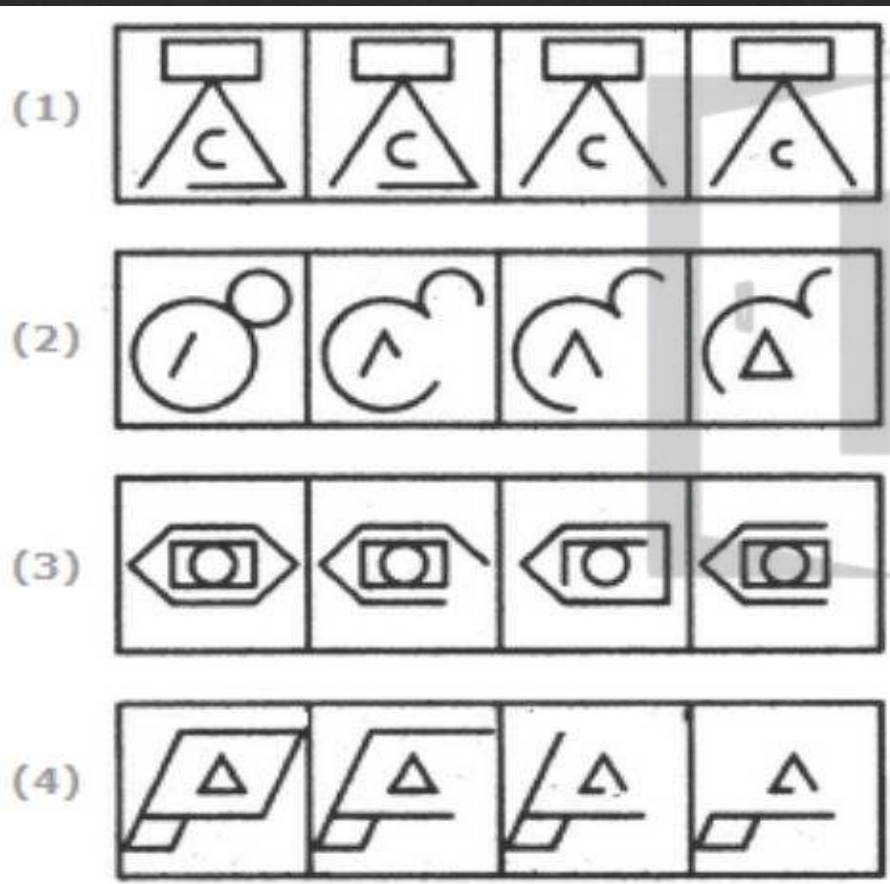


TYPE 14: RULE DETECTION



Q21. Choose the set of figures which follows the given rule.

Rule: Closed figures become more and more open and open figures become more and more closed.

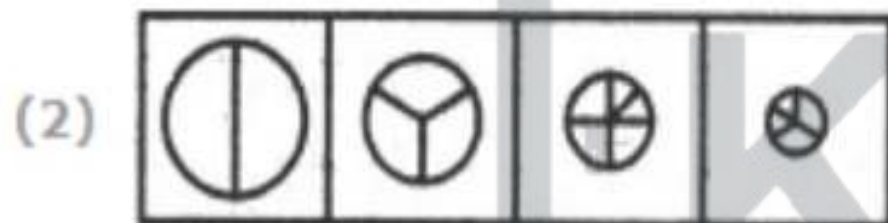
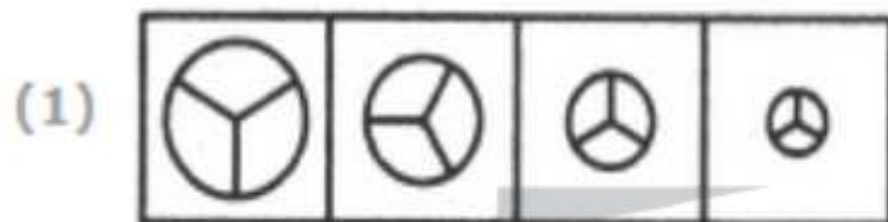


Correct Answer : Option 2



Q22. Choose the set of figures which follows the given rule.

Rule: As the circle decreases in size, its sectors increase in number.



Correct Answer : Option 4



TYPE 15: DOT SITUATION



Q23. Select the figure which satisfies the same conditions of placement of the dots as in Figure-X.



(X)



(1)



(2)



(3)

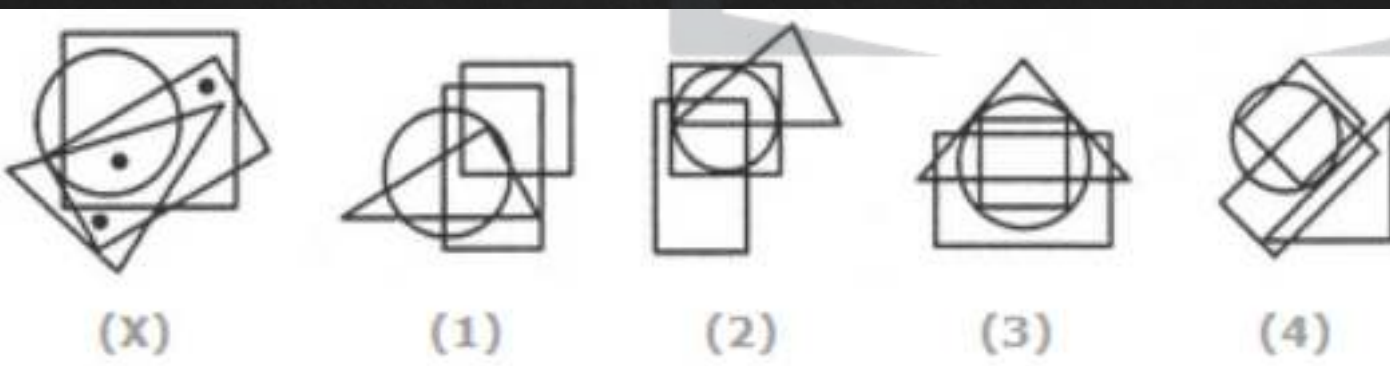


(4)

Correct Answer: Option A

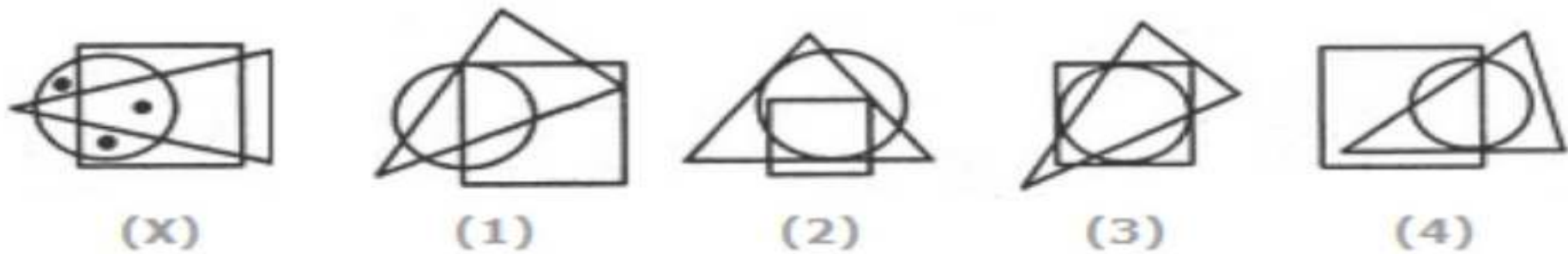
Explanation:

In fig. (X), one of the dots lies in the region common to the square and the rectangle only, another dot lies in the region common to all the four elements - the circle, the square, the triangle and the rectangle and the third dot lies in the region common to the triangle and the rectangle only. In fig. (2) there is no region common to the triangle and the rectangle only. In fig. (3) there is no region common to the square and the rectangle only. In fig. (4) there is no region common to all the four elements - the circle, the square, the triangle and the rectangle. Only fig. (1) consists of all the three types of regions.



Q24. Select the figure which satisfies the same conditions of placement of the dots as in Figure-X.





Correct Answer: Option A

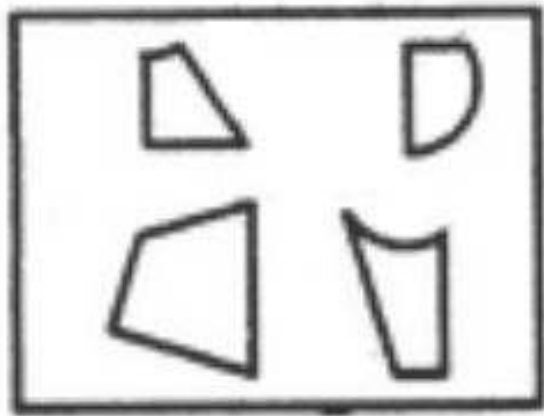
Explanation:

In fig. (X), one of the dots lies in the region common to the circle and the square only, another dot lies in the region common to all the three figures - the circle, the square and the triangle and the third dot lies in the circle alone. In fig. (2) there is no region common to the circle and the square only and in each of the figures (3) and (4) there are regions which lie in the circle alone. Only fig. (1) consists of all the three types of regions.

TYPE 16: IMAGE ANALYSIS



Q25. Find out which of the figures (1), (2), (3) and (4) can be formed from the pieces given in figure (X).



(X)



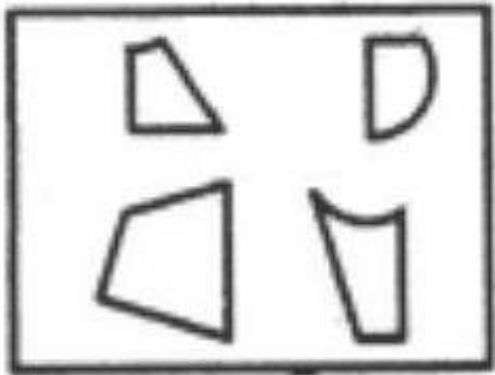
(1)

(2)

(3)

(4)

Correct Answer : Option B



(X)



(1)

(2)

(3)

(4)



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SYLLOGISM



Tujhe Idea Bhi Hai Tu Kya Bol Raha Hai

BAAR
BAAR
DEKHO



Q. Read all the conclusions and then decide which of the given conclusions logically follow from the given statements disregarding commonly known facts.



Statements:

All apples are bananas.
No banana is a mango.
Some mangoes are oranges.

Conclusions:

- I. All oranges can never be bananas.
- II. Some mangoes are apples.

- A. Only conclusion I follows
- B. Only conclusion II follows
- C. Either conclusion I or conclusion II follows.
- D. Neither conclusion I nor conclusion II follows.
- E. Both conclusion I & II follows

BASIC CONCEPTS

The question contains some statements followed by some conclusions.

We need to decide which of the given conclusions logically follow from the given statements, **disregarding commonly known facts**.

Steps:

1. Draw the basic diagram first with **minimum overlap**.
2. For a statement to be false, if it is false in even any one diagram, then also it is false.
3. For a statement to be true, it must be true in all possible diagrams.

PRE-REQUISITES

एक बढ़िया वाला दिमाग

Ability to think about all possibilities.



Basic Concepts (Venn Diagram)

All X's are Y's

No X's are Y's

Some X's are Y's

Some X's are not Y's



Basic Concepts (Venn Diagram)

All X's are Y's





Basic Concepts (Venn Diagram)

No X's are Y's





Basic Concepts (Venn Diagram)

Some X's are Y's





Basic Concepts (Venn Diagram)

Some X's are not Y's



Basic Concepts (Venn Diagram)

Some X's are not Y's



Basic Concepts (Venn Diagram)

All X's are Y's

No X's are Y's

Some X's are Y's

Some X's are not Y's



BASIC CONCEPTS

Steps:

1. Draw the basic diagram first with **minimum overlap**.
2. For a statement to be false, if it is false in even any one diagram, then also it is false.
3. For a statement to be true, it must be true in all possible diagrams.

Minimum & Maximum Overlap Diagram



Basic Concepts (Venn Diagram)

All X's are Y's



Basic Concepts (Venn Diagram)

No X's are Y's



Basic Concepts (Venn Diagram)

Some X's are Y's



Basic Concepts (Venn Diagram)

Some X's are not Y's



Q1. Read all the conclusions and then decide which of the given conclusions logically follow from the given statements disregarding commonly known facts.



Statements:

All apples are bananas.
No banana is a mango.
Some mangoes are oranges.

Conclusions:

- I. All oranges can never be bananas.
- II. Some mangoes are apples.

- A. Only conclusion I follows
- B. Only conclusion II follows
- C. Either conclusion I or conclusion II follows.
- D. Neither conclusion I nor conclusion II follows.
- E. Both conclusion I & II follows

Q1. Read all the conclusions and then decide which of the given conclusions logically follow from the given statements disregarding commonly known facts.



Statements:

All apples are bananas.
No banana is a mango.
Some mangoes are oranges.

Conclusions:

- I. All oranges can never be bananas.
- II. Some mangoes are apples.

- A. Only conclusion I follows
- B. Only conclusion II follows
- C. Either conclusion I or conclusion II follows.
- D. Neither conclusion I nor conclusion II follows.
- E. Both conclusion I & II follows

Correct Answer : Option A

Explanation –

I. All oranges can never be bananas. - (True) - Since some of the part of orange is in mango. Therefore the conclusion is true.

II. Some mangoes are apples. - (False) - It is clear from the diagram. Only conclusion I is follow.

Hence, option A is the correct response.

Q2. Choose the conclusions which logically follow from the given statements.

Statements: All men are vertebrates. Some mammals are vertebrates.



Conclusions:

1. All men are mammals.
2. All mammals are men.
3. Some vertebrates are mammals.
4. All vertebrates are men.

- A. Only (4)
- B. Only (2)
- C. Only (3)
- D. Only (1)

Q2. Choose the conclusions which logically follow from the given statements.

Statements: All men are vertebrates. Some mammals are vertebrates.



Conclusions:

1. All men are mammals.
2. All mammals are men.
3. Some vertebrates are mammals.
4. All vertebrates are men.

- A. Only (4)
- B. Only (2)
- C. Only (3)
- D. Only (1)

Correct Answer : Option C



Q3. Statement :

Some schools are classes.
Some classes are teachers.
All teachers are students.

Conclusions :

Some students are classes.
All schools being students is a possibility.

- A. Only I follow
- B. Both I and II follow
- C. Only II follows
- D. Neither I nor II follow



Q3. Statement :

Some schools are classes.
Some classes are teachers.
All teachers are students.

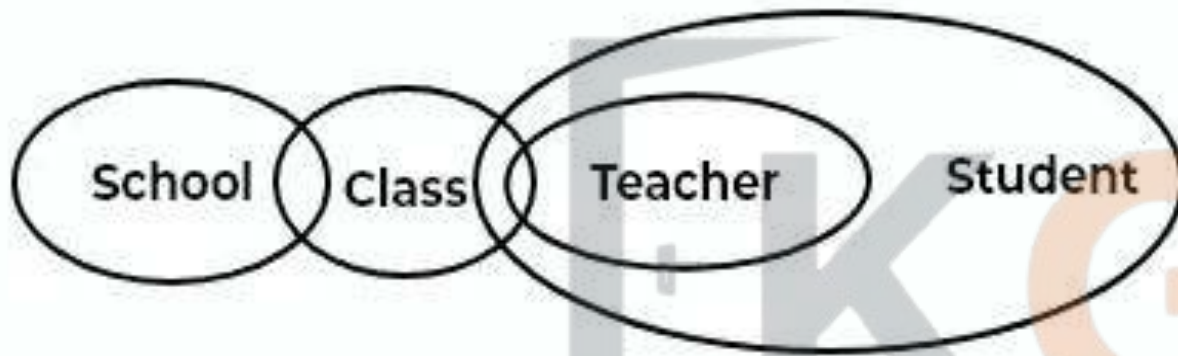
Conclusions :

Some students are classes.
All schools being students is a possibility.

- A. Only I follow
- B. Both I and II follow
- C. Only II follows
- D. Neither I nor II follow

Correct Answer : Option B

Explanation –



From the figure we can both the conclusions are true

Q4. Statements :

Some rude are hate.

All hate is love.

Conclusions :

I. All rude are love.

II. Some rude are not love.

- . Only I follow
- . Only II follow
- . Either I or II follow
- . None follow



Q4. Statements :

Some rude are hate.

All hate is love.

Conclusions :

I. All rude are love.

II. Some rude are not love.

- . Only I follow
- . Only II follow
- . Either I or II follow
- . None follow



Correct Answer : Option C

Explanation –

I. All rude are love. $\rightarrow$ False (It is possible but not definite)

II. Some rude are not love. $\rightarrow$ False (It is possible but not definite)

Both conclusions are false and having the relation of "All" and "some not" between rude and love and hence forms a complementary pair. So in that case either conclusion I or II follows will be correct answer.

Thus, either I or II follows.

Q5. Statements: $A > B$, $D \leq E$, $C \geq F = G$, $B \leq C = D$

Conclusions:

I. $B \geq G$

II. $E \geq G$

- A. Conclusion I follows
- B. Conclusion II follows
- C. Both Conclusions follows
- D. None follows

Q5. Statements: $A > B$, $D \leq E$, $C \geq F = G$, $B \leq C = D$

Conclusions:

I. $B \geq G$

II. $E \geq G$

- A. Conclusion I follows
- B. Conclusion II follows
- C. Both Conclusions follows
- D. None follows

Q5. Statements: $A > B$, $D \leq E$, $C \geq F = G$, $B \leq C = D$

Conclusions:

I. $B \geq G$

II. $E \geq G$

- A. Conclusion I follows
- B. Conclusion II follows
- C. Both Conclusions follows
- D. None follows

Correct Answer : Option B

Explanation –

According to the statements
 $B \geq G$ (false)
 $E \geq G$ (true)
which means conclusion II follows



Q6 (VK). Statements: All cars are cats. All fans are cats

Conclusions:

I. All cars are fans.

II. Some fans are cars.

- A. Only (1) conclusion follows
- B. Only (2) conclusion follows
- C. Either (1) or (2) follows
- D. Neither (1) nor (2) follows

Q6 (VK). Statements: All cars are cats. All fans are cats

Conclusions:

I. All cars are fans.

II. Some fans are cars.

- A. Only (1) conclusion follows
- B. Only (2) conclusion follows
- C. Either (1) or (2) follows
- D. Neither (1) nor (2) follows

Correct Answer : Option D





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BLOOD RELATION

at aptitude class

22. Sonam asked her brother's paternal grandmother's husband's only daughter's son's only maternal uncle's wife to get her a chocolate. Whom did Sonam ask for a chocolate?



Types of Questions

1. General Questions / Puzzle Type Questions
2. Coded Blood Relation
3. Indicating Blood Relation



Concept of Generation



Drawing Blood - Relationship Diagram (BR Diagram)



GENERAL QUESTIONS / PUZZLE TYPE QUESTIONS



Q1: A is father of B, B is the father of C. R is the daughter of A, X is the wife of B

- a) How is X related to A
- b) Number of males
- c) Number of females
- d) How is C related to R
- e) How is A related to C
- f) How is C related to A

Q1: A is father of B, B is the father of C. R is the daughter of A, X is the wife of B



Q1: A is father of B, B is the father of C. R is the daughter of A, X is the wife of B



A is great grandfather of B



A is paternal uncle of B



A is maternal uncle of B



A is daughter in law of B



A is mother in law of B



A is son in law of B



Q2. There are seven members in a family. J is the mother of K and mother in law of E. C is the daughter in law of N. E is the son of C who is married with F. G is spouse of E. There are only two married couple in the family. How is G related with J?



- A. Daughter
- B. Son in law
- C. Daughter in law
- D. Wife



Q2. There are seven members in a family. J is the mother of K and mother in law of E. C is the daughter in law of N. E is the son of C who is married with F. G is spouse of E. There are only two married couple in the family. How is G related with J?



Correct Answer : Option A

Explanation –

C is the daughter in law of N ---> C is the wife of N's Son.

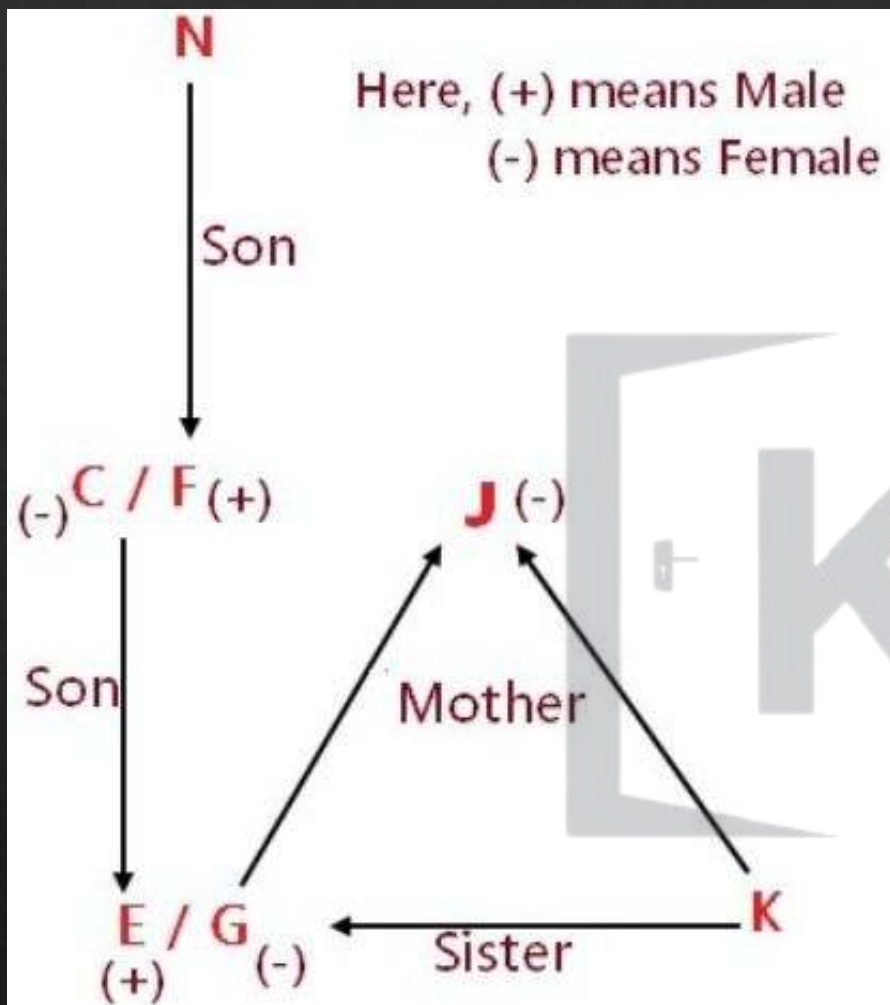
E is the son of C who is married with F ---> F is N's Son as well as C's husband. E is the son of C and F.

G is spouse of E ---> Since E is a male, G must be the wife of E.

J is mother in law of E ---> J is mother of E's wife (ie., J is mother of G).

J is the mother of K ---> Hence, G and K are siblings.

There are only two married couple in the family ---> C and F; E and G.



CODED BLOOD RELATION



Q3.

'X \$ Y' means 'X is father of Y'

'X # Y' means 'X is mother of Y'

'X × Y' means 'X is sister of Y'

How is D related to N in N # A \$ B × D ?

- A. Nephew
- B. Grand son
- C. Grand daughter
- D. Cannot be Determined

Q3.

'X \$ Y' means 'X is father of Y'

'X # Y' means 'X is mother of Y'

'X × Y' means 'X is sister of Y'

How is D related to N in $N \# A \$ B \times D$?



Correct Answer : Option D

Explanation –

$N \# A \$ B \times D$ means N is the mother of A, who is the father of B, who is the sister of D. Thus, D is the son or daughter of A and N is the mother of A. So, D is the grandson or granddaughter of N.



Q4. P is husband of S. Find the correct option:

A + B means A is father of B

A - B means A is mother of B

A * B means A is brother of B

A / B means A is sister of B

A @ B means A is daughter of B

- a) $S * P + R - Q$
- b) $Q - P - R @ S$
- c) $P + R * Q @ S$
- d) $P + Q - S / R$
- e) None of these

Correct Answer : Option C



INDICATING BLOOD RELATIONS



Q5. Pointing to a photograph, Asha said, “His mother's only daughter is my mother.” How is Asha related to the man in the photograph?



- A) Cousin
- B) Wife
- C) Sister
- D) Niece



Correct Answer : Option D

Explanation –

His mother's only daughter is his sister, who is Asha's mother.
So he is Asha's maternal uncle. and thus Asha is his niece.



Q6. Pointing towards the lady, Raj said: “She is the only daughter of my wife’s grand father’s only son”. How is that lady related to Raj?

- A. Mother
- B. Sister
- C. Wife
- D. Daughter



Correct Answer : Option C



Q7. (VK) Six members of a family are travelling these are a, b, c, d, e, and f. 'b' is the son of 'c' but 'c' is not mother of 'b'. 'a' and 'c' are a married couple, 'e' is the brother of 'c'. 'd' is the daughter of 'a'. 'f' is the brother of 'b'.

Which of following is pair of female?

- A. ae
- B. bd
- C. df
- D. ad



Correct Answer : Option D

Explanation –

‘b’ is the son of ‘c’ but ‘c’ is not mothers of ‘b’. Implies ‘c’ is the father of ‘b’. It is given that ‘a’ and ‘c’ are married. Therefore ‘a’ must be a woman. ‘d’ is the daughter of ‘a’. Therefore ‘d’ is also a woman.





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DATA ARRANGEMENT

Q1. Arrange the following words in a meaningful sequence.

1. Cotton
2. Plant
3. Yarn
4. Cloth
5. Shirt

- A) 2, 1, 3, 4, 5
- B) 2, 1, 3, 5, 4
- C) 1, 4, 5, 2, 3
- D) 4, 5, 1, 3, 2



Correct Answer : Option A

Explanation –

The plant provides cotton and cotton provides yarn.
The cloth is produced from yarn and shirt is made from cloth.



Q2. Arrange the following words in a logical sequence.

1. Rain
2. Sun
3. Rainbow
4. Child
5. Happy

- A. 1, 2, 3, 4, 5
- B. 1, 3, 2, 4, 5
- C. 2, 1, 3, 4, 5
- D. 1, 3, 2, 5, 4



Correct Answer : Option A

Explanation –

The rain comes first then Sun, and then rainbow is formed in the sky. When a child sees the rainbow, he feels happy.

Q3. Arrange the given words in a meaningful sequence.

1. Room
2. Building
3. Society
4. Flat
5. City

- A. 5, 3, 2, 1, 4
- B. 3, 5, 2, 4, 1
- C. 3, 1, 2, 4, 5
- D. 5, 3, 2, 4, 1



Correct Answer : Option D

Explanation –

According to size, the correct sequence is a city, society, building, flat, and room.



Q4. Arrange the words given below in a meaningful sequence

1. Never
2. Sometimes
3. Generally
4. Seldom
5. Always

- A. 5, 2, 1, 3, 4
- B. 5, 2, 4, 3, 1
- C. 5, 3, 2, 1, 4
- D. 5, 3, 2, 4, 1



Correct Answer : Option D

Explanation –

Meaning of Seldom: not often, rarely



Q5. Arrange the words given below in a meaningful sequence.

1. Family
2. Community
3. Member
4. Locality
5. Country

- A. 3, 1, 2, 4, 5
- B. 3, 1, 2, 5, 4
- C. 3, 1, 4, 2, 5
- D. 3, 1, 4, 5, 2



Correct Answer : Option A

Explanation –

The correct order is :

Member	Family	Community	Locality	Country
3	1	2	4	5

Q6 (VK). Arrange the words given below in a meaningful sequence.

1. Protect
2. Pressure
3. Relief
4. Rain
5. Flood

- A. 2, 4, 3, 1, 5
- B. 2, 4, 5, 1, 3
- C. 2, 5, 4, 1, 3
- D. 3, 2, 4, 5, 1



Correct Answer : Option B

Explanation –

The correct order is :

Pressure	Rain	Flood	Protect	Relief
2	4	5	1	3



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VISUAL REASONING

Visual Reasoning

1. Pattern Recognition (Figure & Factual Analysis)
2. Cubes & Dices
3. Image & Figure Counting Reasoning (Analytical Reasoning)
4. Venn Diagram
5. Direction Sense (Directions)
6. Seating Arrangement



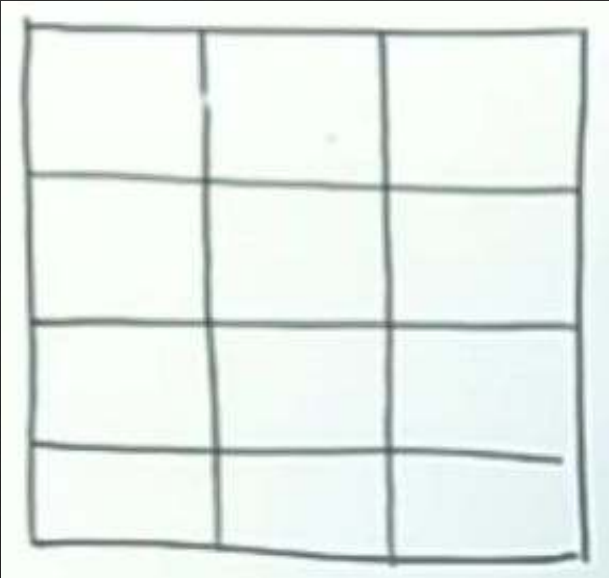
TATA
CONSULTANCY
SERVICES

ANALYTICAL REASONING

(IMAGE & FIGURE COUNTING REASONING)



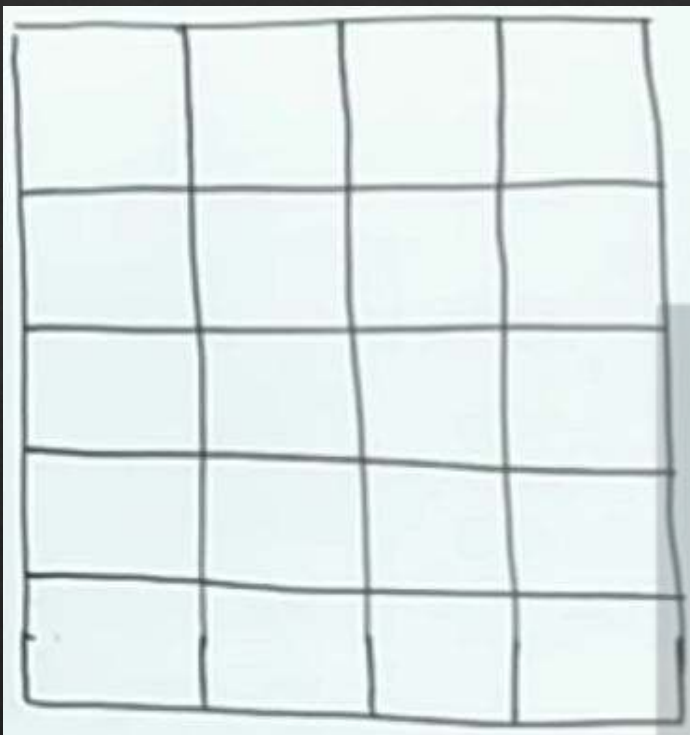
Q1. Count Number of Squares?



Correct Answer: 20



Q2. Count Number of Squares?

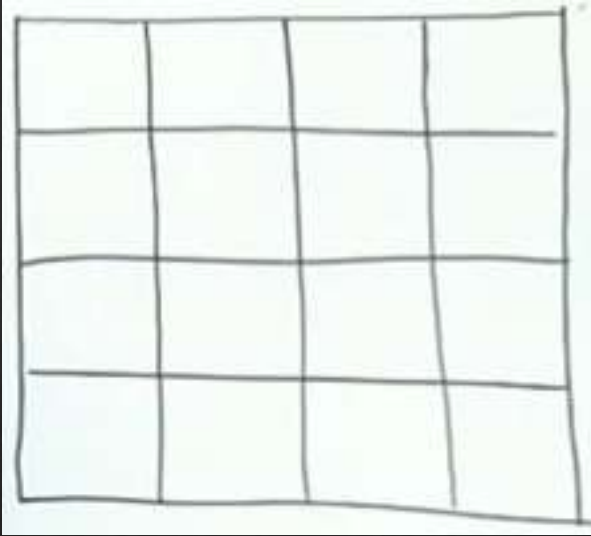


KG

Correct Answer: 40



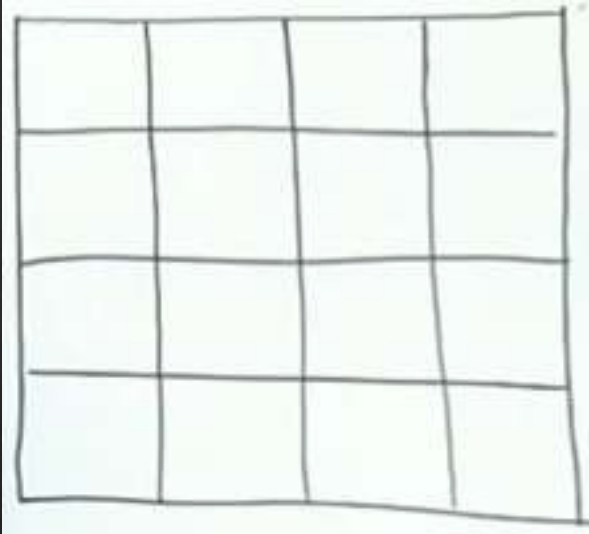
Q3. Count Number of Squares?



Correct Answer: 30

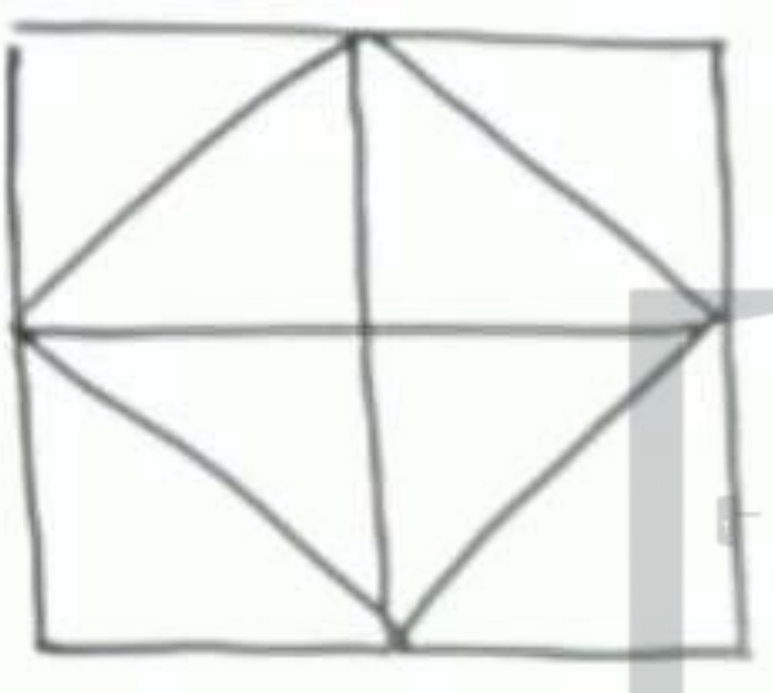


Q3. Count Number of Squares?



$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

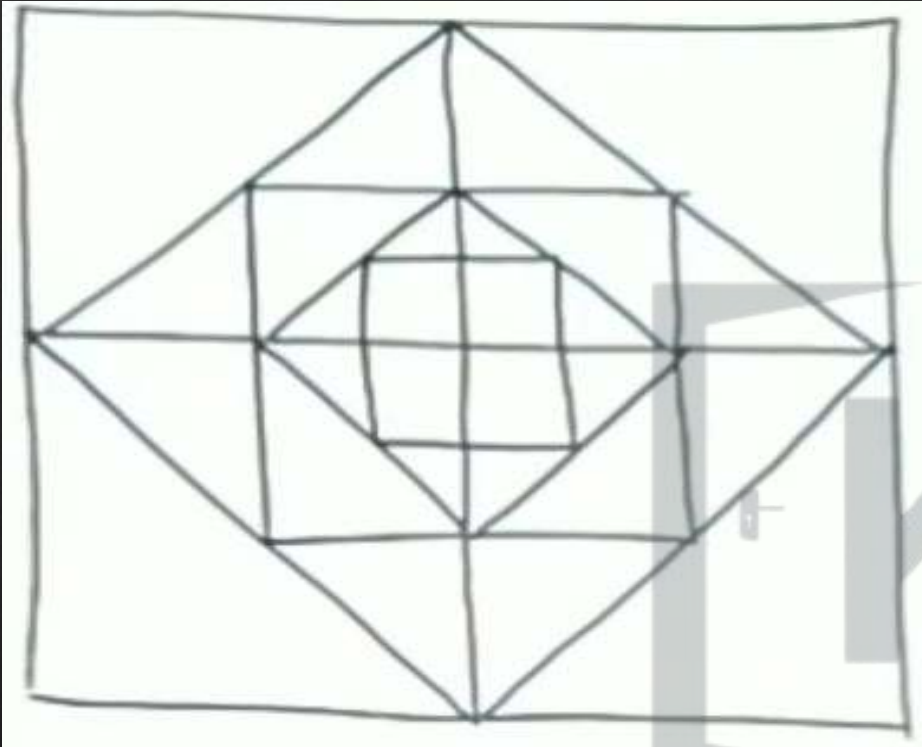
Q4. Count Number of Squares?



Correct Answer: 6



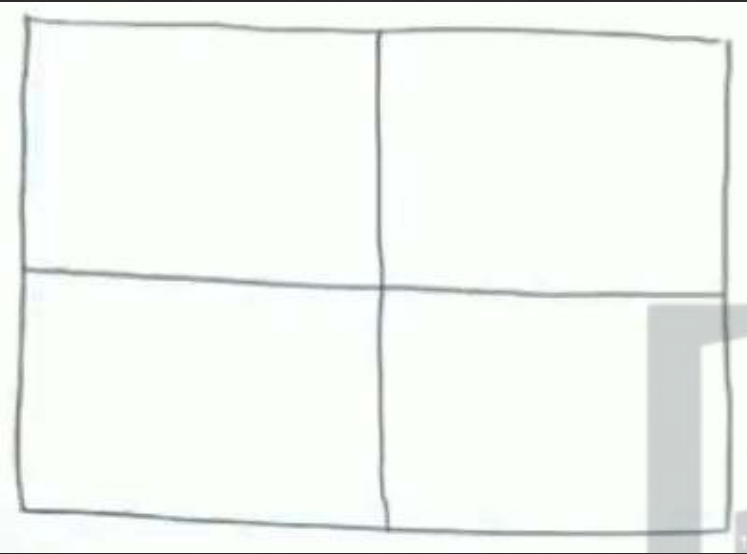
Q5. Count Number of Squares?



Correct Answer: 17



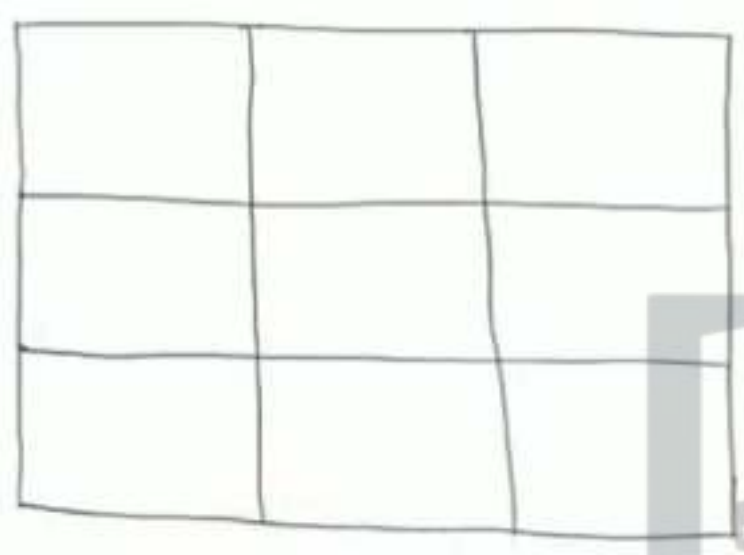
Q6. Count Number of Rectangles?



Correct Answer: 9



Q7. Count Number of Rectangles?

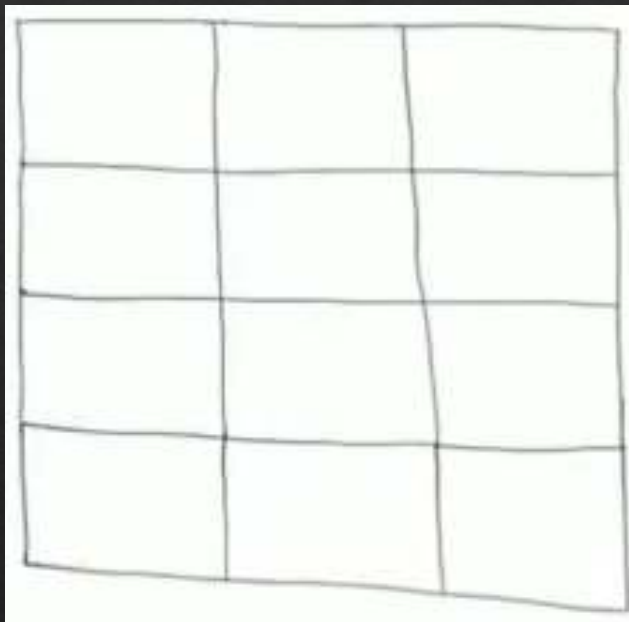


LINK

Correct Answer: 36



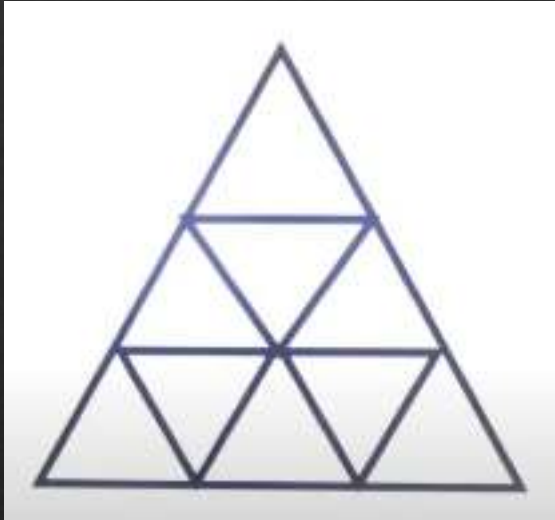
Q8. Count Number of Rectangles?



Correct Answer: 60



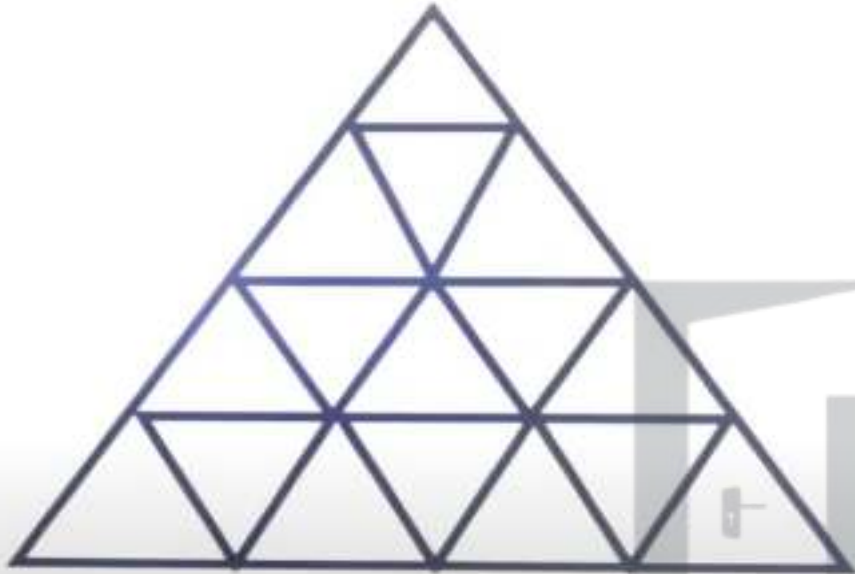
Q9. Count Number of Triangles?



Correct Answer: 13



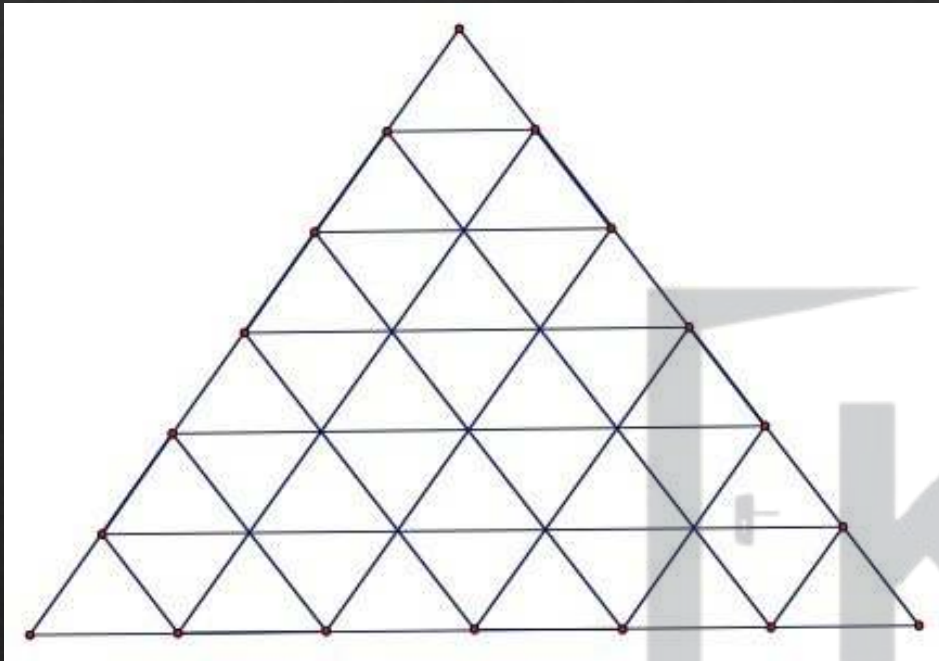
Q10. Count Number of Triangles?



Correct Answer: 27



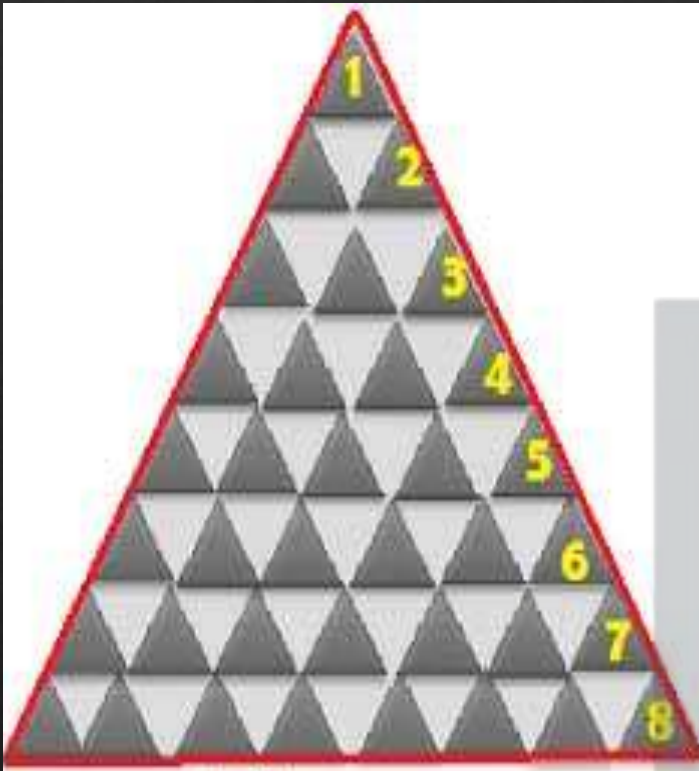
Q11. Count Number of Triangles?



Correct Answer: 78



Q12. Count Number of Triangles?

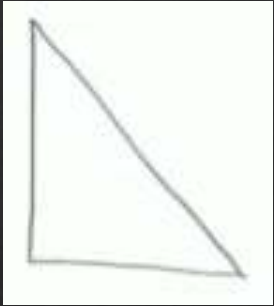


KG

Correct Answer: 170



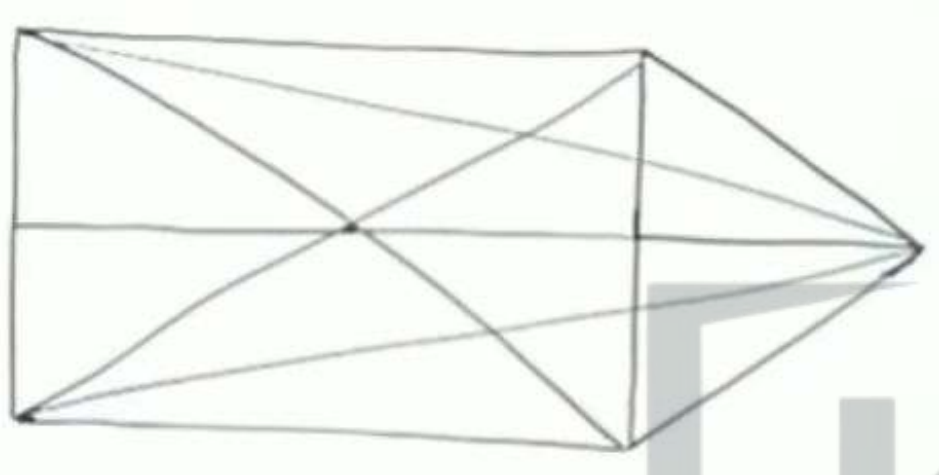
Q13. Count Number of Straight Lines?



Correct Answer: 3



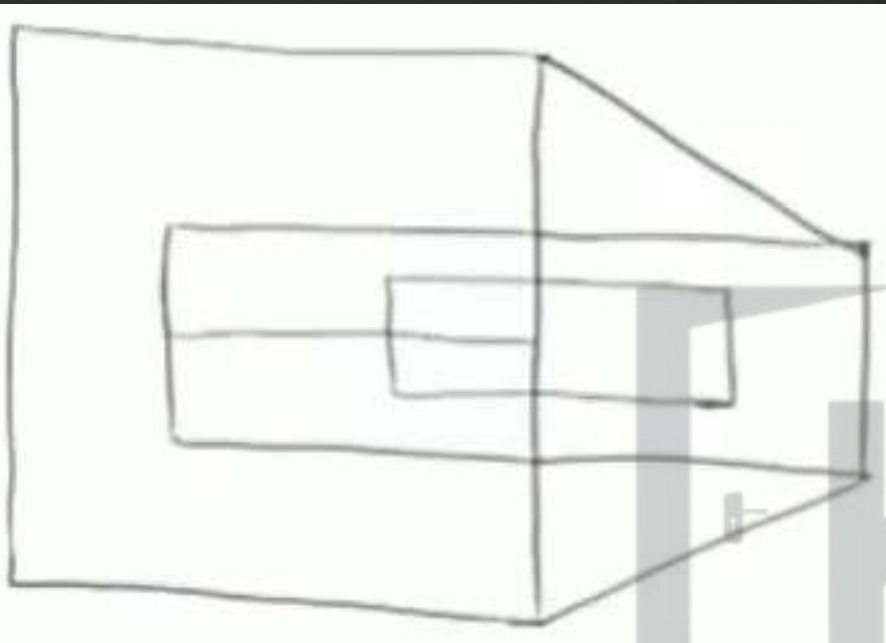
Q14. Count Number of Straight Lines?



Correct Answer: 11



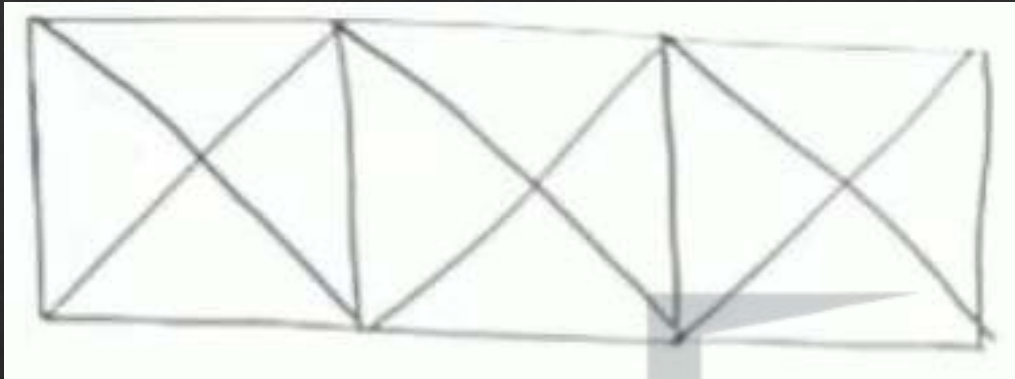
Q15. Count Number of Straight Lines?



Correct Answer: 15



Q16 (VK). Count Number of Straight Lines?



Visual Reasoning

1. Pattern Recognition (Figure & Factual Analysis)
2. Cubes & Dices
3. Image & Figure Counting Reasoning (Analytical Reasoning)
4. Venn Diagram
5. Direction Sense (Directions)
6. Seating Arrangement

Correct Answer: 12





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SPATIAL REASONING

Transformation of Shapes:

1. Translation
2. Scaling
3. Rotation
4. Mirroring
5. Assembling
6. Grouping
7. Paper Cutting
8. Paper Folding
9. Pattern in 2D
10. Pattern in 3D



Transformation of Shapes:

1. Translation – will study
2. Scaling – will study
3. Rotation – will study
4. Mirroring
5. Assembling
6. Grouping
7. Paper Cutting
8. Paper Folding
9. Pattern in 2D
10. Pattern in 3D – will study

TYPES OF QUESTIONS

1. Series
2. Classification
3. Mirror Images
4. Embedded Images
5. Figure Matrix
6. Paper Cutting
7. Grouping of Images
8. Shape Construction
9. Cubes & Dices
10. Analogy
11. Water Images
12. Pattern Completion
13. Paper Folding
14. Rule Detection
15. Dot Situation
16. Image Analysis

The perfect trick to solve Non-Verbal Reasoning Questions is to Eliminate Options.

Translation



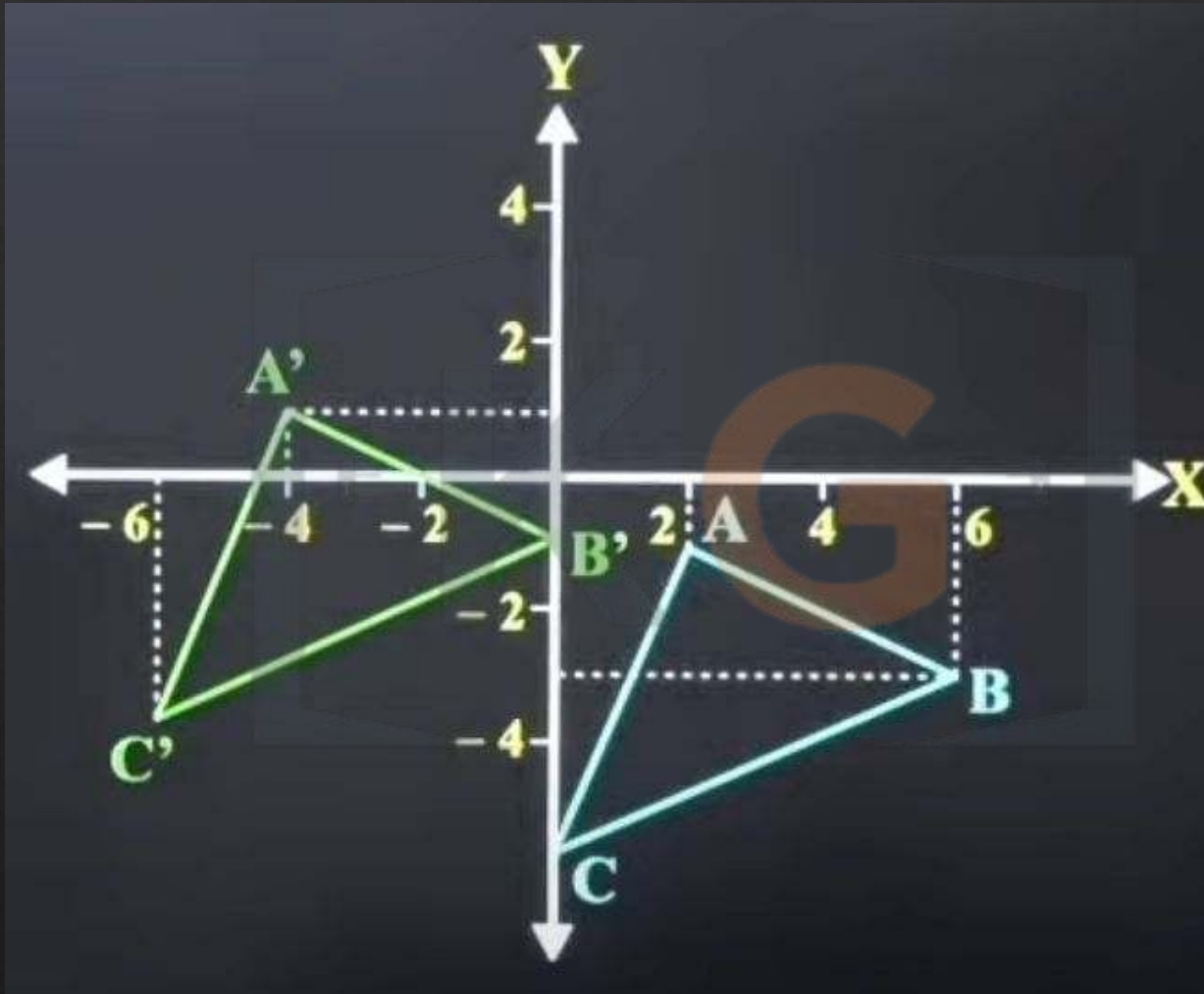
Q1. The Translation Rule for the Image will be

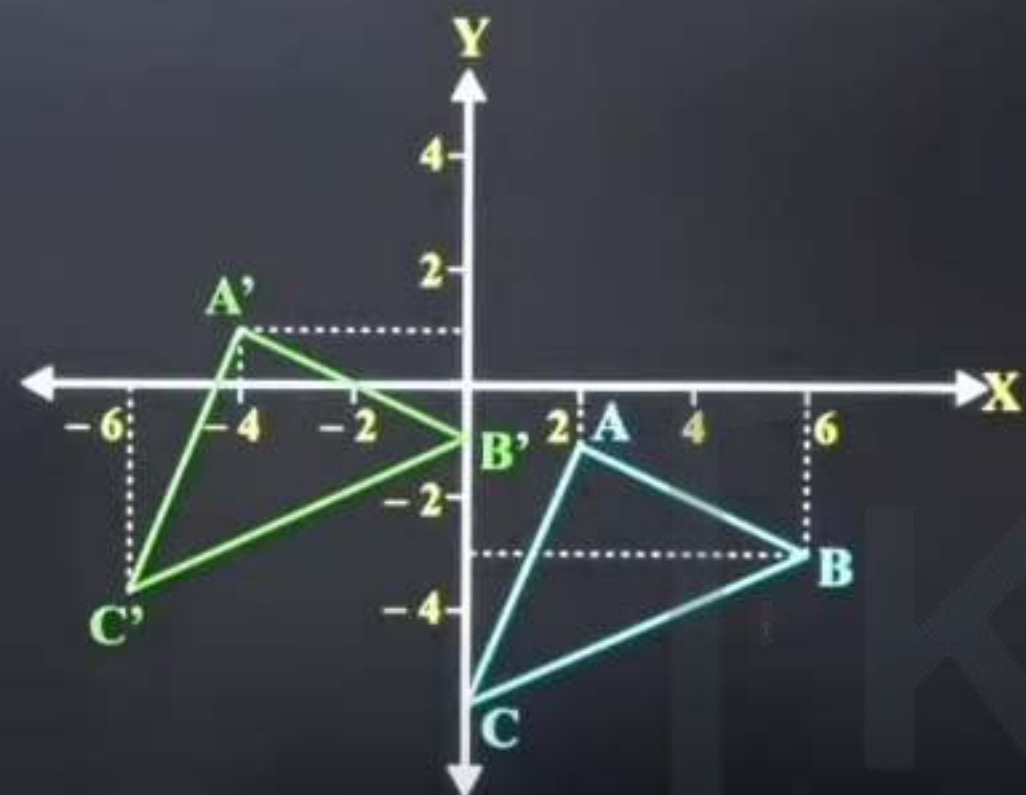
A $(x+6, y+2)$

B $(x-6, y+2)$

C $(x-6, y-2)$

D $(x-2, y+6)$





Correct Answer- Option B



Scaling



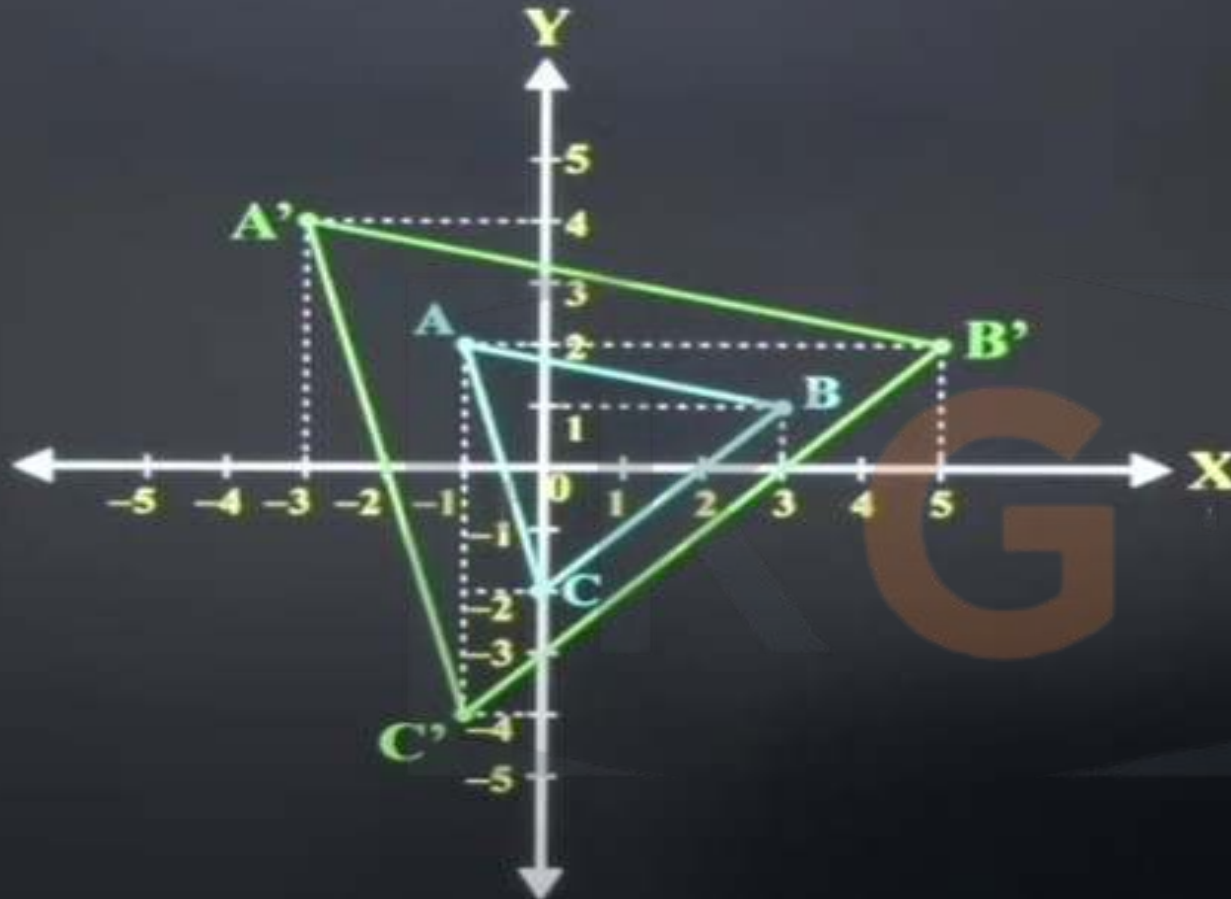
Q2. The scale factor for the dilation will be

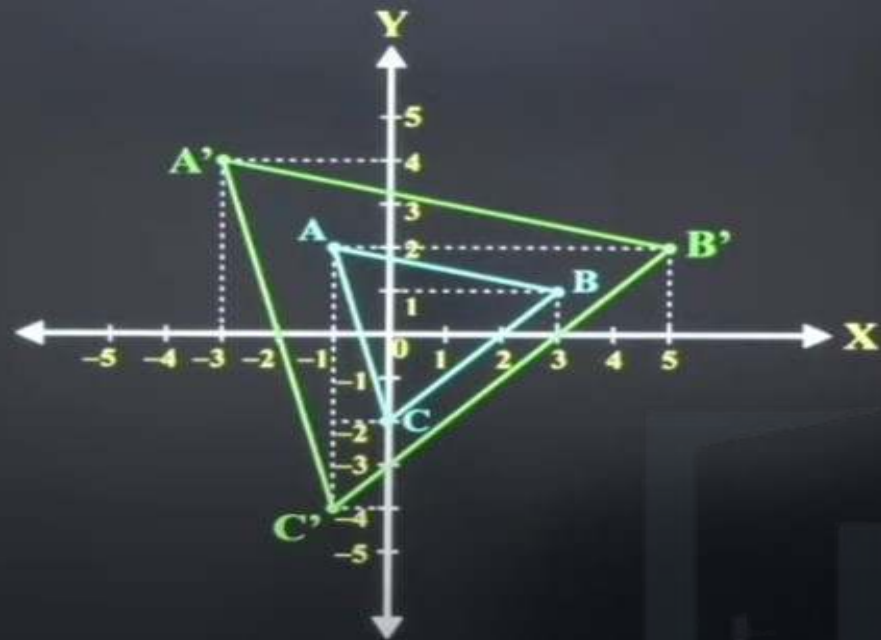
A. 1.3

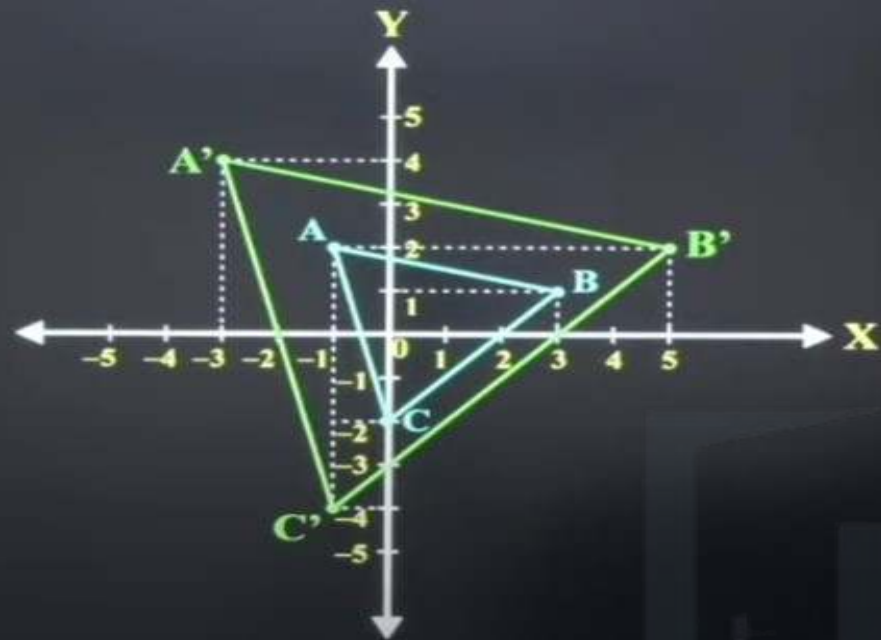
B. 1.5

C. 2

D. 3







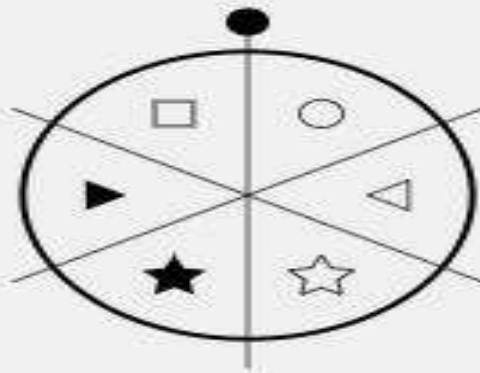
Correct Answer- Option C



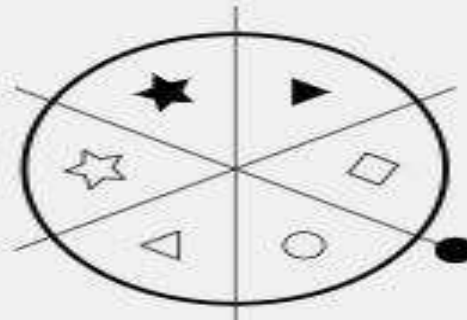
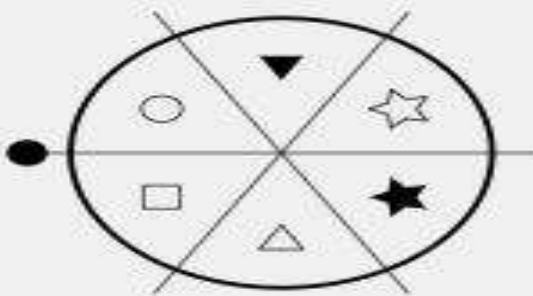
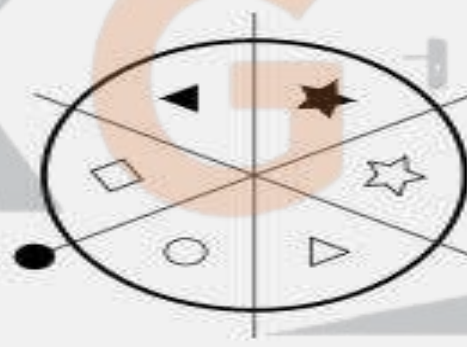
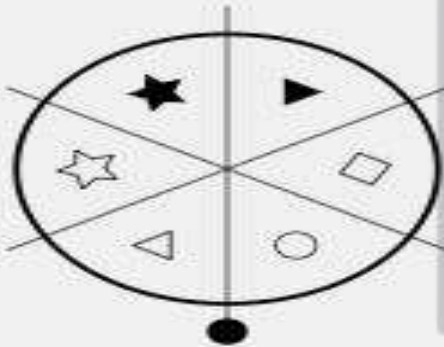
Rotation



Q3.



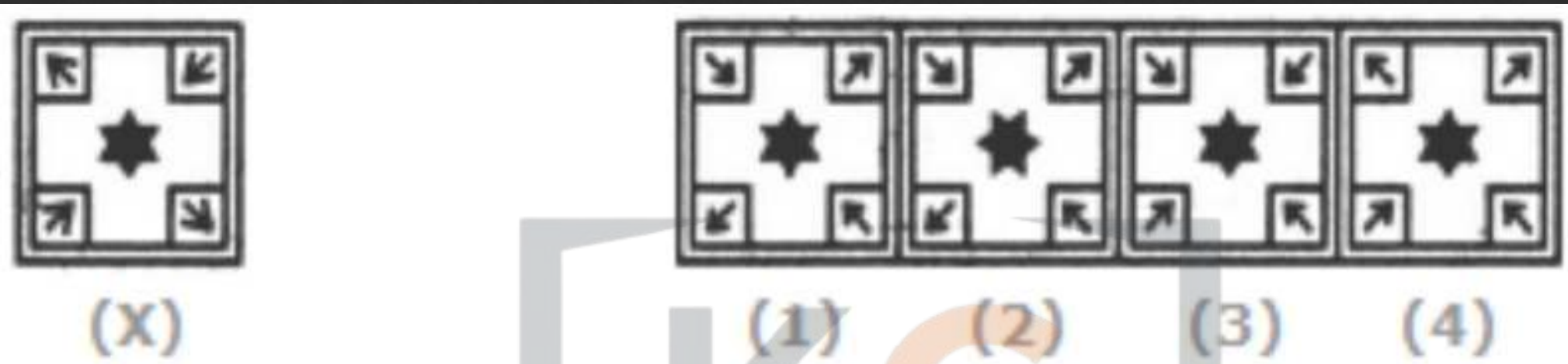
Which figure is a rotation of the object?



Correct Answer- Option D



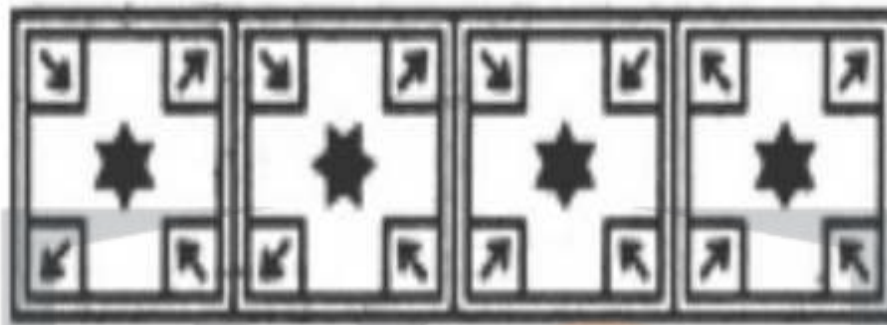
Q4 (VK). Find out how will the key figure (X) look like after rotation.



Correct Answer : Option 2



(X)



(1)

(2)

(3)

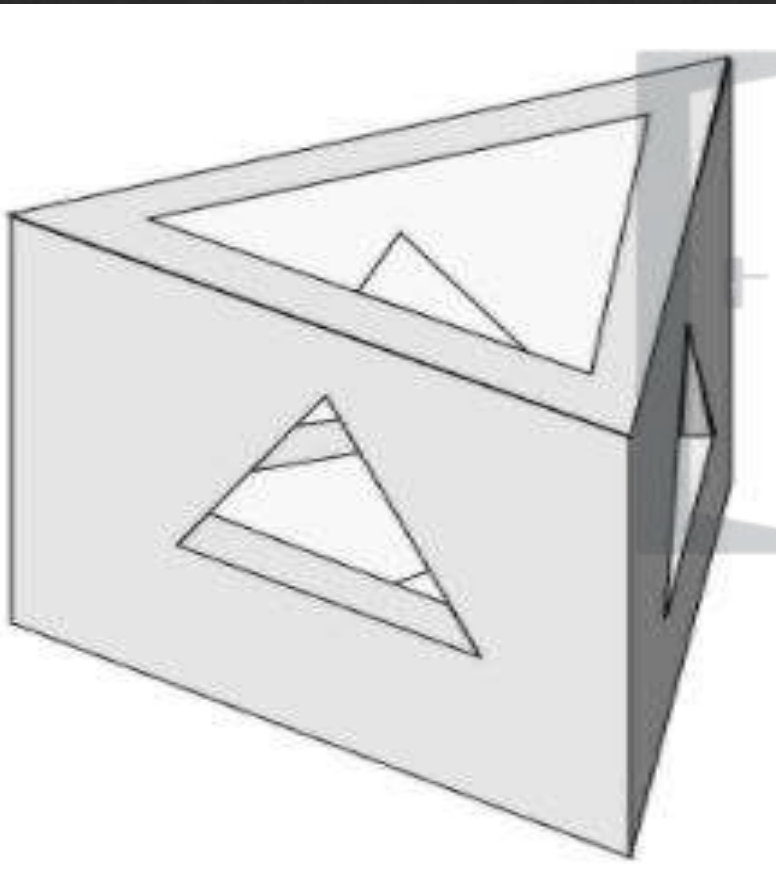
(4)

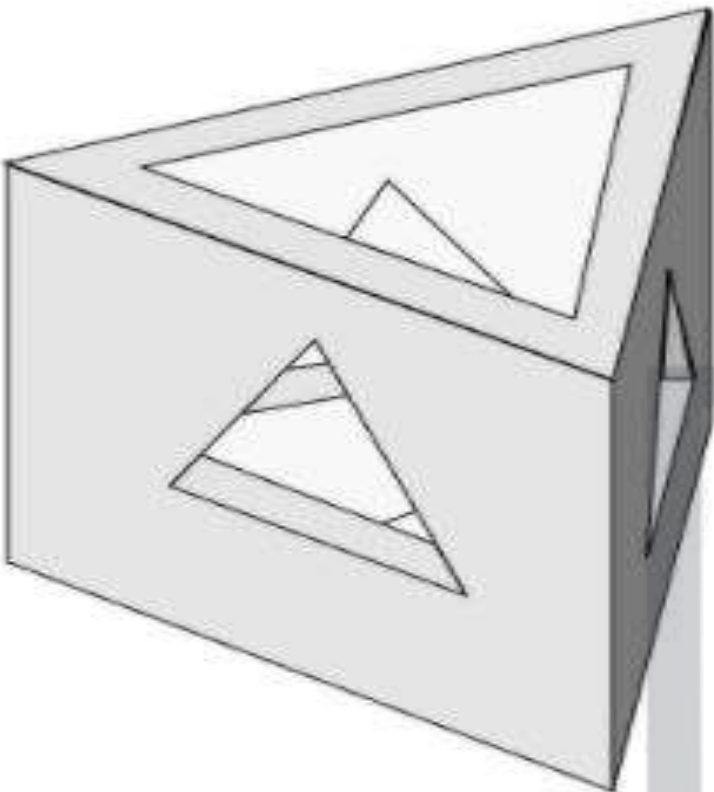
Pattern in 3D



Q5. Which of the shadows on the right can result from casting light on one of the 3D-shape's sides?

a) A b) B c) Both A & B d) Cannot Predict





A



B



Illustration of a side view:



Correct Answer : Option B





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ATTENTION TO DETAIL

Q1. 0452-9858762, 0452-9858762, 0452-9858762



- A, if ALL THREE items given in the question are exactly ALIKE.
- B, if only the FIRST and SECOND items are exactly ALIKE.
- C, if only the FIRST and THIRD items are exactly ALIKE.
- D, if only the SECOND and THIRD items are exactly ALIKE.
- E, if ALL THREE items are DIFFERENT

Correct Answer : Option A

Explanation –

ALL THREE items given in the question are exactly ALIKE.



Q2. If * stands for /
/ stands for -
+ stands for * and
- stands for +

Then $9/8*7+5-10=?$

- A. 13.3
- B. 10.8
- C. 10.7
- D. 11.4



Correct Answer : Option A

Explanation –

$$9 - 8/7 * 5 + 10$$

$$9 - 40/7 + 10$$

$$93/7 = 13.3$$



Q3. NIINIINN, NIININNINN, NIINIINN

- A, if ALL THREE items given in the question are exactly ALIKE.
- B, if only the FIRST and SECOND items are exactly ALIKE.
- C, if only the FIRST and THIRD items are exactly ALIKE.
- D, if only the SECOND and THIRD items are exactly ALIKE.
- E, if ALL THREE items are DIFFERENT

Correct Answer : Option C

Explanation –

Only the FIRST and THIRD items are exactly ALIKE.



Q4. If A stands for '+', B stands for '-', C stands for '×', then

$10 C 4 A 4 C 4 B 6 = ?$

- A) 50
- B) 42
- C) 58
- D) 46



Correct Answer : A

Explanation:

After substituting the symbols in the given expression the new expression will be

$$= 10 \times 4 + 4 \times 4 - 6$$

$$= 40 + 16 - 6$$

$$= 50$$

Hence, option (A) is the correct answer.

Q5. 808088080.8080, 808008080.8080, 808088080.8080

- A, if ALL THREE items given in the question are exactly ALIKE.
- B, if only the FIRST and SECOND items are exactly ALIKE.
- C, if only the FIRST and THIRD items are exactly ALIKE.
- D, if only the SECOND and THIRD items are exactly ALIKE.
- E, if ALL THREE items are DIFFERENT

Correct Answer : Option C

Explanation –

Only the FIRST and THIRD items are exactly ALIKE.



Q6. LLMLLLKLMPUU, LLMLLLKLMPUU, LLMLLLKLMPUU



- A, if ALL THREE items given in the question are exactly ALIKE.
- B, if only the FIRST and SECOND items are exactly ALIKE.
- C, if only the FIRST and THIRD items are exactly ALIKE.
- D, if only the SECOND and THIRD items are exactly ALIKE.
- E, if ALL THREE items are DIFFERENT

Correct Answer : Option A

Explanation –

All the Three items are exactly ALIKE. So correct option is A



Q7 (VK). 4665.8009291, 4665.7999291, 4665.8009291

- A, if ALL THREE items given in the question are exactly ALIKE.
- B, if only the FIRST and SECOND items are exactly ALIKE.
- C, if only the FIRST and THIRD items are exactly ALIKE.
- D, if only the SECOND and THIRD items are exactly ALIKE.
- E, if ALL THREE items are DIFFERENT

Correct Answer : Option C

Explanation –

Only the FIRST and THIRD items are exactly ALIKE.

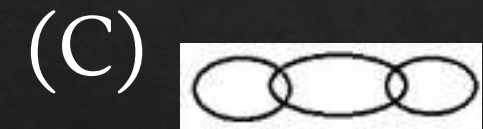




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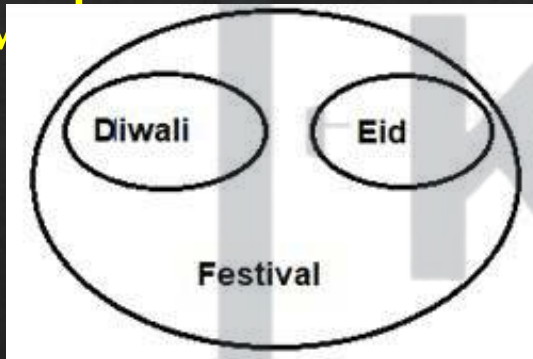
VENN DIAGRAM

Q1. Which of the following Venn diagrams shows the relation among Diwali, Eid, Festival



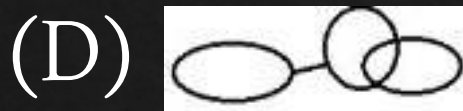
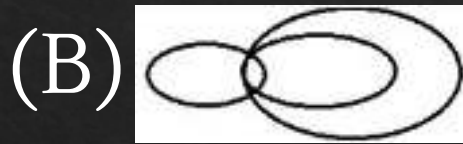
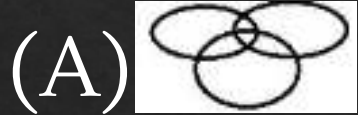
Correct Answer : Option B
Explanation –

Following figure shows the relation among Diwali, Eid, Festival



Hence, the correct option is (B).

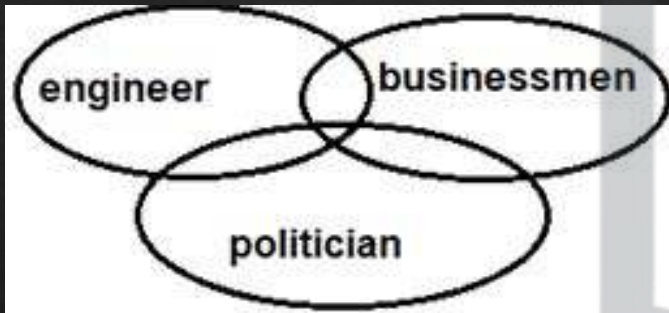
Q2. Which of the following Venn diagrams shows the relation among Engineer, Businessmen, Politician



Correct Answer : Option A

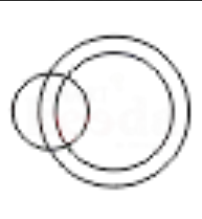
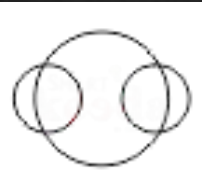
Explanation –

A person can be engineer, businessmen and politician so following Venn diagram can be formed.



Hence, the correct option is (A).

Q3. Which of the following figures best depicts the relationship between thieves, judges and criminals?

- (A) 
- (B) 
- (C) 
- (D) 



Correct Answer : Option B

Explanation –

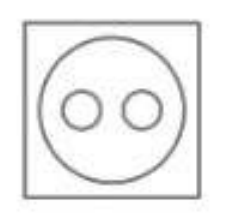
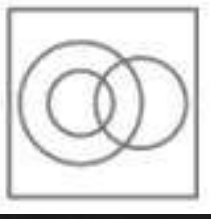
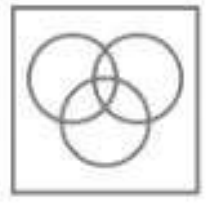
As thieves and criminal belong to one community and judges from another. Below given Venn diagram can be draw



Judge is completely distinct from criminals and thieves.

Hence, the correct option is (B).

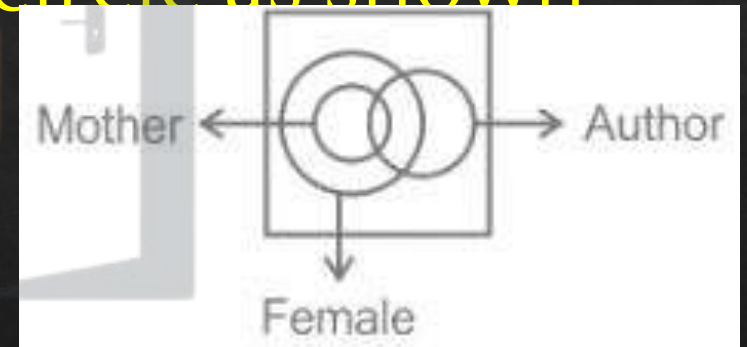
Q4. Identify the diagram that best represents the relationship among the given classes.
Author, Mother, Female

- (A) 
- (B) 
- (C) 
- (D) 

Correct Answer : Option C

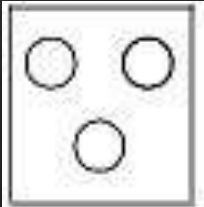
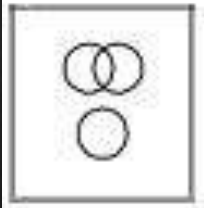
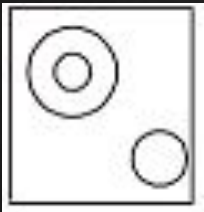
Explanation –

A mother will always be a female. An author can be a mother or a female. Thus, the 'mother' circle will be encircled by the 'female' circle and the 'author' circle will intersect with both the other circle as shown below.



Thus, the third figure represents the relationship between mother, female, and author.

Q5. Which of the following diagrams represents the relationship among Sun, Moon, and Star?

- (A) 
- (B) 
- (C) 
- (D) 



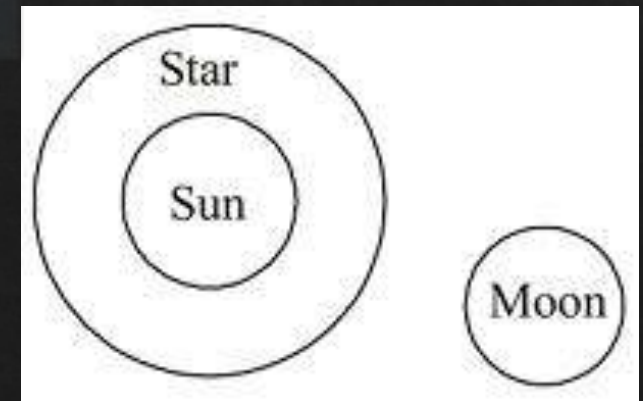
Correct Answer : Option D

Explanation –

As we know that Sun is a star and the Moon is a satellite.

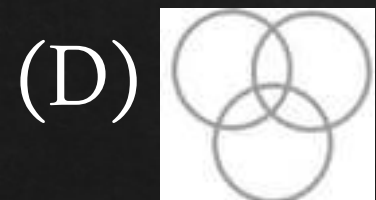
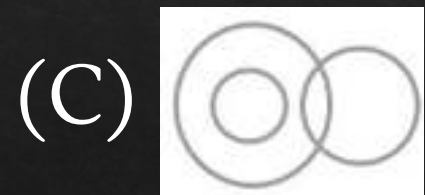
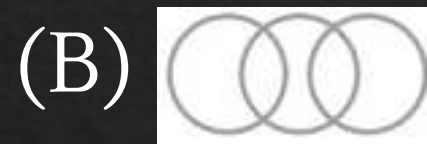
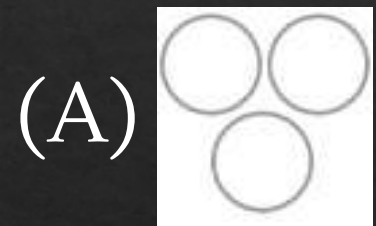
Star will be a big circle and Sun will be a small circle under Star.

Moon is a satellite. So it will be a separate circle. It shows that Sun, Moon, and Star represent the relationship in the given figure.



Q6. Choose the correct options, which gives a suitable relation between the given word.

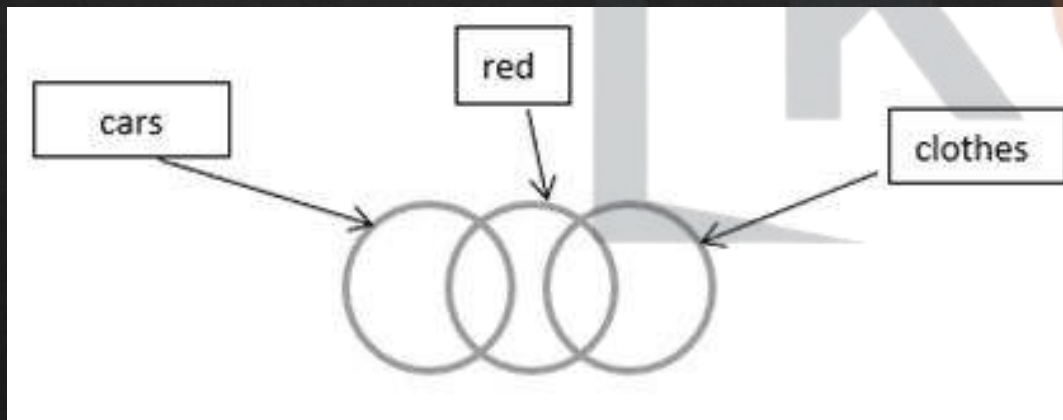
Cars, Clothes, and Red



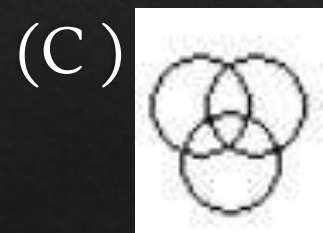
Correct Answer : Option D

Explanation –

From the figure, A car and clothes can be of red color. But cars and clothes are different from each other. Hence, the correct option is (D).



Q7. Which of the following diagrams indicates the best relation between Profit, Dividend and Bonus?



Correct Answer-Option A

Explanation

Bonus and Dividend are different from each other. But both these are parts of profit.

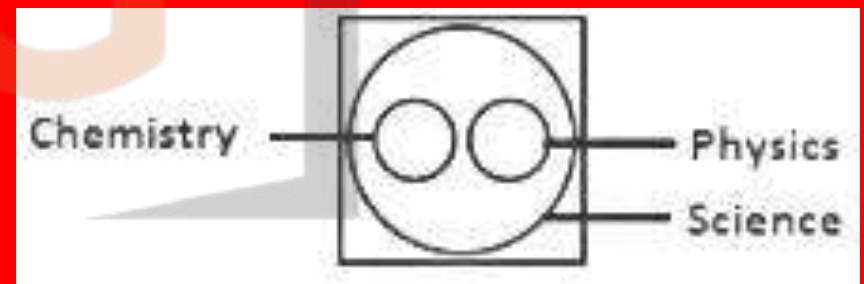


Q8 (VK). Select the option that correctly represents the relationship among the following:



Physics, Chemistry, Science



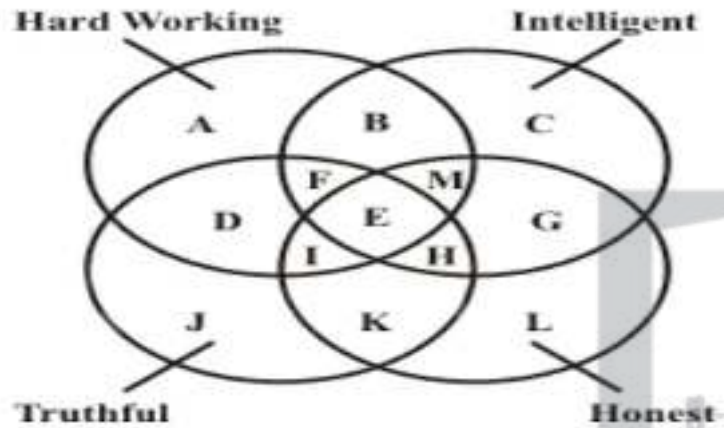




TATA
CONSULTANCY
SERVICES

VENN DIAGRAM BASED DATA INTERPRETATION

Below is given a figure with four intersecting circles, each representing a group of persons having the quality written against it. Study the figure carefully and answer the questions that follow.



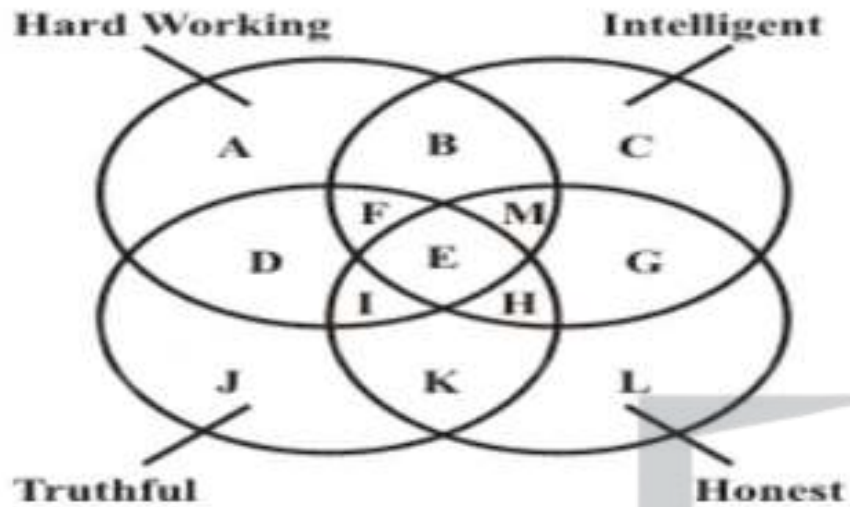
- Q1. The people possessing all the qualities are represented by
- (A) E
 - (B) F
 - (C) H
 - (D) I

Correct Answer-Option A

Explanation:

The required region is the one which is common to all the four circles i.e. E.





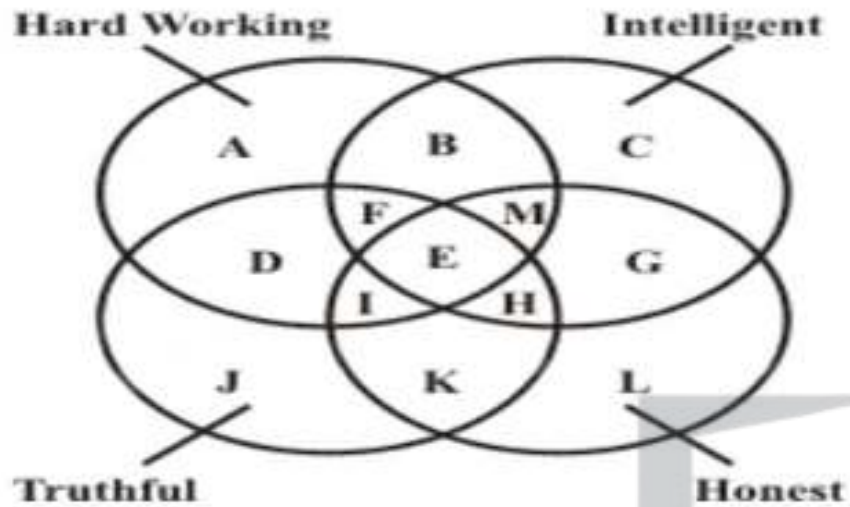
Q2. People who are not honest and truthful but are hard working and intelligent both, are represented by

- (A) E
- (B) B
- (C) M
- (D) I

Correct Answer-Option B

Explanation:

The given conditions are satisfied by the persons denoted by the region which is common to circles 1 and 2 but is not a part of either circle 3 or circle 4. i.e. B.



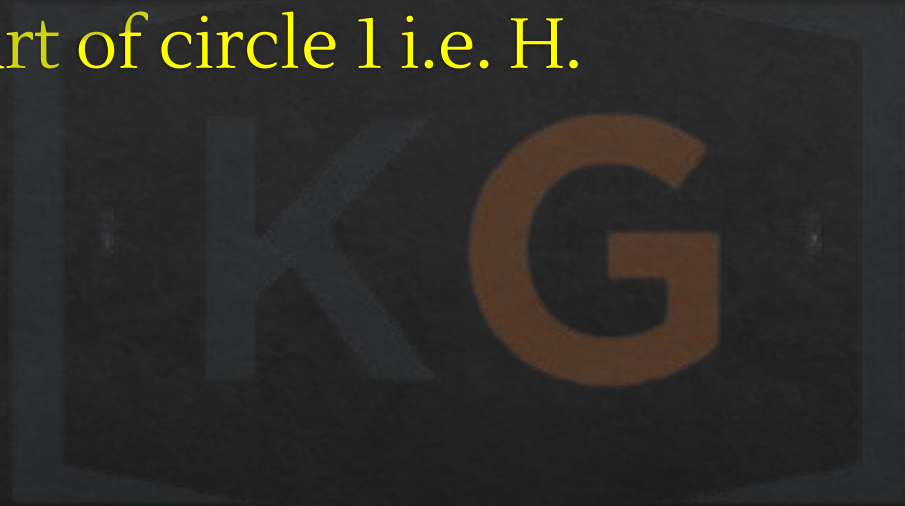
Q3. The region which represents the people who are intelligent, honest and truthful but not hard working is denoted by

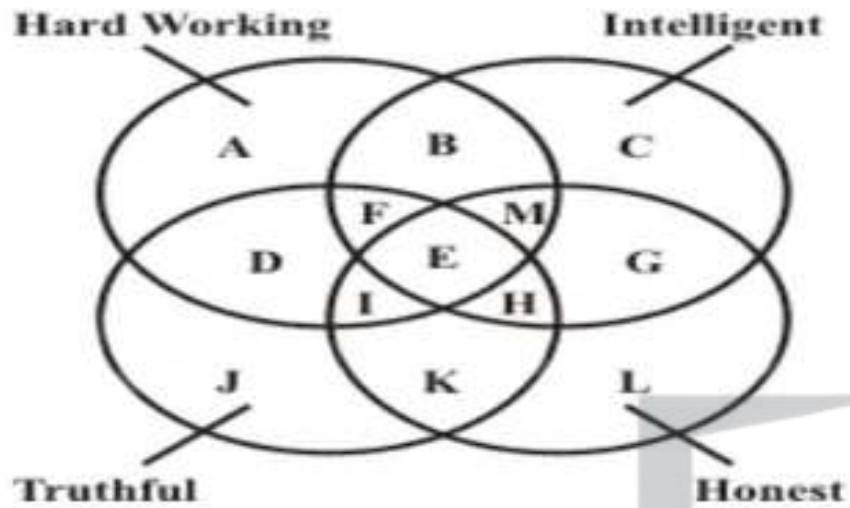
- (A) I
- (B) H
- (C) E
- (D) F

Correct Answer-Option B

Explanation:

The required region is the one which is common to the circles 2, 3 and 4 but is not a part of circle 1 i.e. H.





Q4. People who are not hard working, intelligent and truthful are represented by

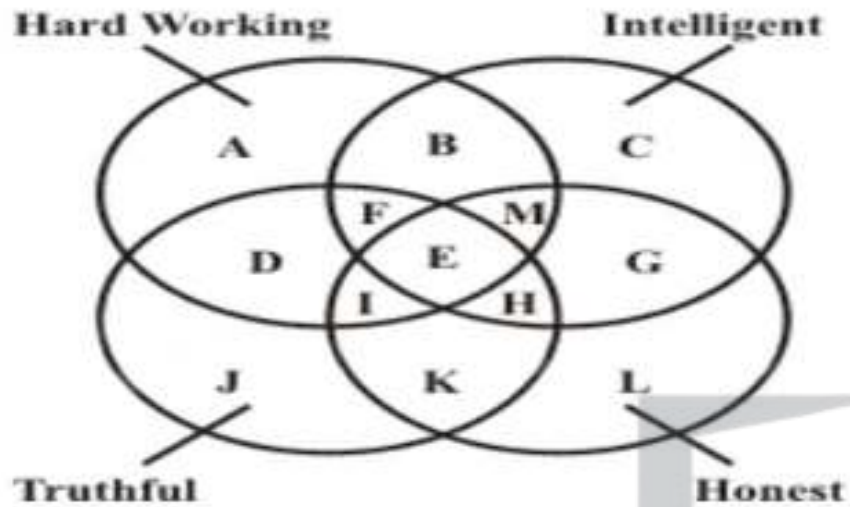
- (A) G
- (B) H
- (C) K
- (D) L

Correct Answer - Option D

Explanation

The required region is the one which does not lie inside circles 1, 2 and 4. i.e. L.





Q5. The region which represents people who are not honest but possess all other three qualities, is denoted by

- (A) B
- (B) D
- (C) F
- (D) I

Correct Answer-Option C

Explanation

The required region is the one which is common to the circles 1, 2 and 4 but lies outside circle 3 i.e. F.





TATA
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CALENDAR

Q1. Which day of the week lied on 18 March 1977?



- a) Monday
- b) Wednesday
- c) Friday
- d) Sunday



Year

Months

Weeks

Days

$\downarrow$ Year = 12 Months = 52 Weeks = 365 or 366 days
Ordinary Year Leap Year

Identifying Leap Years

1996

2019

1600

1700

Handwritten notes on identifying leap years:

	$\div 4$	$\div 400$ for case of century
1996	✓	
2019	X	
1600	✓	
1700	✓	X

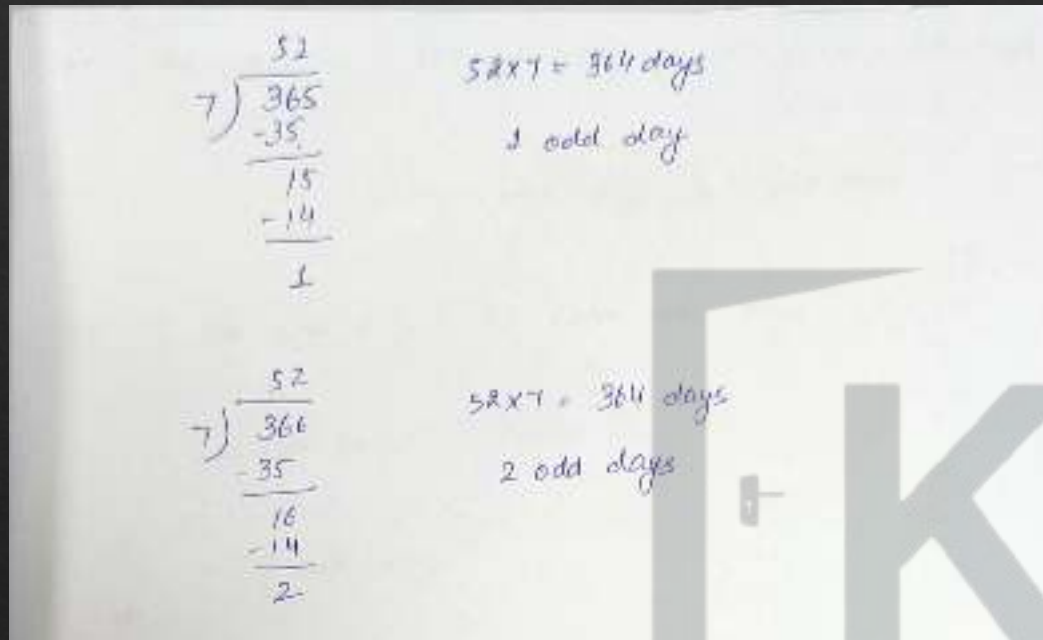
100 years = 76 ordinary years + 24 leap years

1 ordinary year = 1 odd day
1 leap year = 2 odd days

Calculation for 100 years:

$$\frac{100}{4} = 25$$
$$25 - 1 = 24 \text{ leap years}$$

Concept of Odd Days



Year	Number of odd days
100	5
200	3
300	1
400	0
800	0
1200	0
1600	0
2000	0
2400	0
2800	0

100 years $\rightarrow$ 76 ordinary years + 24 leap years

$$\begin{array}{r} 24 \\ \times 2 \\ \hline 48 \end{array}$$

$\Downarrow$
76 odd days + 48 odd days

$\Downarrow$
124 odd days

$\Downarrow$
5 odd days

$$\begin{array}{r} 17 \\ 7 \overline{) 124} \\ \underline{- 7} \\ 54 \\ \underline{- 49} \\ 5 \end{array}$$

or

76 odd days + 48 odd days

$\Downarrow$
6 odd days + 6 odd days

$\Downarrow$
12 odd days

$\Downarrow$
5 odd days

$$\begin{array}{r} 10 \\ 7 \overline{) 76} \\ \underline{- 70} \\ 6 \\ \underline{- 6} \\ 0 \\ 7 \overline{) 48} \\ \underline{- 42} \\ 6 \\ \underline{- 6} \\ 0 \\ 1 \\ 7 \overline{) 12} \\ \underline{- 7} \\ 5 \end{array}$$

300 Years

100 Years = 5 odd days

300 Years = $5 \times 3 = 15$ odd days = 1 odd day

or 300 Years = 228 ordinary Year + 72 leap Year

↓

228 odd days + 144 odd days

↓

372 odd days

↓

1 odd day

400 Years

100 Years = 5 odd day

400 Years = $5 \times 4 = 20$ odd day
= 6 odd day

$6 + 1 = 7$ odd days = 0 odd days

400 is a leap year

100, 200, 300 are not leap years

$$\begin{array}{r} 2 \\ 7 \overline{) 15} \\ \underline{-14} \\ 1 \end{array}$$

$$\begin{array}{r} 900 \\ 4 \overline{) 3600} \\ \underline{3600} \\ 0 \end{array}$$

$$\begin{array}{r} 75 \\ 3 \overline{) 225} \\ \underline{-225} \\ 0 \end{array}$$

$$\begin{array}{r} 300 \\ 72 \overline{) 21600} \\ \underline{-21600} \\ 0 \end{array}$$

$$\begin{array}{r} 53 \\ 7 \overline{) 372} \\ \underline{-35} \\ 22 \\ \underline{-21} \\ 1 \end{array}$$

$$\begin{array}{r} 2 \\ 7 \overline{) 20} \\ \underline{-14} \\ 6 \end{array}$$

$$\begin{array}{r} 1 \\ 7 \overline{) 7} \\ \underline{-7} \\ 0 \end{array}$$

Q1. Which day of the week lied on 18 March 1977?



18 March 1977

5 Dec 1976 [Year 0 to 1976]

$$1976 = 1600 + 800 + 76$$

$$= 0 + 1 + \begin{array}{l} \swarrow \searrow \\ 57 \quad 19 \\ \downarrow \quad \downarrow \\ 57 \text{ odd days} \quad 19 \text{ odd days} \end{array}$$

$$\frac{76}{4} = 19$$

$$\begin{array}{r} 19 \text{ leap} \\ \text{year} \\ -16 \\ \hline 3 \\ -19 \\ \hline 57 \end{array}$$

$$\text{Total odd days} = 1 + 57 + 38 = 96 \text{ odd days}$$

$$\text{Jan 1977} \rightarrow 31 \rightarrow 3 \text{ odd days}$$

$$\text{Feb 1977} \rightarrow 28 \rightarrow 0 \text{ odd days}$$

$$\text{Mar 1977} \rightarrow 31 \rightarrow 4 \text{ odd days}$$

$$7 \text{ odd days}$$

$$\text{Total odd days} = 96 + 7 = 103 \text{ odd days}$$

↓

5 odd days

2

Friday

Sun → 0	Fri → 5
Mon → 1	Sat → 6
Tue → 2	
Wed → 3	
Thu → 4	

$$\begin{array}{r} 1 \\ 3 \overline{) 31} \\ \underline{-26} \\ 5 \end{array}$$

$$\begin{array}{r} 2 \\ 7 \overline{) 14} \\ \underline{-14} \\ 0 \end{array}$$

$$\begin{array}{r} 14 \\ 7 \overline{) 103} \\ \underline{-7} \\ 33 \\ \underline{-28} \\ 5 \end{array}$$

Q1. Which day of the week lied on 18 March 1977?



18 march 1977 day

Images

Zodiac

In india

Videos

Bir

About 21,70,00,000 results (0.38 seconds)

Friday, 18 March 1977

Q1. Which day of the week lied on 18 March 1977?



Q2. On what day does 22 October 1964 lie ?

**DESH VIDESH MAI KAHI BHI SARKAR BANANE KE
LIYE SAMPARK KARE - MOTA BHAI, GUJARAT VALE**

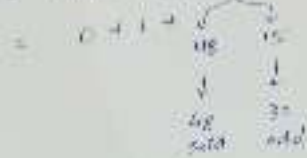


Q2. On what day does 22 October 1964 lie ?

22 October 1964

31 Dec 1963 [Year 0 to Year 1963]

$$1963 = 1600 + 300 + 63$$



Total odd days = $1 + 48 + 30 = 79$ odd days

Jan 1964 $\rightarrow 31 \rightarrow 3$

Feb 1964 $\rightarrow 28 \rightarrow 1$

Mar 1964 $\rightarrow 31 \rightarrow 3$

Apr 1964 $\rightarrow 30 \rightarrow 2$

May 1964 $\rightarrow 31 \rightarrow 4$

Jun 1964 $\rightarrow 30 \rightarrow 2$

July 1964 $\rightarrow 31 \rightarrow 3$

Aug 1964 $\rightarrow 31 \rightarrow 3$

Sep 1964 $\rightarrow 30 \rightarrow 2$

Oct 1964 $\rightarrow 31 \rightarrow 1$

23 odd days

Total odd days = $79 + 23 = 102$ odd days

102 odd days $\Rightarrow$ Thursday

$$\frac{63}{15} = 4.2$$

$$\frac{102}{7} = 14.57$$

$$\frac{102}{7} = 14.57$$

Q2. On what day does 22 October 1964 lie ?



Q2. On what day does 22 October 1964 lie ?



22 October 1964 day

Images

Zodiac sign

Birthday

Astrology

About 11,80,00,000 results (0.49 seconds)

Thursday

22 October 1964

Q3. On what dates of April, 2001 did Wednesday fall?



- A. 1st, 8th, 15th, 22nd, 29th
- B. 2nd, 9th, 16th, 23rd, 30th
- C. 3rd, 10th, 17th, 24th
- D. 4th, 11th, 18th, 25th



Q3. On what dates of April, 2001 did Wednesday fall?



Q3. On what dates of April, 2001 did Wednesday fall?

1 April 2001

31 Dec 2000 [Year 0 to Year 2000]

2000 $\Rightarrow$ 0 odd days

Jan 2001 $\Rightarrow$ 31 $\rightarrow$ 3

Feb 2001 $\Rightarrow$ 28 $\rightarrow$ 0

Mar 2001 $\Rightarrow$ 31 $\rightarrow$ 3

Apr 2001 $\Rightarrow$ 1 $\rightarrow$ 1

7 odd days

Total $\Rightarrow$ 0 + 7 $\Rightarrow$ 7 odd days $\Rightarrow$ 0 odd days
(Sunday)

1 Apr 2001 $\rightarrow$ Sun

2 $\rightarrow$ Mon

3 $\rightarrow$ Tue

4 $\rightarrow$ Wed

4 $\xrightarrow{+7}$ 11 $\xrightarrow{+7}$ 18 $\xrightarrow{+7}$ 25 Wednesday

Correct Answer : Option D

Explanation –

We shall find the day on 1st April, 2001.

1st April, 2001 = (2000 years + Period from 1.1.2001 to 1.4.2001)

Odd days in 1600 years = 0, Odd days in 400 years = 0

Jan. Feb. March April $(31 + 28 + 31 + 1) = 91$ days 0 odd days.

Total number of odd days = $(0 + 0 + 0) = 0$

On 1st April, 2001 it was Sunday.

In April, 2001 Wednesday falls on 4th, 11th, 18th and 25th.

Finding Day on a particular Date
when Day on some other Date is

given to us

1st October 2010 Monday

25th June 2015 ???

Four Categories

a) Month and Year Same, Date Different

1st March 2011 Monday

31st March 2011 ???

b) Date and Month Same, Year Different

5th January 2001 Monday

5th January 2007 ???

Four Categories

c) Date and Year Same, Month Different

23st March 2015 Tuesday

23st November 2015 ???

d) All three different

10th March 2013 Monday

27th August 2017 ???

Category - 1

a) Month and Year Same, Date Different

1 st March 2011	Monday
31 st March 2011	???



day on 1 march 2011

 All  News  Videos  Images

About 3,00,00,00,000 results (0.79 seconds)

Tuesday

1 March 2011

1st March 2011

Monday

31st March 2011

???

1 March 2011 → Monday

31 March 2011 → Wednesday

$$\begin{array}{r} 31 \\ - 1 \\ \hline 30 \end{array}$$

$$\begin{array}{r} 4 \\ 7 \overline{) 30} \\ - 28 \\ \hline 2 \end{array}$$

2 odd days

Monday + 2 = Wednesday

31st March 2011

Wednesday

1st March 2011

???



31st March 2011

Wednesday

1st March 2011

???

31 March 2011 → Wednesday

1 March 2011 → ??

$$\begin{array}{r} 31 \\ - 1 \\ \hline 30 \end{array}$$

$$\begin{array}{r} 4 \\ 7 \overline{) 30} \\ - 28 \\ \hline 2 \end{array}$$

2 odd days

Wednesday - 2 = Monday

Category - 2

b) Date and Month Same, Year
Different

5th January 2001 Monday

5th January 2007 ???

15 th March 1848	Monday
15 th March 1862	???



15th March 1848 Monday
15th March 1862 ???

15 March 1848 → Monday
15 March 1862 → Thursday

$$\begin{array}{r} 1862 \\ - 1848 \\ \hline 14 \end{array}$$

14 years

11 ordinary
3 Leap

11 odd day 6 odd day

1852
1856
1860 } 3 Leap Years

$$\begin{array}{r} 2 \\ 7 \overline{) 17} \\ - 14 \\ \hline 3 \end{array}$$

$11 + 6 = 17 \text{ odd day} \rightarrow 3 \text{ odd day}$

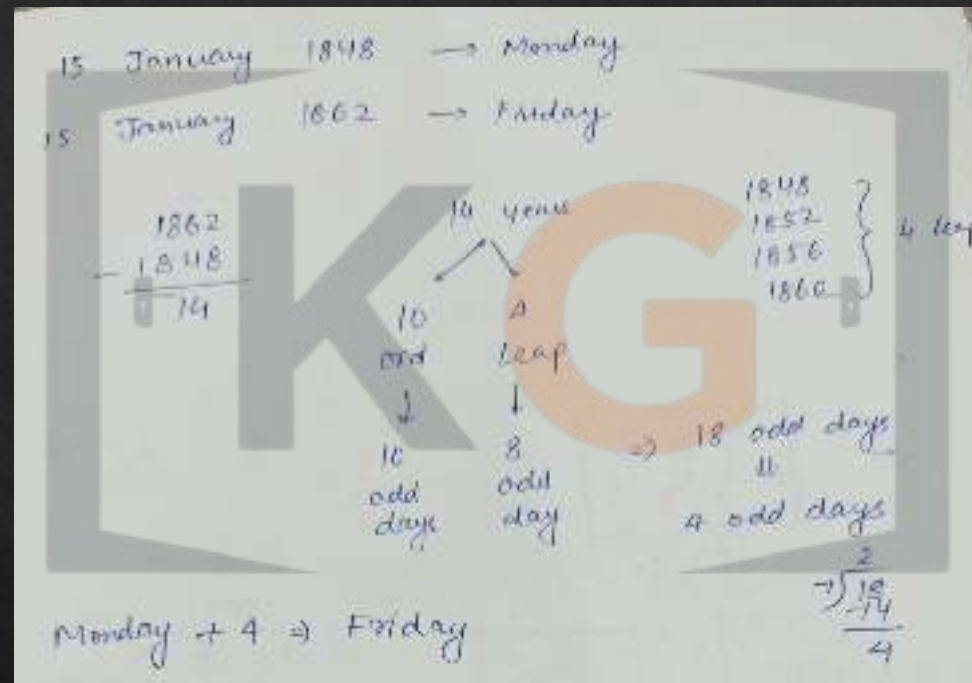
Monday + 3 = Thursday

15th January 1848

Monday

15th January 1862

???



Category - 3

c) Date and Year Same, Month
Different

23 st March 2015	Tuesday
23 st November 2015	???

23st March 2015 Tuesday
23st November 2015 ???

23 March 2015 → Tuesday
23 November 2015 → Tuesday

~~23 March 2015 → Tuesday~~
~~23 October 2015 → ??~~

March → 31 → 3
April → 30 → 2
May → 31 → 3
June → 30 → 2
July → 31 → 3

August → 31 → 3
Sept → 30 → 2
October → 31 → 3

21 odd days
↓
0 odd day

Tuesday + 0 = Tuesday

$$\begin{array}{r} 3 \\ 7 \overline{) 21} \\ \underline{-21} \\ 0 \end{array}$$

2019 Calendar

1. January — October
2. February — March — November
3. April — July
4. September — December

January 2019

S	M	T	W	Th	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

February 2019

S	M	T	W	Th	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28		

March 2019

S	M	T	W	Th	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

April 2019

S	M	T	W	Th	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

May 2019

S	M	T	W	Th	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

June 2019

S	M	T	W	Th	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

July 2019

S	M	T	W	Th	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

August 2019

S	M	T	W	Th	F	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

September 2019

S	M	T	W	Th	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

October 2019

S	M	T	W	Th	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

November 2019

S	M	T	W	Th	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

December 2019

S	M	T	W	Th	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

2020 Calendar

For a leap year the results would be:

1. January — April—July
2. February — August
3. March — November
4. September — December

January						
S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

February						
S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29

March						
S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

April						
S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

May						
S	M	T	W	T	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

June						
S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

July						
S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

August						
S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

September						
S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

October						
S	M	T	W	T	F	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

November						
S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

December						
S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

1. January — October

2. February — March — November

3. April — July

4. September — December

For a leap year the results would be:

1. January — April—July
2. February — August
3. March — November
4. September — December

23st March 2015 Tuesday
23st November 2015 ???



23st March 2015 Tuesday
23st November 2015 ???

March & November has same calendar for non
leap year.

23 March 2015 = 23 Nov 2015 = Tuesday

1. January — October
2. February — March — November
3. April — July
4. September — December

16th January 1992

Monday

16th December 1992

???



16th January 1992
16th December 1992

Monday
???

16 Jan 1992 → Monday
16 Dec 1992 → ??

16 July 1992 → Monday
16 Sept 1992 → ??

July → 31 → 3
Aug → 31 → 3
6 odd days

Monday + 6 = Sunday

16 Sept 1992 → Sunday ⇒ 16 Dec 1992 → Sunday

For a leap year the results would be:

1. January — April—July
2. February — August
3. March — November
4. September — December

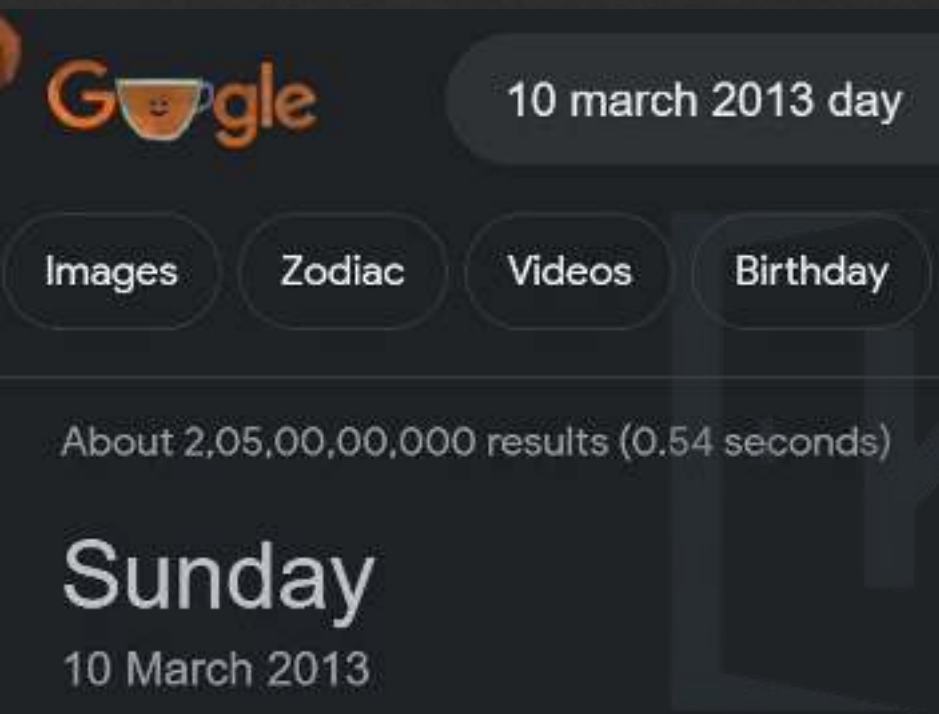
Category - 4

d) All three different

10th March 2013 Monday

27th August 2017 ???

Category - 4



10 March 2013 → Monday

10 March 2017 → ??

Monday + 5 = Saturday

10 March 2017 → Saturday

27 March 2017 → ?

$$\begin{array}{r} 27 \\ -10 \\ \hline 17 \end{array}$$

$$\begin{array}{r} 2 \\ 7 \overline{) 17} \\ \underline{-14} \\ 3 \end{array}$$

3 odd days

Saturday + 3 = Tuesday

27 March 2017 → Tuesday

27 Aug 2017 → ?

Mar → 31 → 3

Apr → 30 → 2

May → 31 → 3

June → 30 → 2

July → 31 → 3

13 odd days

4

6 odd days

Tuesday + 6 = Monday

∴ 27 Aug 2017 = Monday

$$\begin{array}{r} 2017 \\ -2013 \\ \hline 4 \end{array}$$

4

3 ord

2 leap

(2016)

3 odd days

2 odd days

3 + 2 = 5 odd days

$$\begin{array}{r} 1 \\ 7 \overline{) 13} \\ \underline{-7} \\ 6 \end{array}$$

Q4. In 2003, there are 28 days in February and there are 365 days in the year. In 2004, there are 29 days in February and there are 366 days in the year. If the date March 11, 2003 is a Tuesday, then which one of the following would be the date March 11, 2004 be?

- A. Wednesday
- B. Tuesday
- C. Thursday
- D. Monday



Category - 2

b) Date and Month Same, Year
Different

5th January 2001 Monday

5th January 2007 ???

Q4. In 2003, there are 28 days in February and there are 365 days in the year. In 2004, there are 29 days in February and there are 366 days in the year. If the date March 11, 2003 is a Tuesday, then which one of the following would be the date March 11, 2004 be?

Handwritten solution for Q4:

11 March 2003 → Tuesday

11 March 2004 →

Tuesday + 2 = Thursday

11 March 2004 → Thursday

Calculation for the number of days between 2003 and 2004:

$$\begin{array}{r} 2004 \\ - 2003 \\ \hline 1 \end{array}$$

[2004]

1 leap year

↓

2 odd days

Correct Answer : Option C

Explanation –

Every year day is increased by 1 odd day.

Or in leap year it is increased by 2 odd days.

So 11 March 2003 is Tuesday.

11 March 2004 is Thursday

Q5. Munna Bhaiya's birthday is on Monday 18th June. On what day of the week will be Guddu Bhaiya's birthday in the same year if Guddu Bhaiya was born on 12th October?



- (A) Tuesday
- (B) Friday
- (C) Saturday
- (D) Wednesday



Q5. Munna Bhaiya's birthday is on Monday 18th June. On what day of the week will be Guddu Bhaiya's birthday in the same year if Guddu Bhaiya was born on 12th

18 June xxxx → Monday
12 October xxxx → ??

~~18 Oct~~

18 June xxxx → Monday
18 Oct xxxx → ??

June → 30 → 2
July → 31 → 3
Aug → 31 → 3
Sept → 30 → 2

10 odd days
↓
3 odd days

Monday + 3 ⇒ Thursday

18 Oct xxxx = Thursday

12 Oct xxx ⇒
Thursday - 6 = Friday

12 Oct xxx = Friday

18
-12
6 odd days

Correct Answer : Option B

Explanation –

18th June = Monday

Number of days between 18th June to 12th October = $12 + 31 + 31 + 30 + 12 = 116$

Number of odd days = $\frac{116}{7} = 4$ days

So, Monday + 4 = Friday

Hence, the correct option is (B).

Q6 (VK). Valentine day 14 Feb. 2005 was celebrated by Boyfriend and Girlfriend on Monday. They were very happy. What day of the week would come on 14 Feb. 2010?

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- A. Monday
- B. Sunday
- C. Tuesday
- D. Wednesday



Correct Answer : Option B

Explanation –

The given day 14th Feb, 2005 falls on Monday. We have to find the day of the week on 14th February, 2010.

From 14 - 2 - 2005 to 14 - 2 - 2010, there are 5 years.

Of these 5 years, 4 are ordinary years and 1 is leap year.

Each ordinary year has 1 odd day and leap year has 2 odd days

So there are $4 + 2 = 6$ odd days

The day of the week on 14 th February 2010 = Monday + 6 = Sunday



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CODING DECODING

What Coding Decoding Actually Is ??

Q1: CAT is coded as DBU, then MAN is coded as _____



What Coding Decoding Actually Is ??

Q1: CAT is coded as DBU, then MAN is coded as _____



A	B	C	D	E	F	G	H
1	2	3	4	5	6	7	8
I	J	K	L	M	N	O	P
9	10	11	12	13	14	15	16
Q	R	S	T	U	V	W	
17	18	19	20	21	22	23	
	X	Y	Z				
	24	25	26				



Types of Questions

1. Letters / Alphabets
2. Numbers
3. Symbols
4. Grouping
5. Miscellaneous



Q2: In a certain language, if SUNSHINE is coded as TVOTIJOF, then how will MOON be coded?

- a) NPPQ
- b) NPPO
- c) PPON
- d) NQQP



Q2: In a certain language, if SUNSHINE is coded as TVOTIJOF, then how will MOON be coded?

- a) NPPQ
- b) NPPO
- c) PPON
- d) NQQP



Q3. If in a certain code “RANGE” is coded as 12345 and “RANDOM” is coded as 123678. Then the code for the word “MANGO” would be?

- A. 89343
- B. 84629
- C. 82347
- D. 83274



Q3. If in a certain code “RANGE” is coded as 12345 and “RANDOM” is coded as 123678. Then the code for the word “MANGO” would be?

- A. 89343
- B. 84629
- C. 82347
- D. 83274



Correct Answer : Option C

Explanation –

R $\rightarrow$ 1 R $\rightarrow$ 1
A $\rightarrow$ 2 A $\rightarrow$ 2
N $\rightarrow$ 3 N $\rightarrow$ 3
G $\rightarrow$ 4 D $\rightarrow$ 6
E $\rightarrow$ 5 O $\rightarrow$ 7
M $\rightarrow$ 8

Now MANGO is coded as:

M $\rightarrow$ 8
A $\rightarrow$ 2
N $\rightarrow$ 3
G $\rightarrow$ 4
O $\rightarrow$ 7

Thus MANGO is coded as '82347'.

Q4. In a certain code language COMPUTER is written as RFUVQNPC.
How will MEDICINE be written in that code language?



- A. MFEDJJOE
- B. EOJDEJFM
- C. MFEJDJOE
- D. EOJDJEFM



Q4. In a certain code language COMPUTER is written as RFUVQNPC.
How will MEDICINE be written in that code language?



- A. MFEDJJOE
- B. EOJDEJFM
- C. MFEJDJOE
- D. EOJDJEFM



Correct Answer : Option D

Explanation –

There are 8 letters in the word.

– The coded word can be obtained by taking the immediately following letters of word, except the first and the last letters of the given word but in the reverse order. That means, in the coded form the first and the last letters have been interchanged while the remaining letters are coded by taking their immediate next letters in the reverse order.

Types of Questions

1. Letters / Alphabets
2. Numbers
3. Symbols
4. Grouping
5. Miscellaneous



Q5: In a certain language, if SNOW is coded as 7100, then how will WALL be coded?

- a) 5000
- b) 4700
- c) 4800
- d) 4000



Q5: In a certain language, if SNOW is coded as 7100, then how will WALL be coded?

- a) 5000
- b) 4700
- c) 4800
- d) 4000



Q6 (VK): In a certain language, if SNOW is coded as 7100, then how will GIRLFRIEND be coded?



Q7: In a certain language, if CAP is coded as 61, then how will PEN be coded?

- a) 40
- b) 66
- c) 80
- d) 46



Q7: In a certain language, if CAP is coded as 61, then how will PEN be coded?

- a) 40
- b) 66
- c) 80
- d) 46



Q7: In a certain language, if CAP is coded as 61, then how will PEN be coded?

- a) 40
- b) 66
- c) 80
- d) 46



Q8. If $5+3+2 = 151022$, $9+2+4=183652$, $8+6+3 = 482466$ and $5+4+5 = 202541$, then $7+2+5 =$



- A. 143547
- B. 132234
- C. 2577224
- D. 112321



Q8. If $5+3+2 = 151022$, $9+2+4=183652$, $8+6+3 = 482466$ and $5+4+5 = 202541$, then $7+2+5 =$



- A. 143547
- B. 132234
- C. 2577224
- D. 112321



Correct Answer : Option A



Q9. If $9 \times 7 = 3545$ and $4 \times 3 = 1520$ then $6 \times 8 = ?$

- A. 5040
- B. 6050
- C. 4030
- D. 3040



Q9. If $9 \times 7 = 3545$ and $4 \times 3 = 1520$ then $6 \times 8 = ?$

- A. 5040
- B. 6050
- C. 4030
- D. 3040



Correct Answer : Option C

Explanation –

The pattern followed in the given equations is:

$$9 \times 7 = 7 \times 9 \rightarrow (7 \times 5)(9 \times 5) = 3545$$

$$4 \times 3 = 3 \times 4 \rightarrow (3 \times 5)(4 \times 5) = 1520$$

$$\text{Similarly, } 6 \times 8 = 8 \times 6 \rightarrow (8 \times 5)(6 \times 5) = 4030$$

Hence, 4030 is the answer.

Types of Questions

1. Letters / Alphabets
2. Numbers
3. Symbols
4. Grouping
5. Miscellaneous



Q10: In a certain language, if RADAR is coded as *?*?# and DOOR is coded as ?%%# then how will DAM be coded?

- a) ?#%
- b) *#?
- c) *#/
- d) ??#



Q10: In a certain language, if RADAR is coded as *?*?# and DOOR is coded as ?%%# then how will DAM be coded?

- a) ?#%
- b) *#?
- c) *#/
- d) ??#



Q11. If '-' means '+',
'+' means '-',
'*' means '÷' and
'÷' means '*' then

which of the following will be the correct equation?

- A. $10 + 5 - 14 \div 10 * 15 = 158$
- B. $30 + 5 + 14 - 10 * 15 = 122$
- C. $30 - 5 + 14 \div 10 * 15 = 162$
- D. $30 * 5 - 4 \div 10 + 15 = 31$

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- B. $30 + 5 + 14 - 10 * 15 = 122$
- C. $30 - 5 + 14 \div 10 * 15 = 162$
- D. $30 * 5 - 4 \div 10 + 15 = 31$

Correct Answer : Option D

Explanation –

$$30 * 5 - 4 \div 10 + 15 = 31$$

$$= 30 \div 5 + 4 * 10 - 15$$

$$= 31$$



Types of Questions

1. Letters / Alphabets
2. Numbers
3. Symbols
4. Grouping
5. Miscellaneous



Q12: CATACOMBA is coded as MBACATACO then how will BLACKJACK be coded ____



Q12: CATACOMBA is coded as MBACATACO then how will BLACKJACK be coded ____



Types of Questions

1. Letters / Alphabets
2. Numbers
3. Symbols
4. Grouping
5. Miscellaneous



Q13. In a certain code language,

‘speak nicely to all’ is coded as “ka cu ma he”

‘all are like us’ is coded as “si fo he to”

‘teach us lesson nicely’ is coded as “po ma fo re”

‘lesson like all humans’ is coded as “he re gu si”

What would be the code for “lesson”?

- A. re
- B. he
- C. si
- D. fo

Q13. In a certain code language,

‘speak nicely to all’ is coded as “ka cu ma he”

‘all are like us’ is coded as “si fo he to”

‘teach us lesson nicely’ is coded as “po ma fo re”

‘lesson like all humans’ is coded as “he re gu si”

What would be the code for “lesson”?

- A. re
- B. he
- C. si
- D. fo

Correct Answer : Option A

Explanation –

‘speak nicely to all’ ———-> “ka cu ma he”(1)

‘all are like us’ —-> ” si fo he to”(2)

‘teach us lesson nicely’ —-> ” po ma fo re”(3)

‘lesson like all humans’ ———-> “he re gu si”(4)

Inference:

From (1) and (2) we get
all —> he(5)

From (1) and (3) we get
nicely —→ ma.....(6)

From (2) and (4) we get
like —→ si(7)

From (2) and (3) we get
us —→ fo(8)

From (3) and (4) we get
lesson —→ re.....(9)

From (1), (5) and (6) we get
speak to —→ ka cu

From (2), (5), (7) and (8) we get
are —→ to

From (3), (6), (8) and (9) we get
teach —→ po

From (4), (5), (7) and (9) we get
humans —→ gu

Q14: If 'dog' is called 'lion', 'lion' is called 'bison', 'bison' is called 'snake', 'snake' is called 'mangoose', 'mangoose' is called 'crocodile', then which one is reared as pet?

- a) Lion
- b) Dog
- c) Bison
- d) Snake



Q14: If 'dog' is called 'lion', 'lion' is called 'bison', 'bison' is called 'snake', 'snake' is called 'mangoose', 'mangoose' is called 'crocodile', then which one is reared as pet?

- a) Lion
- b) Dog
- c) Bison
- d) Snake



Q15 (VK): In a certain language, if CAP is coded as 61, then how will GIRL be coded?

Girl: Baby I'm having a headache, it's hurting

Boys: Aww baby mmuaaahh

Girl: Yeah it's better now

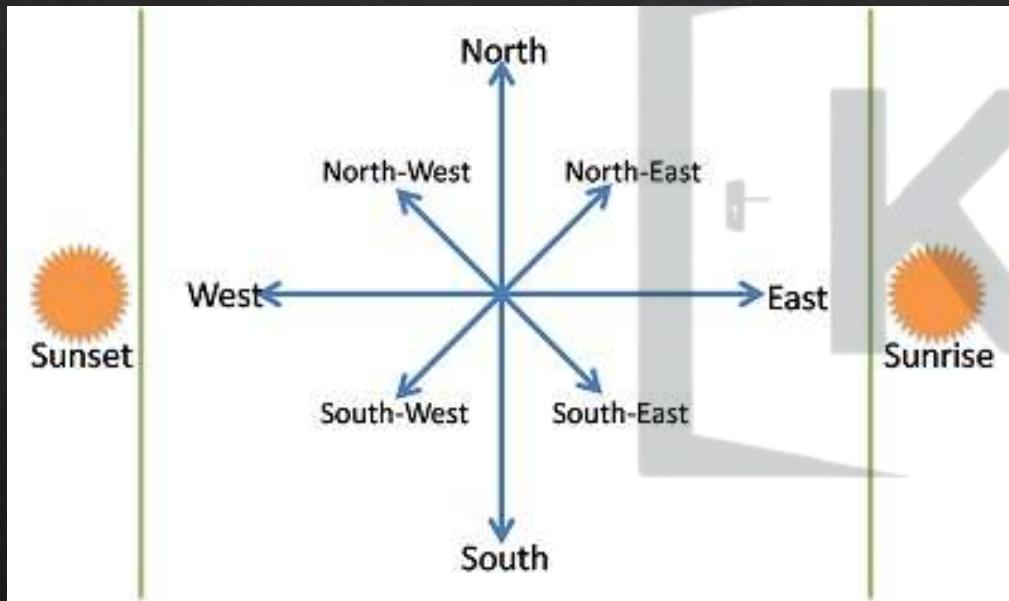




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DIRECTION SENSE

Direction sense - There are four main directions East, West, North and South. We will denote it as E, W, N and S respectively. There are four Cardinal Directions that exist between four main directions namely North-East, North-West, South-East and South-West. We will represent them as NE, NW, SE and SW.

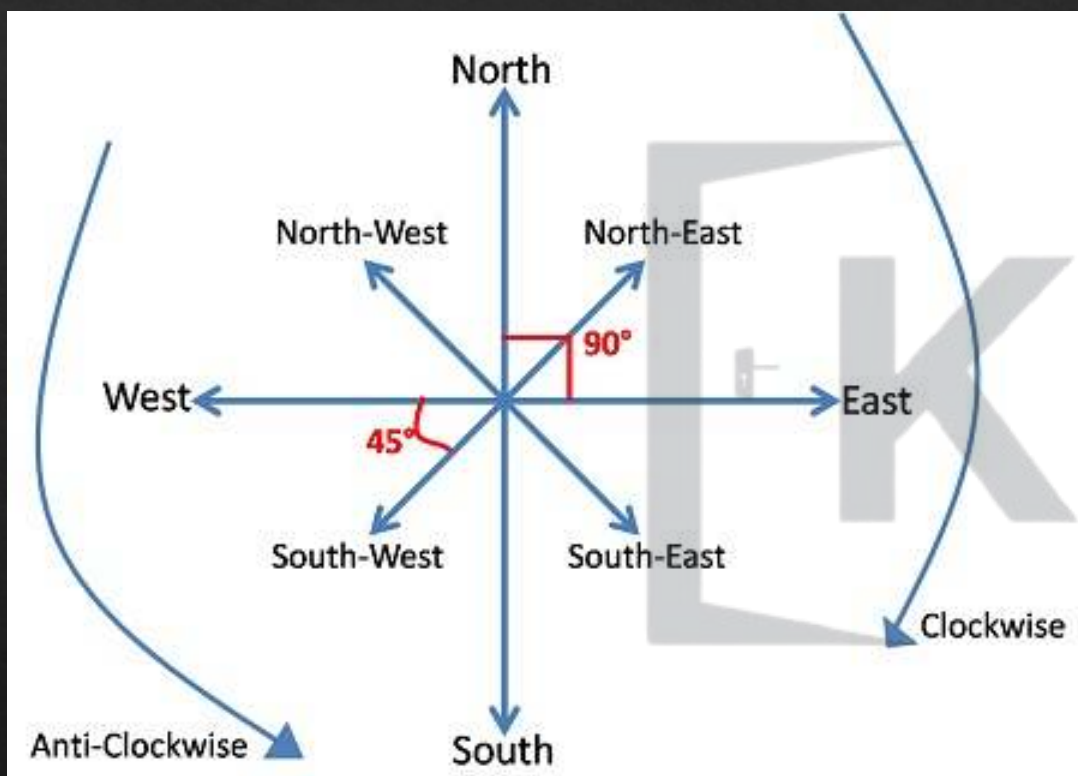


Angle - Now one thing we have to remember is that angle between two main directions is always 90° and angle between main direction and its adjacent cardinal direction is always 45° .

Selection of right or left direction depends on whether it is clockwise (Right turn) or anti-clockwise (Left turn).

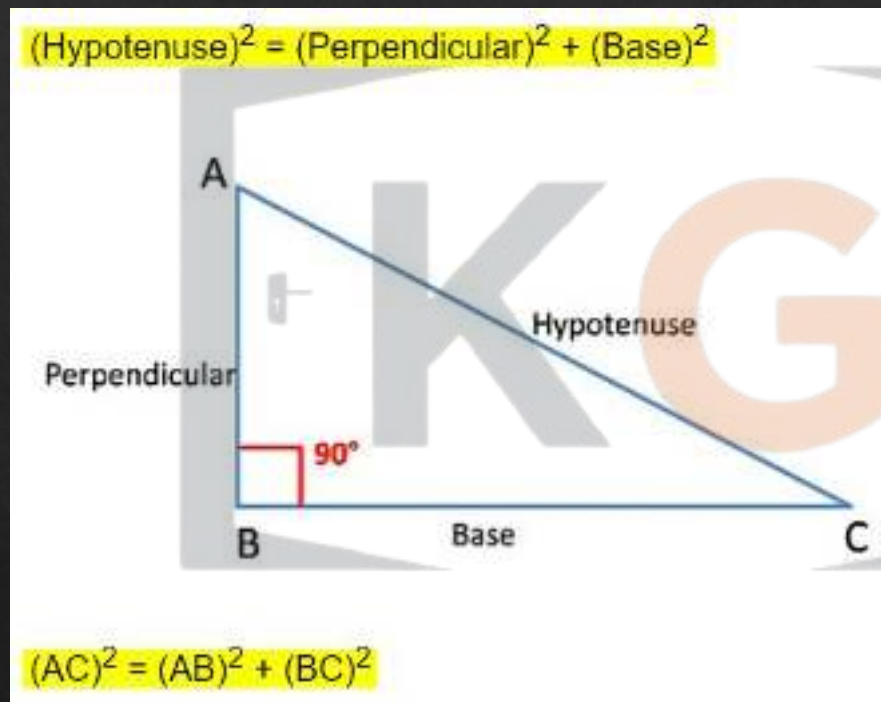


Example: A boy is facing towards west and turns 45° clockwise.
So in this case Boy will turn on his right side 45° .
If it was 45° ANTI-CLOCKWISE then Boy will turn on his left 45° .



Pythagoras Theorem

In Pythagoras Theorem, relation between three sides of the right angle triangle is defined. This theorem is helpful in measuring distance between given points.



Right and Left turn always make 90° angle.



Types of Questions

- 1) Direction Based Questions
- 2) Distance Based Questions
- 3) Direction-Distance Based Questions
- 4) Shadow Based Questions



Type 1 : DIRECTION BASED QUESTIONS

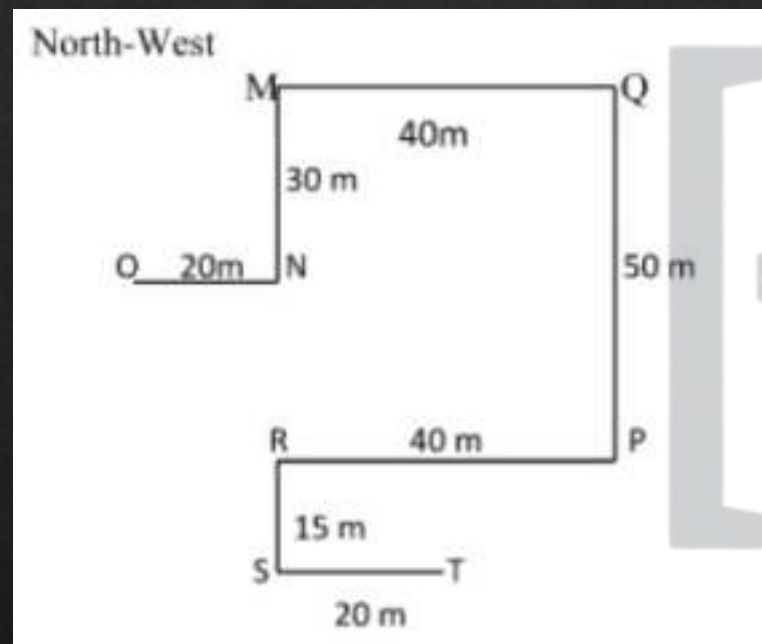


Q1. Eight persons M to T are standing in such a way that O is 20 m apart from N towards West, N is 30 m South with respect to M. M is 40 m towards West with respect to Q. P is 50 m towards South with respect to Q. R is 15 m apart from S towards North. T is 20 m towards East with respect to S. R is 40 m towards West with respect to P. In which direction is O standing with respect to R?

- A. North-West
- B. North
- C. North-East
- D. Cannot be determined

Correct Answer : Option A

Explanation –



Type 2 : DISTANCE BASED QUESTIONS



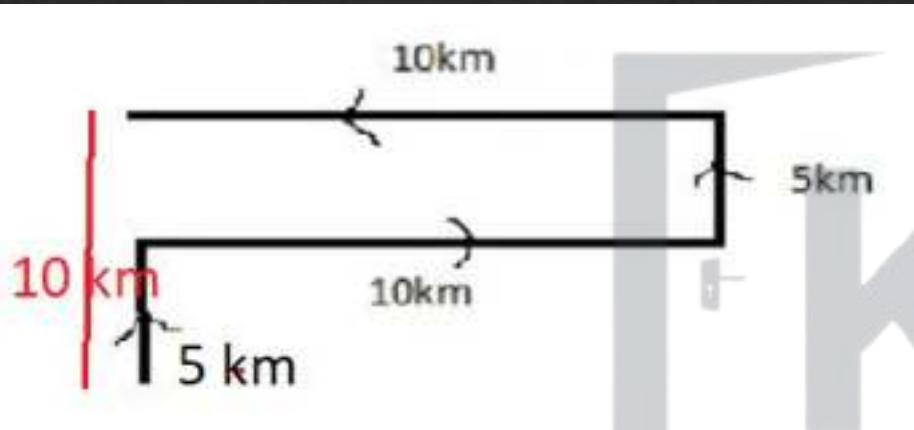
Q2. One day raj left home and walked 5 km northwards, turned right and walked for 10km and turned left and walked 5 km more and finally turned left and walked 10km. How many kilometers will he have to walk to reach his home straight?

- A. 15 Km
- B. 10 Km
- C. 25 Km
- D. 5 Km



Correct Answer : Option B

Explanation –

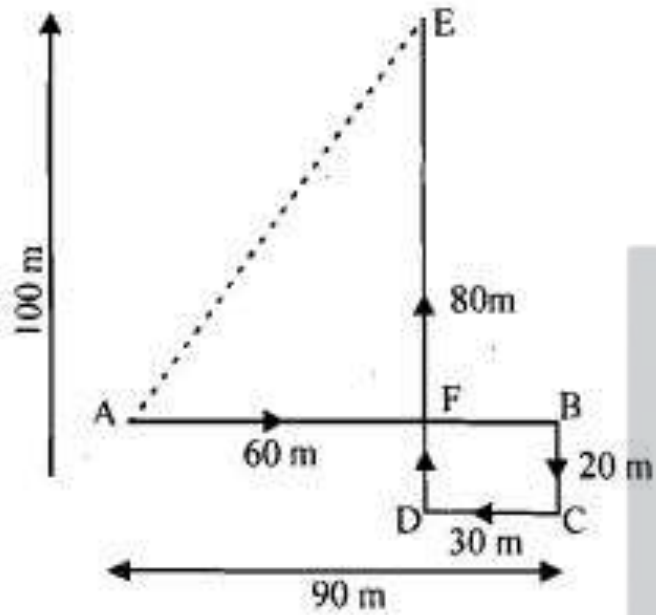


Q3. A child was looking for his father. He went 90 m in the East before turning to his right. He went 20 m before turning to his right again to look for his father at his uncle's place 30 m from this point. His father was not there. From here he went 100 m to the North before meeting his father in a street. How far did the son meet his father from the starting point?

- A. 80 m
- B. 30 m
- C. 90 m
- D. 100 m



Correct Answer : Option D



Type 3 : DIRECTION - DISTANCE BASED QUESTIONS



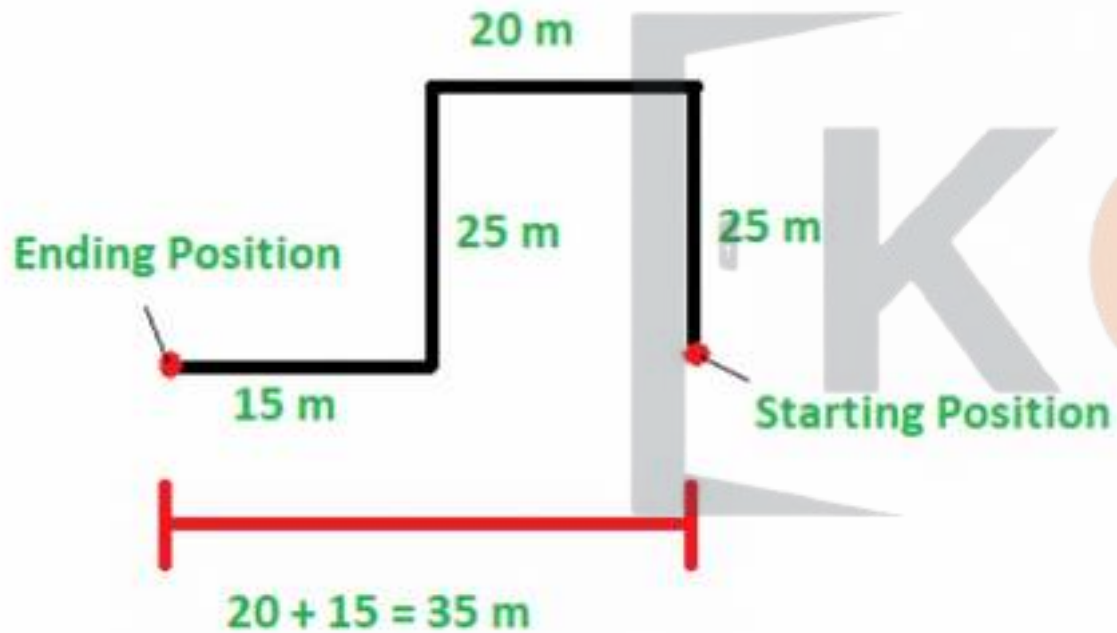
Q4. Salman walked 25 m towards North. Then he turned to his left and walked 20 m. He then turned to his left and walked 25 m. He again turned to his right and walked 15 m. At what distance is he from the starting point and in which direction?

- A. 35m, west
- B. 31m, north
- C. 32m, south
- D. 35m, east



Correct Answer : Option A

Explanation –



Type 4 : SHADOW BASED QUESTIONS



Concept of Shadow (Sunrise)



Concept of Shadow (Sunset)



Key Points

- 1) There would be no Shadow at 12 Noon.
- 2) After sun rise till 12 PM, shadow would be towards west.
- 3) After 12 PM and before sun set, shadow would be towards east.
- 4) If a person faces towards North before 12 pm, his/her shadow would be towards his/her left side and after 12 pm it would be towards his/her right side.
- 5) If a person faces towards South before 12 pm, his/her shadow would be towards his/her right side and after 12 pm it would be towards his/her left side.
- 6) Right and Left turn always make 90° angle.

Q5: One morning after sunrise, Anup Ji was standing facing a pole. The shadow of the pole fell exactly to his right. To which direction was he facing?

- a) East
- b) South
- c) West
- d) Data is inadequate



Correct Answer: Option
B



Q6: Kaleen Bhaiya was standing facing to the Pole at 1:48PM. Shadow of the Pole fell towards his right. To which direction was Kaleen Bhaiya facing?

- a) West
- b) South
- c) East
- d) North



Correct Answer: Option
D



Q7: James and Henry were standing facing each other at 8AM. Shadow of the Henry fell exactly to his right. To which direction was James facing?



- a) East
- b) South
- c) North
- d) Data is inadequate



Correct Answer: Option
C



Q8. If a man stands in front of sun what is the first letter of the direction which is left to him:

- A. N
- B. S
- C. W
- D. Cannot be determined



Correct Answer : Option D

Explanation –

It is given that the man is standing in front of sun, but it is not mentioned whether its morning or evening.

If its morning then he is standing facing East direction then the direction left to him will surely be North direction and first letter of North is 'N'.

But if its evening then he is standing facing West direction then the direction left to him will be South direction and first letter of South is 'S'. Hence answer cannot be determined.

Q9 (VK). From his house, Lokesh went 15 km to the South. Then he turned west and covered 10 km. Then he turned south and covered 5 km. Finally turning to the east, he covered 10 km. In which direction is he from his house?

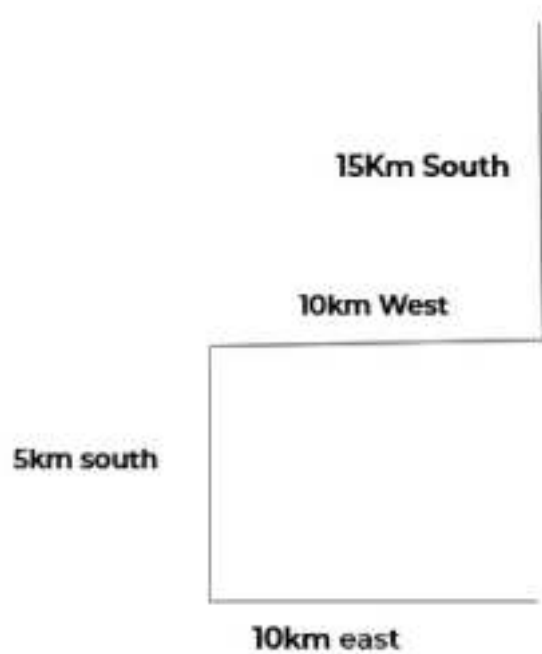


- A. North
- B. North-West
- C. East
- D. South



Correct Answer : Option D

Explanation –





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SEATING ARRANGEMENT

BASIC CONCEPTS

In this topic, there is a lot of raw data given. Based on this data certain conclusions are to be drawn. The given raw data will help us to arrange the objects or people in a predetermined manner after performing necessary analysis over it.



Note:

If a person is facing North, then "Your left and right" will be the same as the "Person facing North".

Left

Right

If a person is facing South, then it will be a mirror image i.e. "Person's Left" will be "Your Right" and vice-versa.

Right

Left

TRICKS TO SOLVE QUESTIONS

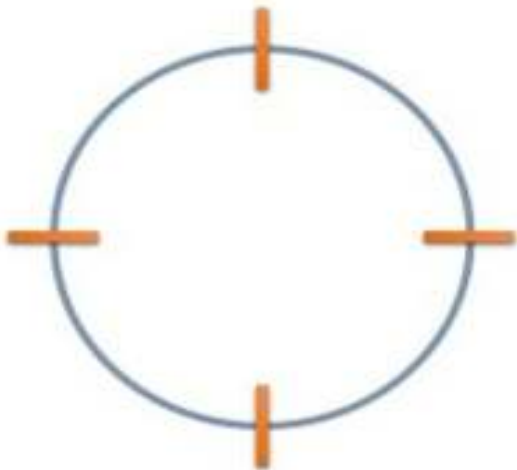
Step-1 : First go through the entire question and count total number of people and prepare a "Seating Frame".

Example

1. If there are 7 persons in a single row.



2. If there are 4 persons on a circular table.



Creating such a "Seating Frame" will help to identify and mark the positions of the people from the given questions in an easy manner

TRICKS TO SOLVE QUESTIONS

Step-2: Solve individual statements and then merge all the solved small sets into a final complete set.



TYPES OF SEATING ARRANGEMENTS

In seating arrangement, questions are mainly based on 4 main types:

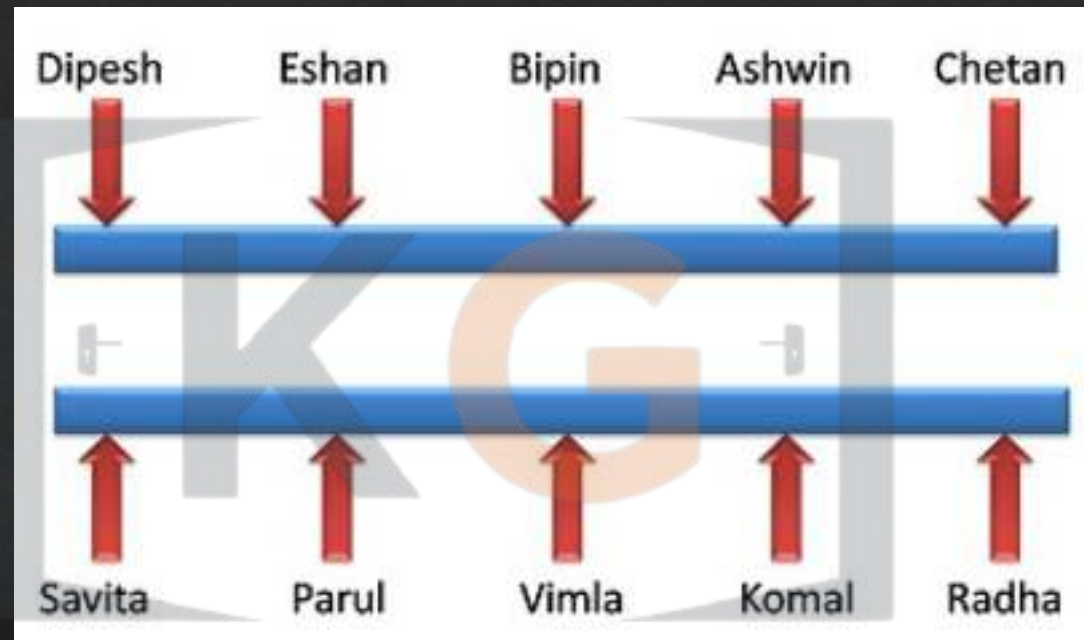
- Single Row
- Double Row
- Circular Table
- Polygonal Table



TYPES OF SEATING ARRANGEMENTS

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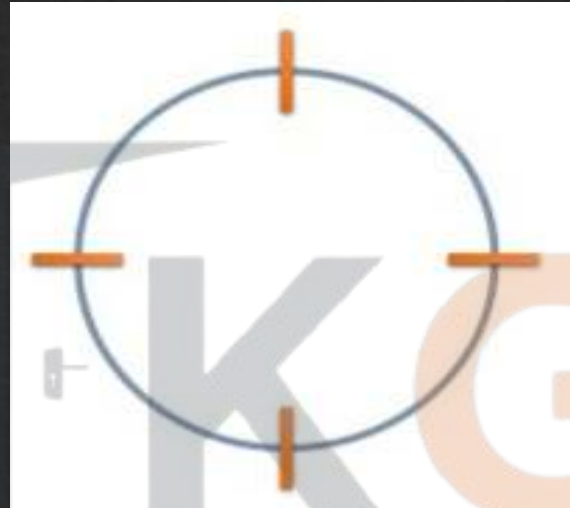
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TYPES OF SEATING ARRANGEMENTS

In seating arrangement, questions are mainly based on 4 main types:

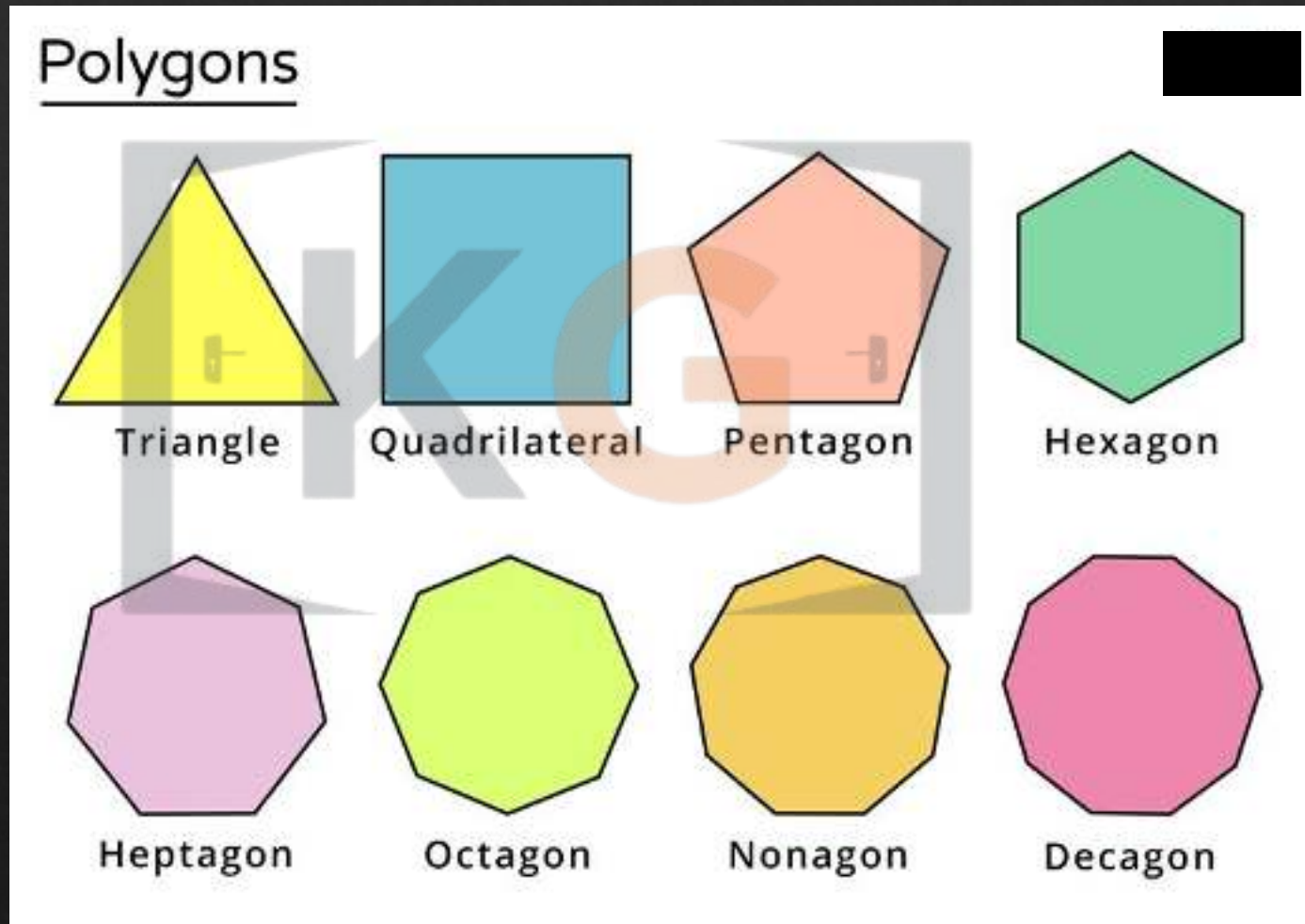
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- Double Row
- Circular Table
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TYPES OF SEATING ARRANGEMENTS

In seating arrangement, questions are mainly based on 4 main types:

- Single Row
- Double Row
- Circular Table
- Polygonal Table



TYPE 1 : SINGLE ROW SEATING ARRANGEMENT



Q1. A,B, C, D and E are sitting on a bench. A is sitting next to B, C is sitting next to D, D is not sitting with E who is on the left end of the bench. C is on the second position from the right. A is to the right of B and E. A and C are sitting together. In which position A is sitting ?

- A. Between B and D
- B. Between B and C
- C. Between E and D
- D. Between C and E



Q1. A, B, C, D and E are sitting on a bench. A is sitting next to B, C is sitting next to D, D is not sitting with E who is on the left end of the bench. C is on the second position from the right. A is to the right of B and E. A and C are sitting together. In which position A is sitting?

- A. Between B and D
- B. Between B and C
- C. Between E and D
- D. Between C and E



Correct Answer : Option B

Explanation –

•
E •
B •
A •
C •
D •

Therefore, A is sitting in between B and C.

TYPE 2 : DOUBLE OR TWO ROW SEATING ARRANGEMENT



Q2. Five boys Ashwin, Dipesh, Eshan, Chetan and Bipin and five girls Parul, Komal, Radha, Savita and Vimla sit in two rows facing towards each other.

- All the boys are in one row and all the girls in the other row.
- Eshan who is to the immediate right of Bipin and opposite to Parul is not at any end.
- Radha, who is immediate to the right of Komal and opposite to Chetan, is at one of the ends.
- Ashwin is opposite to Komal who is the third to the right of Savita.
- Dipesh and Vimla are not opposite each other.

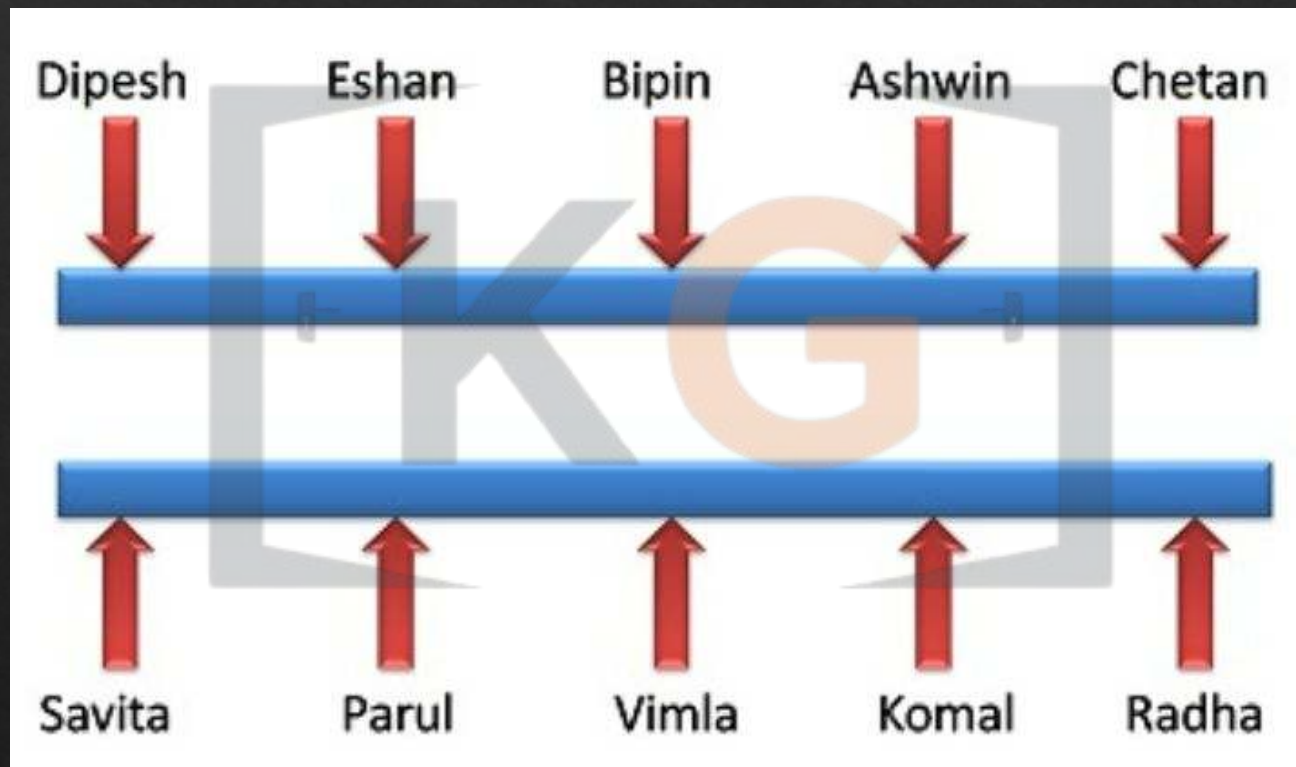
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- Ashwin is opposite to Komal who is the third to the right of Savita.
- Dipesh and Vimla are not opposite each other.

- Eshan who is to the immediate right of Bipin and opposite to Parul is not at any end.
- Radha, who is immediate to the right of Komal and opposite to Chetan, is at one of the ends.
- Ashwin is opposite to Komal who is the third to the right of Savita.
- Dipesh and Vimla are not opposite each other.



- Eshan who is to the immediate right of Bipin and opposite to Parul is not at any end.
- Radha, who is immediate to the right of Komal and opposite to Chetan, is at one of the ends.
- Ashwin is opposite to Komal who is the third to the right of Savita.
- Dipesh and Vimla are not opposite each other.



i) Who is in the middle of the row of boys?

- a) Ashwin
- b) Bipin
- c) Dipesh
- d) None of these

ii) Who is immediately to the right of Parul?

- a) Radha
- b) Savita
- c) Vimla
- d) Komal



i) Who is in the middle of the row of boys?

- a) Ashwin
- b) Bipin
- c) Dipesh
- d) None of these

ii) Who is immediately to the right of Parul?

- a) Radha
- b) Savita
- c) Vimla
- d) Komal



iii) Who is opposite of Dipesh?

- a) Komal
- b) Savita
- c) Parul
- d) Radha

iv) How many boys are there between Ashwin and Bipin?

- a) 0
- b) 1
- c) 2
- d) 3



iii) Who is opposite of Dipesh?

- a) Komal
- b) Savita
- c) Parul
- d) Radha

iv) How many boys are there between Ashwin and Bipin?

- a) 0
- b) 1
- c) 2
- d) 3



Answers:

- i) Bipin
- ii) Vimla
- iii) Savita
- iv) 0



TYPE 3 : CIRCULAR TABLE SEATING ARRANGEMENT

This type has three sub types:

1. Facing Inwards
2. Facing Outwards
3. Facing Inwards-Outwards





TYPE 3-A : FACING INWARD



Q3. Six friends (Rupesh, Priti, Ashwini, Sachin, Dipen, Rishi) are sitting in a circle facing inwards.

1. Priti and Ashwini are exactly opposite to each other.
2. Sachin is in between Priti and Dipen.
3. Dipen is immediately to the left of Ashwini.
4. Rishi is not exactly opposite to Dipen.



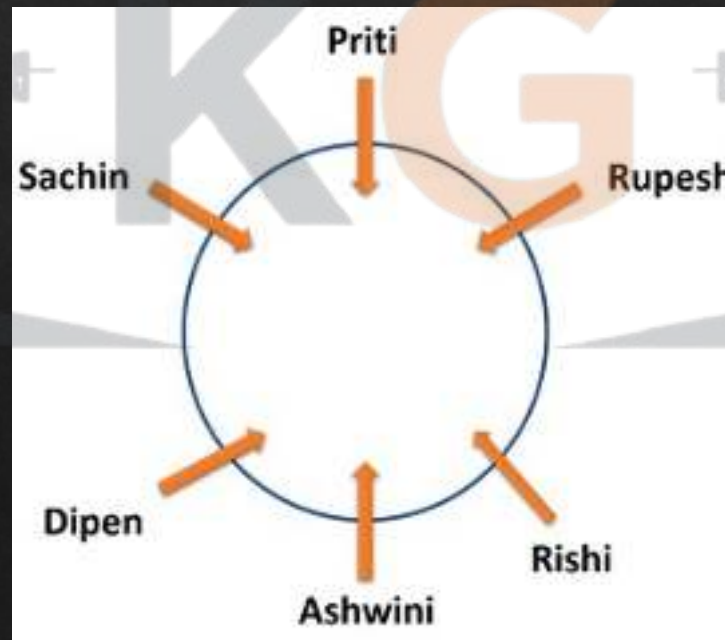
Q3. Six friends (Rupesh, Priti, Ashwini, Sachin, Dipen, Rishi) are sitting in a circle facing inwards.

1. Priti and Ashwini are exactly opposite to each other.
2. Sachin is in between Priti and Dipen.
3. Dipen is immediately to the left of Ashwini.
4. Rishi is not exactly opposite to Dipen.



Q3. Six friends (Rupesh, Priti, Ashwini, Sachin, Dipen, Rishi) are sitting in a circle facing inwards.

1. Priti and Ashwini are exactly opposite to each other.
2. Sachin is in between Priti and Dipen.
3. Dipen is immediately to the left of Ashwini.
4. Rishi is not exactly opposite to Dipen.



1. Who are the neighbors of Dipen?

- a. Rishi and Rupesh
- b. Ashwini and Sachin
- c. Priti and Sachin
- d. None of these

2. Who is sitting opposite to Dipen?

- a. Rupesh
- b. Ashwini
- c. Sachin
- d. Rishi

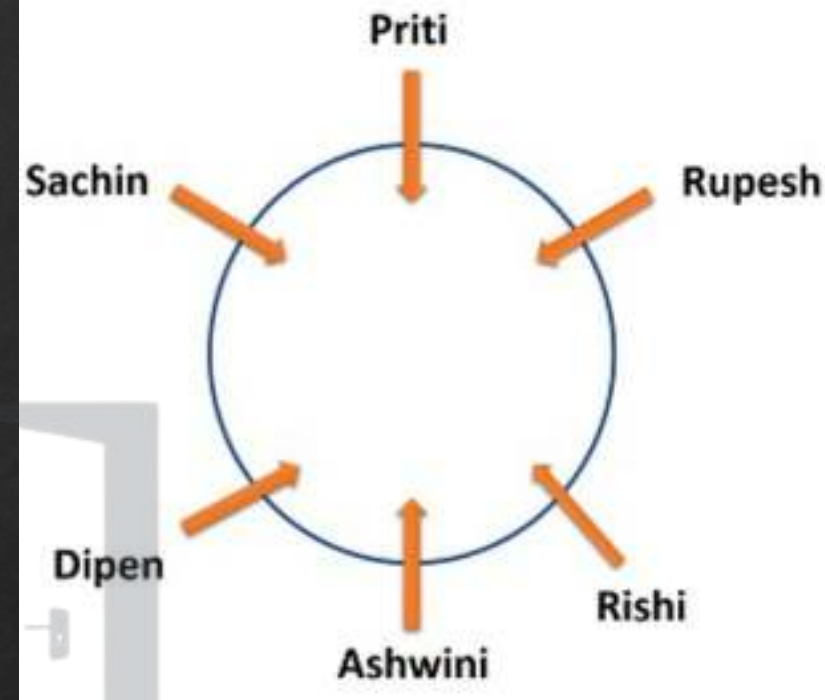


1. Who are the neighbors of Dipen?

- a. Rishi and Rupesh
- b. Ashwini and Sachin
- c. Priti and Sachin
- d. None of these

2. Who is sitting opposite to Dipen?

- a. Rupesh
- b. Ashwini
- c. Sachin
- d. Rishi



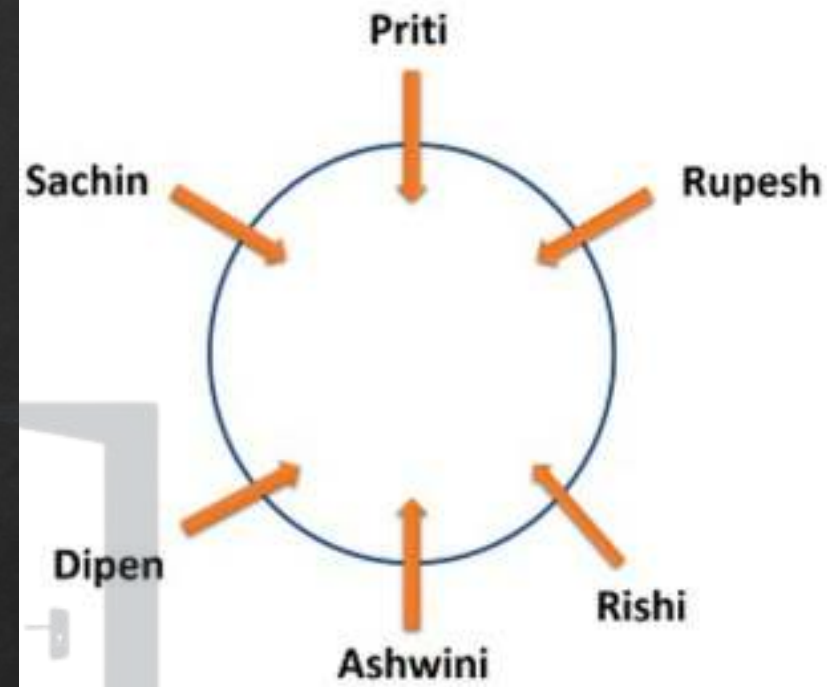
3. Who is just right to Ashwini?

- a. Rishi
- b. Sachin
- c. Ashwini
- d. Rupesh



3. Who is just right to Ashwini?

- a. Rishi
- b. Sachin
- c. Ashwini
- d. Rupesh



TYPE 3-B : FACING OUTWARD



Q4. Six Friends A, B, C, D, E and F are sitting on a circular table facing outwards.

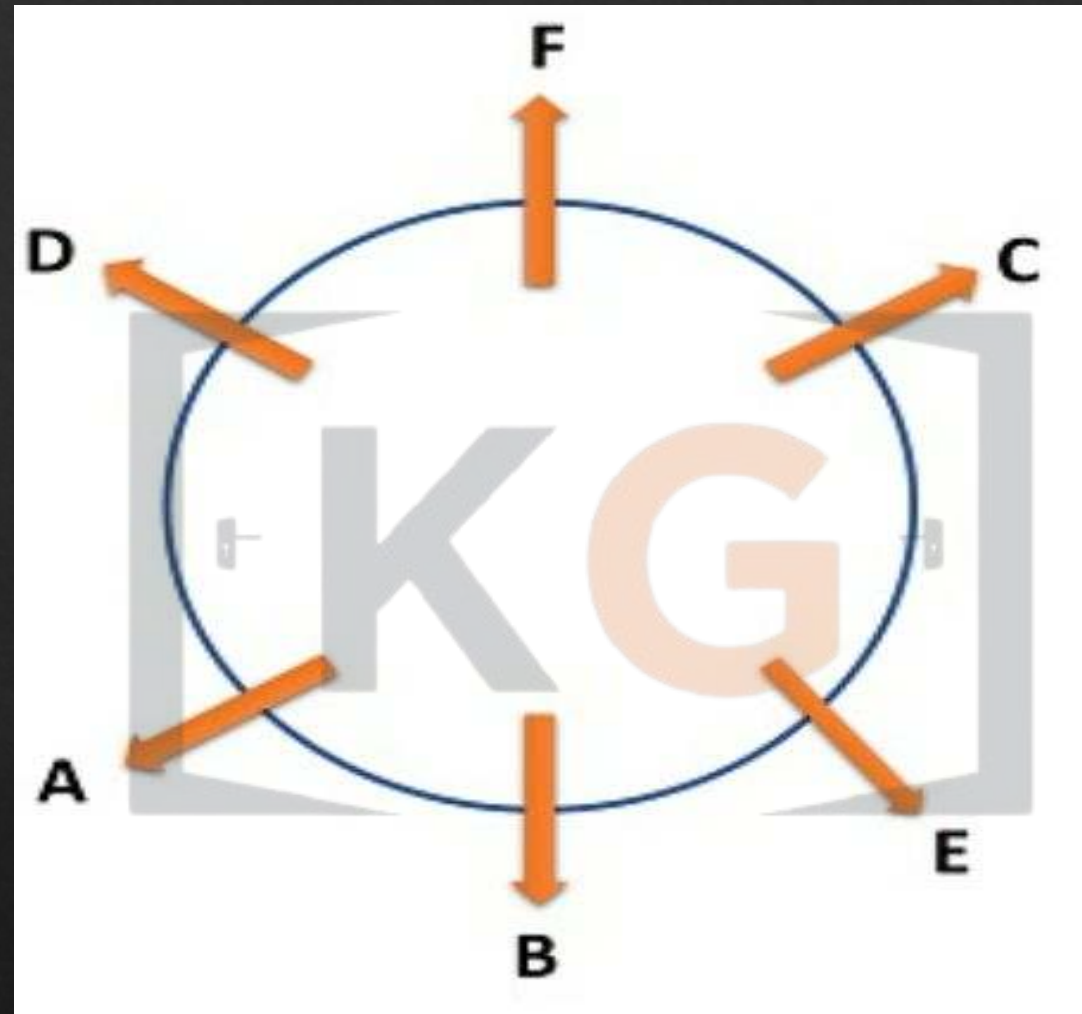
- E is between B and C.
- B and F are exactly opposite to each other.
- A is exactly to the right of B.



Q4. Six Friends A, B, C, D, E and F are sitting on a circular table facing outwards.

- E is between B and C.
- B and F are exactly opposite to each other.
- A is exactly to the right of B.





1. Who is in between D and B?

- a. E
- b. A
- c. F
- d. C

2. Who is exactly to the right of D?

- a. A
- b. C
- c. F
- d. E

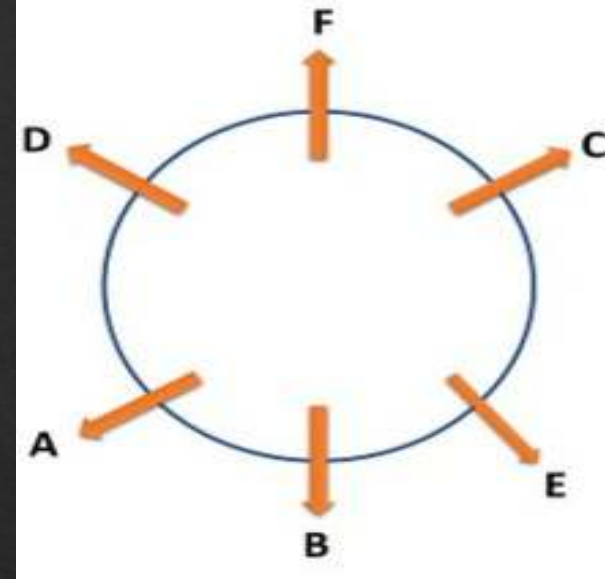


1. Who is in between D and B?

- a. E
- b. A
- c. F
- d. C

2. Who is exactly to the right of D?

- a. A
- b. C
- c. F
- d. E



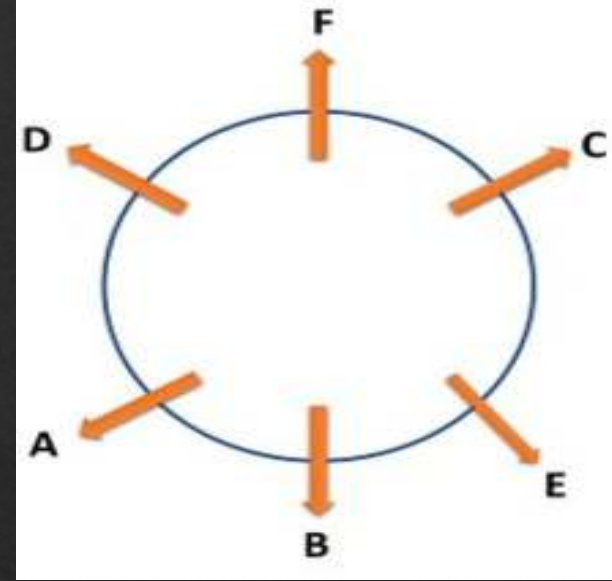
3. Who are the neighbors of A?

- a. B and D
- b. B and E
- c. F and E
- d. F and C



3. Who are the neighbors of A?

- a. B and D
- b. B and E
- c. F and E
- d. F and C



TYPE 3-C : FACING INWARDS - OUTWARDS



Q5. Six Friends A, B, C, D, E and F are sitting on a circular table 2 facing inwards and are exactly opposite to each other and rest facing outwards.

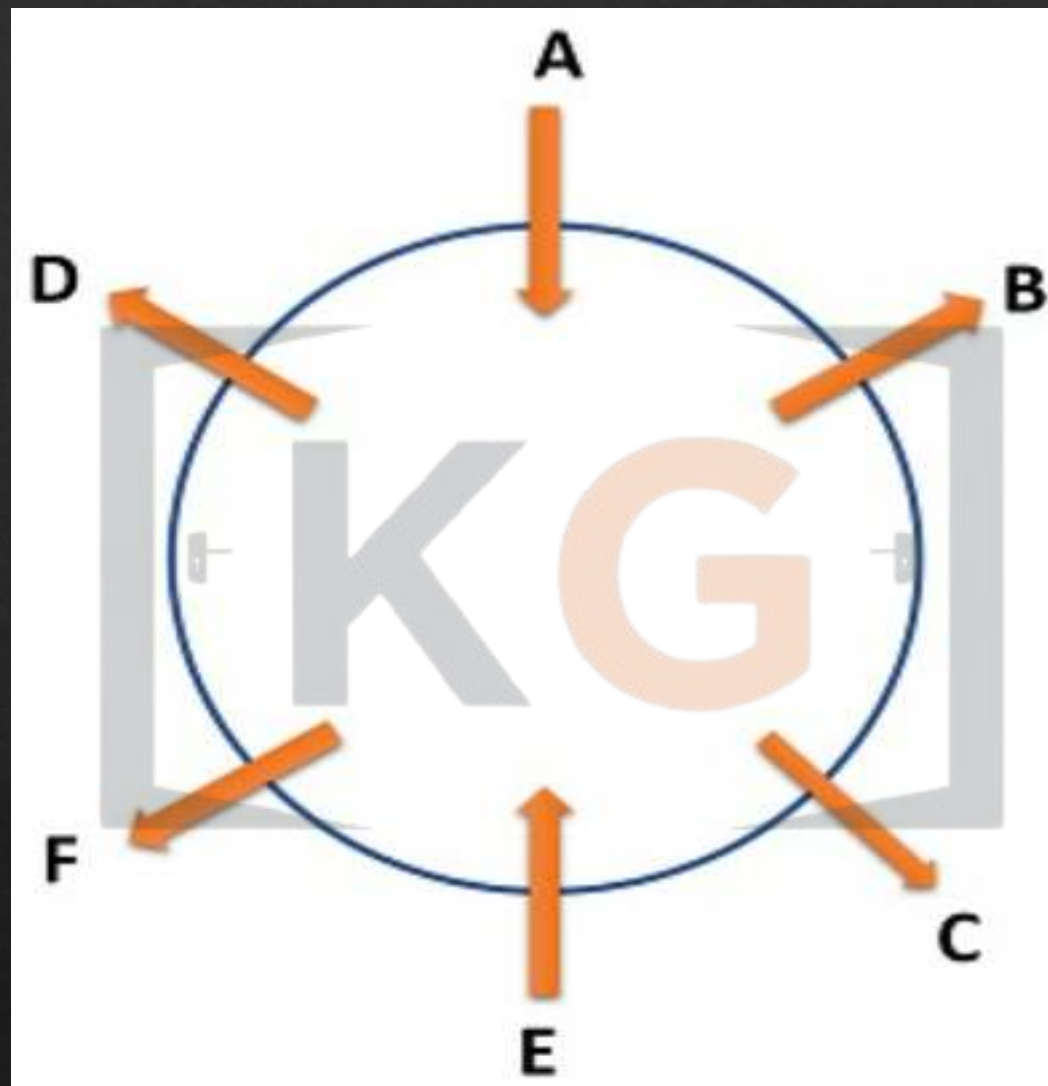
- B is facing outwards and is left to C.
- F is in between E and D.
- E is facing inwards.
- A is 2nd to the left of C.
- C is not facing inwards.



Q5. Six Friends A, B, C, D, E and F are sitting on a circular table 2 facing inwards and are exactly opposite to each other and rest facing outwards.

- B is facing outwards and is left to C.
- F is in between E and D.
- E is facing inwards.
- A is 2nd to the left of C.
- C is not facing inwards.





1. 2 persons exactly opposite to each other and facing center are:

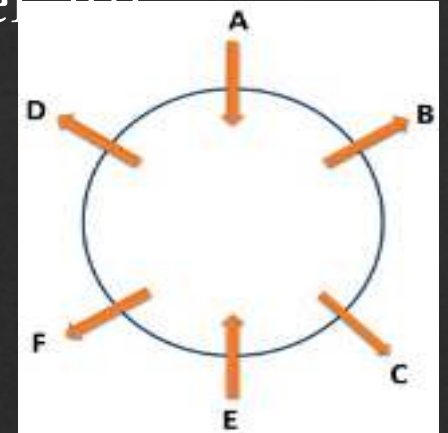
- a. B and F
- b. B and C
- c. A and E
- d. E and B

2. If positions of B and D and F and C are interchanged, then who is 2 places left to B

- a. C
- b. E
- c. A
- d. D

1. 2 persons exactly opposite to each other and facing center are:

- a. B and F
- b. B and C
- c. A and E
- d. E and B



2. If positions of B and D and F and C are interchanged, then who is 2 places left to B

- a. C
- b. E
- c. A
- d. D

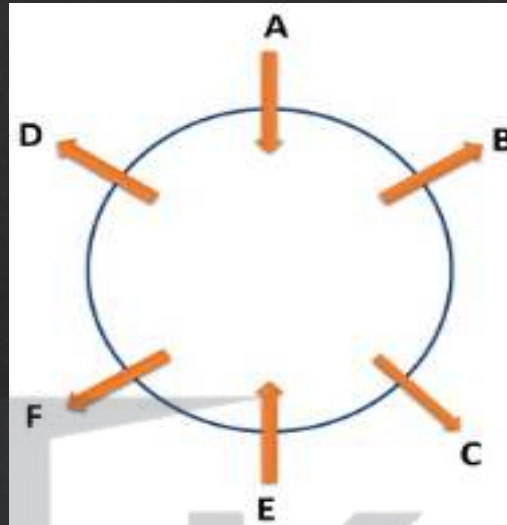
3. E is in between

- a. B and D
- b. F and C
- c. B and C
- d. None of the above



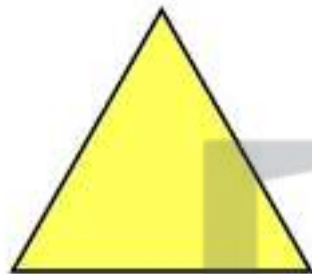
3. E is in between

- a. B and D
- b. F and C
- c. B and C
- d. None of the above



TYPE 4 : POLYGON SEATING ARRANGEMENT

Polygons



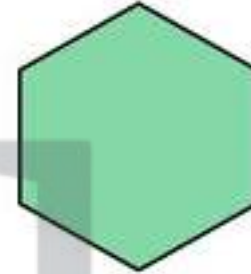
Triangle



Quadrilateral



Pentagon



Hexagon



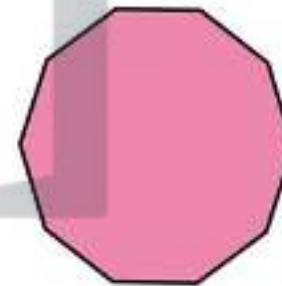
Heptagon



Octagon



Nonagon



Decagon

Q6. Eight boy's A, B, C, D, E, F, G, H are seating around a square table, they seating in such way that four sits in middle of each four sides and facing outsides, and another four boy's sits at the four corners facing center.

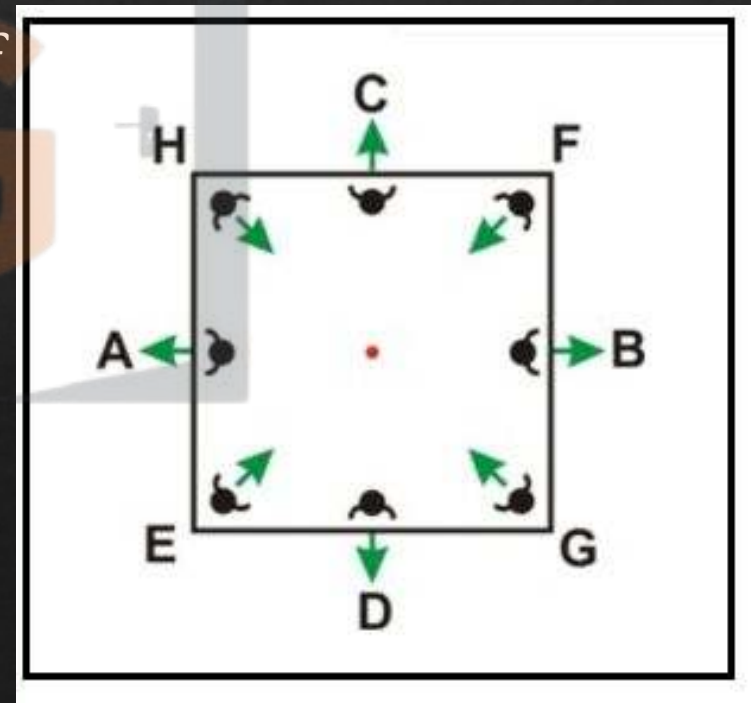
- E who faces the center sits 3rd to the right of B.
- H faces center who is not immediate neighbor of B.
- One boy sits between B and D.
- G faces center, F sits second to the right of G.
- C is not immediate neighbor of E.

Q6. Eight boy's A, B, C, D, E, F, G, H are seating around a square table, they seating in such way that four sits in middle of each four sides and facing outsides, and another four boy's sits at the four corners facing center.

- E who faces the center sits 3rd to the right of B.
- H faces center who is not immediate neighbor of B.
- One boy sits between B and D.
- G faces center, F sits second to the right of G.
- C is not immediate neighbor of E.

Q6. Eight boy's A, B, C, D, E, F, G, H are seating around a square table, they seating in such way that four sits in middle of each four sides and facing outsides, and another four boy's sits at the four corners facing center.

- E who faces the center sits 3rd to the right of B.
- H faces center who is not immediate neighbor of B.
- One boy sits between B and D.
- G faces center, F sits second to the right of
- C is not immediate neighbor of E.



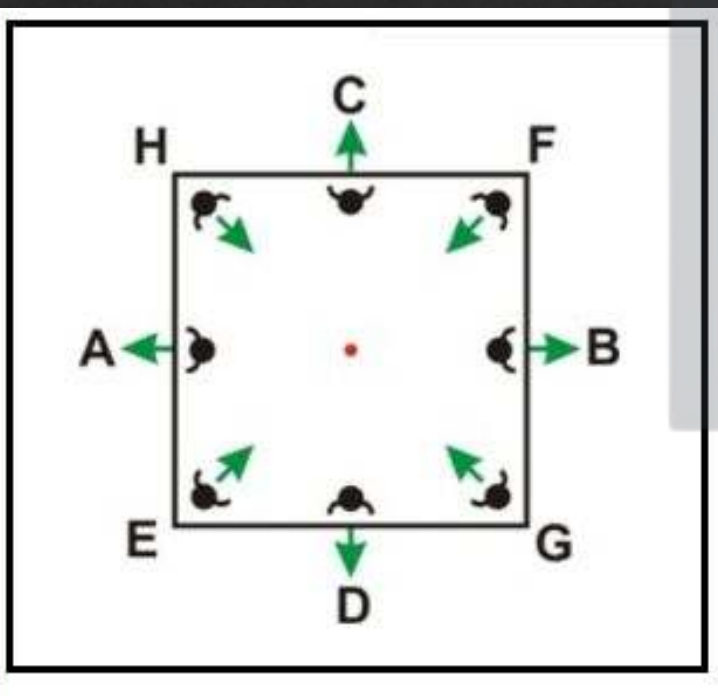
i) Who sits diagonally of E ?

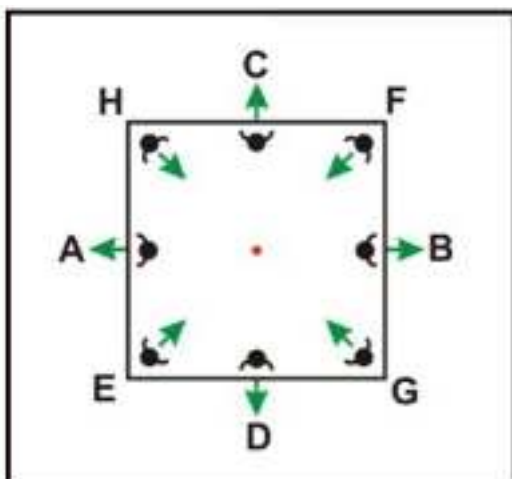
- A) G
- B) C
- C) D
- D) F



i) Who sits diagonally of E ?

- A) G
- B) C
- C) D
- D) F





We can easily see that F sits diagonally of E

Hence the correct option is D.

ii) What should be the places in the question mark ?

EF GH CD ?

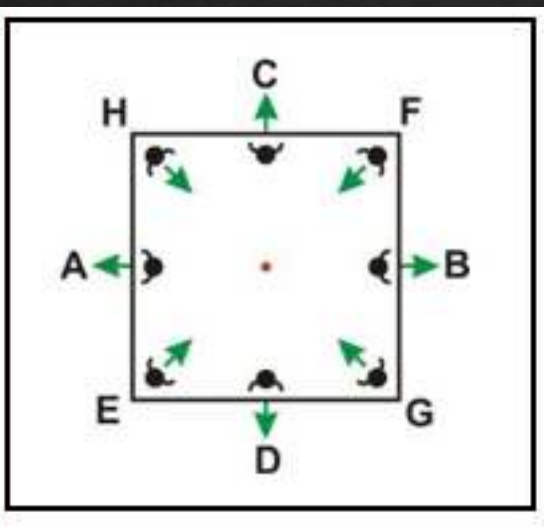
- A) ED
- B) AB
- C) HC
- D) BD

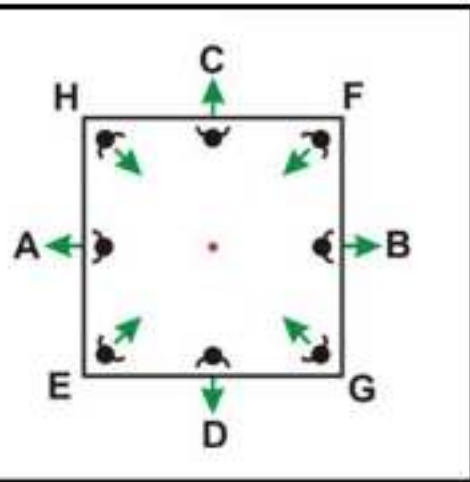


ii) What should be the places in the question mark ?

EF GH CD ?

- A) ED
- B) AB
- C) HC
- D) BD





We can easily see that EF, GH, CD sit opposite to each other, Similarly, AB sits opposite to each other.

Hence the correct option is B.

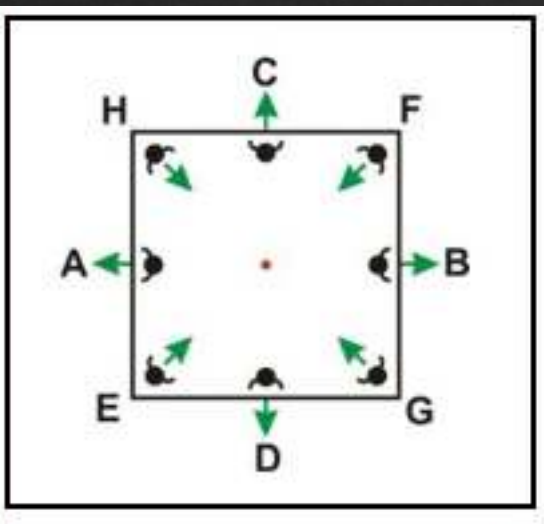
iii) Who sits second to the left of G ?

- A) F
- B) E
- C) A
- D) C



iii) Who sits second to the left of G ?

- A) F
- B) E
- C) A
- D) C



Q7 (VK). A, P, R, X, S and Z are sitting in a row. S and Z are in the center. A and P are at the ends. R is sitting to the left of A. Who is to the right of P ?



- A. A
- B. X
- C. S
- D. Z



Correct Answer : Option B

Explanation –

Explanation: The seating arrangement is as follows:

•	•	•	•	•	•
P	X	S	Z	R	A

Therefore, right of P is X.